

Supplementary Material for
Random-walk SIR models of indoor epidemics:
reproduction numbers, discrete individuals and
pre-specified tests against outbreaks, contact records
and tracer measurements

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References to equations, figures, tables, sections and statements without the prefix S are to the main text. The repository named in the main text holds the code and data behind every number; each number in the LaTeX source of this document carries a comment naming the file and key it comes from.

Contents

S1	Scenes and notation	1
S2	The basic reproduction number: proofs and further results	3
S3	Room size, leaks and hotspot maps: proofs and worked examples	8
S4	Pair-level theory: proofs and further results	14
S5	Simulations of the scenes and of interventions: details	18
S6	The outbreak records	33
S7	The first outbreak test in detail	40
S8	Protocol of the first outbreak test, sensitivity analyses and post-hoc log	47
S9	The second outbreak test and the further analyses in detail	55
S10	Protocols of the second outbreak test and of the further analyses; register of all tests	68
S11	Verification of the theorems, the simulator and the analyses	77

S1 Scenes and notation

Table S2 lists the symbols used in more than one section. Time is in days and lengths are in lattice cells in Sections 2–6 of the main text; Section 7 uses hours and metres. Labels of criteria and outcomes are local to their analysis: A1–A4 and W1–W4 belong to the first outbreak test, B1–B3 and V1–V5 to the second, A0 (the lowest outcome of its ladder) to the analysis of contact records, D1–D3 and S1–S5 to the tracer analysis; the amendments V1–V7 and data gaps D1–D6

Table S1: Zones of the four scenes of Table 2: number of cells, mobility multiplier $d = D/D_0$ and infection efficiency q . The metro multipliers are $d = c n_d^{-\eta}$ evaluated at the crowd density $n_d = 4.59$ people per m^2 . All values are illustrative assumptions; none is a measurement.

Zone	cells	d	q	Zone	cells	d	q
<i>Office</i>				<i>Classroom</i>			
workstations	156	0.3	1.4	desks, window side	40	0.05	1.2
corridor	39	1.5	0.8	desks, inner	40	0.05	2.0
meeting room	26	0.5	1.5	platform	18	0.8	1.5
kitchen	13	1.0	1.2	aisles	50	0.2	0.8
barriers	26	–	–	obstacle	2	–	–
<i>Supermarket</i>				<i>Metro carriage</i>			
shelf lanes	84	0.4	1.0	seats	60	0.0044	2.5
main aisles	142	2.0	0.6	standing area	132	0.0047	2.8
fresh produce	24	0.3	1.8	doors	60	0.11	2.0
checkout queue	16	0.1	2.0	grab rails	18	0.011	2.5
entrance and exit	12	1.5	0.8				
shelf racks, counters	97	–	–				

of Table S24, the records of Section S6 (for example S1, a supermarket) and the events W1, W2 of the second dataset are unrelated to them. In Section 7, K also denotes the number of cases of an event, and P with a subscript a set-based infectious pressure (Section S10.5).

Table S2: Notation. Inequalities between vectors or matrices are entrywise; the column index of a matrix is the source cell.

Symbol	Meaning	Symbol	Meaning
Ω, n	walkable cells and their number	R_0	basic reproduction number, $\rho(\mathbf{K})$
x, y, z	cells	λ_1	growth rate, $s(\mathbf{J})$
a	lattice spacing (m)	$s(\cdot), \rho(\cdot)$	spectral abscissa, spectral radius
$D_x = D_0 d_x$	mobility of cell x ; scale and zone multiplier	$n_{\text{off}}(x)$	offspring number, column sum of $\mathbf{K}$
$\omega_{x \rightarrow y}$	hop rate from x to a neighbour y	$\mathbf{u}, \ell$	right and left Perron vectors of $\mathbf{J}$
$\mathbf{M}, \mathbf{M}_0$	movement generator; at $D_0 = 1$	$\mathbf{w}, \mathbf{z}$	right and left Perron vectors of $\mathbf{K}$
$\boldsymbol{\pi}$	stationary distribution of $\mathbf{M}$	$\mathbf{e}$	elasticities $\partial \ln R_0 / \partial \ln q_x$
$P_t(y x)$	propagator, $(e^{\mathbf{M}t})_{yx}$	μ_k, ϕ_k	eigenvalues and orthonormal eigenvectors of $-\Pi^{-1/2} \mathbf{M} \Pi^{1/2}$, $\Pi = \text{diag } \boldsymbol{\pi}$ (of $-\mathbf{M}$ for the symmetric rule)
$q_x, \mathbf{Q}$	infection efficiency; $\text{diag } q$	$\tilde{q}_{jk}$	$\phi_j^T \mathbf{Q} \phi_k$
$\langle q \rangle_\pi, q_{\max}, q_{\min}$	$\sum_x \pi_x q_x$; largest and smallest q_x	Z, μ_Z	zone; its exit rate $-s(\mathbf{M}[Z, Z])$
β, γ	transmission and recovery rates	$\kappa_x, \mathbf{M}_\kappa$	exit rate; $\mathbf{M} - \text{diag } \kappa$
$\mathbf{P}$	contact kernel, $P_{xy} = P(y x)$	μ_{leak}	leak rate, $-s(\mathbf{M}_\kappa)$
N	number of people	ν	stationary law of the walk conditioned to stay
S_x, I_x, R_x	expected numbers in cell x	R_0^{room}	reproduction number of a leaky room
$\mathbf{F}, \mathbf{V}$	$\beta \mathbf{P}^T \mathbf{Q}; \gamma \mathbf{1} - \mathbf{M}$	$\bar{\rho}$	other people per cell, $(N - 1)/n$
$\mathbf{K}$	next-generation matrix $\mathbf{FV}^{-1}$	$h(x, y)$	pair hazard $\beta q_x P_{xy} / \bar{\rho}$
$\mathbf{J}$	$\mathbf{M} + \mathbf{F} - \gamma \mathbf{1}$	$p(x, y)$	pair infection probability
τ	infectious period (exponential, rate γ)	$R_1(x), \bar{R}_1$	pair-level secondary cases; uniform average
$f, \chi(t), \theta(t)$	occupied fraction of the day; schedule multiplier; cumulative occupied time	$\mathbf{K}_1$	pair-level next-generation matrix
$\hat{R}$	counted secondary cases in simulation	g_0	co-location Green's function of a pair started together (on the torus: return Green's function of the relative walk)
FD, DD	frequency- and density-dependent calibration	$\mathbf{M}_0, \mathbf{M}_1, \mathbf{M}_2$	models of Section 7
$P_{\text{maj}}, P_{\text{br}}$	simulated and branching probabilities of a major outbreak	$D_p, D_{\text{air}}, \Lambda$	mobility of people, mixing coefficient and removal rate of quanta
ϱ	near/far ratio of the constant-ratio model (of hazards in the second outbreak test, of risks otherwise); also an eigenvalue in a proof (Section S2)	Δ_0, Δ_C	log score of a model minus that of $\mathbf{M}_0$ and of the constant-ratio model
f_{iz}	fraction of the time of group i spent in zone z (Section 7.9)	ε_k	infectiousness parameter of event k

S2 The basic reproduction number: proofs and further results

This section proves the statements of Section 3 of the main text and gives the variational and modal forms of R_0 . Throughout, Assumption A holds, $\mathbf{V} = \gamma \mathbf{1} - \mathbf{M}$, the column index of a matrix is the source cell, and inequalities between vectors or matrices are entrywise. The proofs use only the standard facts of Lemma S2.1, which are cited and not re-proved. On attribution (Section 1 and Table 1 of the main text): the monotone decrease of the growth bound with the dispersal rate is Lemma 3.4 of Allen et al. [1], as cited by Gao and Dong [15], whose Theorem 2.3 shows that R_0 is strictly decreasing and strictly convex in the dispersal rate unless all patch reproduction numbers coincide; theorem numbers are those of the published version of Gao and Dong [15]. Our proof of Theorem 3.3(c) follows the reduction-principle argument of Altenberg [2], which rests on the convexity theorem of Cohen [12]. The fast-mixing expansion of Proposition 3.4(i) has the form of the first-order correction obtained by Tien et al. [68] (their Section 4.2) for a network on which the pathogen moves: with equal decay rates their generalised group inverse is

the group inverse of the movement generator, and their correction term becomes $\beta C/(\langle q \rangle_\pi D_0)$ (their eqs. (29)–(31) with transmissibilities βq_x and stationary weights π_x).

S2.1 Standard facts

Lemma S2.1 (Standard facts). *A real square matrix is Metzler if its off-diagonal entries are non-negative.*

- (P1) *If A is Metzler and irreducible, $s(A)$ is a real, algebraically simple eigenvalue with strictly positive right and left eigenvectors, and no other eigenvalue has a non-negative eigenvector.*
- (P2) *If A is Metzler, $v \geq 0$, $v \neq 0$ and $Av \geq \mu v$, then $s(A) \geq \mu$. In particular $s(A) \geq s(A[Z, Z])$ for every principal submatrix.*
- (P3) *If A is Metzler and irreducible and $E \geq 0$, $E \neq 0$, then $s(A + E) > s(A)$.*
- (P4) *If $0 \leq A \leq B$ entrywise then $\rho(A) \leq \rho(B)$, with strict inequality if moreover B is irreducible and $A \neq B$; $\rho(A)$ is an eigenvalue of $A \geq 0$ with non-negative right and left eigenvectors.*
- (P5) *Let A have non-positive off-diagonal entries. Then A is a non-singular M-matrix iff $s(-A) < 0$ iff A^{-1} exists and $A^{-1} \geq 0$. If $A \leq B$ are non-singular M-matrices then $A^{-1} \geq B^{-1} \geq 0$. If all column sums of A are positive, A is a non-singular M-matrix.*
- (P6) *For a Metzler matrix A , the map $v \mapsto s(A + \text{diag } v)$ is convex on $\mathbb{R}^n$. If A is irreducible it is real-analytic, with gradient $\ell_x u_x / (\ell^\top \mathbf{u})$, where $\mathbf{u}, \ell$ are the right and left Perron vectors of $A + \text{diag } v$.*

References. For a Metzler matrix A and c large, $A + c\mathbf{1} \geq 0$ and $s(A) = \rho(A + c\mathbf{1}) - c$, so (P1)–(P4) are the Perron–Frobenius theorem and its comparison corollaries [6, Chapter 2]; in (P1) every eigenvalue $\lambda \neq s(A)$ has $\text{Re } \lambda < s(A)$, because $|\lambda + c| \leq \rho(A + c\mathbf{1}) = s(A) + c$. (P5) collects equivalent characterisations of non-singular M-matrices [6, Chapter 6]; the inverse inequality follows from $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$. The convexity in (P6) is the theorem of Cohen [12] (for another proof see Kato [32]); analyticity and the gradient are first-order perturbation theory of a simple eigenvalue [33, Chapter II, §1]. $\square$

S2.2 Next-generation matrix, threshold and offspring numbers

Proof of Theorem 3.1. (a) Since $\mathbf{1}^\top \mathbf{M} = 0$ and $\mathbf{M}$ is irreducible, $s(\mathbf{M}) = 0$ by (P1); hence $s(\mathbf{M} - \gamma \mathbf{1}) = -\gamma < 0$ and $\mathbf{V}^{-1} = \int_0^\infty e^{(\mathbf{M} - \gamma \mathbf{1})t} dt$. The integrand is positive because $e^{\mathbf{M}t} = e^{-ct} e^{(\mathbf{M} + c\mathbf{1})t}$ and all entries of the exponential series of the irreducible non-negative matrix $\mathbf{M} + c\mathbf{1}$ are positive for $t > 0$. Recall $(e^{\mathbf{M}t})_{zx} = P_t(z | x)$. An infective introduced in x is still infectious and located in z at time t with probability $e^{-\gamma t} P_t(z | x)$; while there it makes infectious contacts at rate βq_z , each of which lands in y with probability P_{zy} and, at the disease-free state, meets a susceptible with probability $S_y/N_y^* = 1$.

(b), (c) Some row y of $\mathbf{F}$ is non-zero and $\mathbf{V}^{-1} > 0$, so $K_{yy} > 0$ and $R_0 \geq K_{yy} > 0$ by (P4). For $c \geq 0$ put $\psi(c) = s(\mathbf{M} + c\mathbf{F} - \gamma \mathbf{1})$. The matrix is Metzler and irreducible, so ψ is continuous and, by (P3), strictly increasing. *Claim: for $c > 0$, $\psi(c) < 0$ if and only if $cR_0 < 1$.* Let $A_c = \mathbf{V} - c\mathbf{F}$, a matrix with non-positive off-diagonal entries and $s(-A_c) = \psi(c)$; note $A_c = (\mathbf{1} - c\mathbf{K})\mathbf{V}$. If $cR_0 < 1$, then $(\mathbf{1} - c\mathbf{K})^{-1} = \sum_{k \geq 0} (c\mathbf{K})^k \geq 0$, so $A_c^{-1} = \mathbf{V}^{-1}(\mathbf{1} - c\mathbf{K})^{-1} \geq 0$, and $\psi(c) < 0$ by (P5). Conversely, if $\psi(c) < 0$, then $A_c^{-1} \geq 0$ by (P5) and $(\mathbf{1} - c\mathbf{K})^{-1} = \mathbf{V}A_c^{-1} = (A_c + c\mathbf{F})A_c^{-1} = \mathbf{1} + c\mathbf{F}A_c^{-1} \geq \mathbf{1}$. Let $\mathbf{z} \geq 0$, $\mathbf{z} \neq 0$ with $\mathbf{z}^\top \mathbf{K} = R_0 \mathbf{z}^\top$ (P4). Then $\mathbf{z}^\top = (1 - cR_0) \mathbf{z}^\top (\mathbf{1} - c\mathbf{K})^{-1}$, where $\mathbf{z}^\top (\mathbf{1} - c\mathbf{K})^{-1} \geq \mathbf{z}^\top$ is non-negative and non-zero; hence $1 - cR_0 > 0$. This proves the claim. Therefore $\{c \geq 0 : \psi(c) < 0\} = [0, 1/R_0)$, and by continuity and strict monotonicity $1/R_0$ is the unique zero of ψ : this is (b) with $R = 1/c$. Statement (c) follows because $\lambda_1 = \psi(1)$.

(d) By (3) and $0 \leq S_y \leq N_y^*$, $r(t) := \mathbf{J}I(t) - \dot{I}(t) \geq 0$. Variation of constants gives $e^{\mathbf{J}t}I(0) - I(t) = \int_0^t e^{\mathbf{J}(t-s)}r(s) ds \geq 0$, because $e^{\mathbf{J}t} \geq 0$ for a Metzler matrix. Let $\ell > 0$ be the left Perron vector of $\mathbf{J}$ (P1). Then $\ell^\top I(t) \leq \ell^\top e^{\mathbf{J}t}I(0) = e^{\lambda_1 t} \ell^\top I(0)$, which is the stated bound with $C = \max_x \ell_x / \min_x \ell_x$. Finally, by (P1), $e^{\mathbf{J}t}I(0) = e^{\lambda_1 t}[(\ell^\top I(0)/\ell^\top \mathbf{u}) \mathbf{u} + o(1)]$ with $\mathbf{u} > 0$ the right Perron vector, and $\lambda_1 > 0$ when $R_0 > 1$ by (c). $\square$

Remark S2.2 (What the threshold does not say). In a closed SIR model the infectives vanish for every value of R_0 : $\mathbf{1}^\top R(t)$ is non-decreasing and bounded by N , so $\gamma \int_0^\infty \mathbf{1}^\top I dt \leq N$, and $\mathbf{1}^\top I(t) \rightarrow 0$ because its derivative is bounded. The content of (d) is that for $R_0 < 1$ there is no growth above $Ce^{\lambda_1 t} \mathbf{1}^\top I(0)$ at any time, whereas for $R_0 > 1$ a small seed grows, at rate λ_1 in the linearised model, until the depletion of susceptibles stops it. We therefore do not say that the infection ‘dies out if and only if $R_0 < 1$ ’.

Remark S2.3 (Offspring map). The column sums of $\mathbf{K}$ are the expected numbers of secondary cases of an index case introduced in cell x (the offspring number $n_{\text{off}}(x)$; the subscript distinguishes it from the number of cells n). For every row-stochastic kernel,

$$\begin{aligned} n_{\text{off}}(x) &= \sum_y K_{yx} = \beta [\mathbf{q}^\top (\gamma \mathbf{1} - \mathbf{M})^{-1}]_x = \beta \mathbb{E}_x \int_0^\tau q(X_t) dt, \\ \frac{\beta q_{\min}}{\gamma} &\leq n_{\text{off}}(x) \leq \frac{\beta q_{\max}}{\gamma}, \quad \sum_x \pi_x n_{\text{off}}(x) = \frac{\beta \langle q \rangle_\pi}{\gamma}, \end{aligned} \tag{S1}$$

where X_t is the walk started at x and τ its exponential infectious period (rate γ). Indeed $\mathbf{1}^\top \mathbf{P}^\top = \mathbf{1}^\top$ gives $\mathbf{1}^\top \mathbf{F} = \beta \mathbf{q}^\top$, the third expression is (a) summed over y , the bounds follow from $\mathbf{1}^\top \mathbf{V}^{-1} = \gamma^{-1} \mathbf{1}^\top$ and $\mathbf{V}^{-1} \geq 0$, and the mean from $\mathbf{V}^{-1} \boldsymbol{\pi} = \boldsymbol{\pi}/\gamma$. In the office scene at $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$, $n_{\text{off}}(x)$ ranges from 3.995 to 5.335 over the cells, with uniform mean 4.643.

S2.3 Reversible movement: Rayleigh and modal formulas

Theorem S2.4 (Rayleigh formula; reversible movement, same-cell kernel). *Let $\mathbf{F} = \beta \mathbf{Q}$ and let $\mathbf{M}$ satisfy detailed balance, $\pi_x \omega_{x \rightarrow y} = \pi_y \omega_{y \rightarrow x}$. With $c_{xy} = \pi_x \omega_{x \rightarrow y} = c_{yx}$ and the Dirichlet form $\mathcal{E}(v) = \frac{1}{2} \sum_{x \neq y} c_{xy} (v_x / \sqrt{\pi_x} - v_y / \sqrt{\pi_y})^2 \geq 0$,*

$$R_0 = \max_{v \neq 0} \frac{\beta v^\top \mathbf{Q} v}{\gamma |v|^2 + \mathcal{E}(v)}, \quad \lambda_1 = \max_{|v|=1} [\beta v^\top \mathbf{Q} v - \mathcal{E}(v)] - \gamma. \tag{S2}$$

Proof. Let $\Pi = \text{diag } \boldsymbol{\pi}$ and $A = \Pi^{-1/2} \mathbf{M} \Pi^{1/2}$, so that $A_{yx} = c_{xy} / \sqrt{\pi_x \pi_y}$ for $y \neq x$ and $A_{xx} = -\sum_{y \neq x} c_{xy} / \pi_x$. Detailed balance makes A symmetric, and with $f_x = v_x / \sqrt{\pi_x}$, $-v^\top A v = \sum_{x \neq y} c_{xy} f_x^2 - \sum_{x \neq y} c_{xy} f_x f_y = \mathcal{E}(v)$. Hence $B = \gamma \mathbf{1} - A$ is symmetric positive definite with $v^\top B v = \gamma |v|^2 + \mathcal{E}(v)$. Since $\mathbf{Q}$ commutes with Π , $\Pi^{-1/2} \mathbf{K} \Pi^{1/2} = \beta \mathbf{Q} B^{-1}$, which is similar to the symmetric positive semi-definite matrix $B^{-1/2} \beta \mathbf{Q} B^{-1/2}$. The spectral radius of the latter is its largest eigenvalue, $\max_{u \neq 0} u^\top B^{-1/2} \beta \mathbf{Q} B^{-1/2} u / |u|^2$, which becomes the first formula with $v = B^{-1/2} u$. The second is the Rayleigh–Ritz theorem [26] for the symmetric matrix $A + \beta \mathbf{Q} - \gamma \mathbf{1}$, which is similar to $\mathbf{J}$. $\square$

Corollary S2.5 (Modal formula). *In the setting of Theorem S2.4 let $0 = \mu_0 < \mu_1 \leq \dots \leq \mu_{n-1}$ and $\phi_0 = \sqrt{\boldsymbol{\pi}}, \phi_1, \dots$ be the eigenvalues and orthonormal eigenvectors of $-\Pi^{-1/2} \mathbf{M} \Pi^{1/2}$, $\Pi = \text{diag } \boldsymbol{\pi}$, and $\tilde{q}_{jk} = \phi_j^\top \mathbf{Q} \phi_k$. Then*

$$R_0 = \lambda_{\max}(\mathbf{R}), \quad \mathbf{R}_{jk} = \frac{\beta \tilde{q}_{jk}}{\sqrt{(\gamma + \mu_j)(\gamma + \mu_k)}}, \quad j, k = 0, \dots, n-1. \tag{S3}$$

If $R_0^{(m)}$ is the largest eigenvalue of the leading $m \times m$ block, then $\beta \langle q \rangle_\pi / \gamma = R_0^{(1)} \leq R_0^{(2)} \leq \dots \leq R_0^{(n)} = R_0$, and every diagonal entry satisfies $\beta \tilde{q}_{jj} / (\gamma + \mu_j) \leq R_0$.

Proof. Insert $v = \sum_j c_j \phi_j$ in (S2): $v^\top \mathbf{Q} v = \sum_{j,k} c_j \tilde{q}_{jk} c_k$ and $\gamma|v|^2 + \mathcal{E}(v) = \sum_j (\gamma + \mu_j) c_j^2$. With $d_j = \sqrt{\gamma + \mu_j} c_j$ the quotient is the Rayleigh quotient $d^\top \mathbf{R} d / |d|^2$ of the symmetric matrix $\mathbf{R}$. The largest eigenvalue of a leading block is non-decreasing in its size by Cauchy interlacing [26]; a diagonal entry is the Rayleigh quotient at a coordinate vector; and $\mathbf{R}_{00} = \beta \sum_x \pi_x q_x / \gamma$. $\square$

The largest diagonal entry of the modal matrix is thus a lower bound on R_0 , equal to R_0 when q is constant, because truncating the matrix to its diagonal drops the coupling between modes. In the office scene at $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$ the largest diagonal term is 4.961, the blocks of 3 and 10 modes give 5.080 and 5.106, and $R_0 = 5.107$ (Fig. S24c). The mode expansion is therefore a rapidly convergent computational device once the coupling matrix $(\tilde{q}_{jk})$ is kept.

S2.4 Bounds, mobility, zones and edges

Proof of Theorem 3.2. Upper bounds. $\mathbf{1}^\top \mathbf{V} = \gamma \mathbf{1}^\top$ gives $\mathbf{1}^\top \mathbf{V}^{-1} = \gamma^{-1} \mathbf{1}^\top$: the positive matrix $\mathbf{V}^{-1}$ has a positive left eigenvector with eigenvalue $1/\gamma$, so $\rho(\mathbf{V}^{-1}) = 1/\gamma$ by (P1). Since $0 \leq \mathbf{K} \leq \beta q_{\max} \mathbf{V}^{-1}$, (P4) gives $R_0 \leq \beta q_{\max} / \gamma$, strictly if $q_x < q_{\max}$ for some x , because $\mathbf{V}^{-1}$ is irreducible. For the growth rate, $\mathbf{J} + E = \mathbf{M} + (\beta q_{\max} - \gamma) \mathbf{1}$ with $E = \beta(q_{\max} \mathbf{1} - \mathbf{Q}) \geq 0$, so $\lambda_1 \leq \beta q_{\max} - \gamma$, strictly if $E \neq 0$ (P3).

Constant $q \equiv q_0$. $\mathbf{K} = \beta q_0 \mathbf{V}^{-1}$ and $\mathbf{J} = \mathbf{M} + (\beta q_0 - \gamma) \mathbf{1}$.

Lower bounds. Let $f(v) = s(\mathbf{M} + \text{diag } v)$. By (P6) f is convex and differentiable; $f(0) = 0$ with right and left Perron vectors $\boldsymbol{\pi}$ and $\mathbf{1}$, so $\nabla f(0) = \boldsymbol{\pi}$ and $f(v) \geq \sum_x \pi_x v_x =: \langle v \rangle_\pi$ for all v . With $v = \beta \mathbf{q} - \gamma \mathbf{1}$ this is $\lambda_1 \geq \beta \langle q \rangle_\pi - \gamma$. With $v = \beta \mathbf{q} / R_0 - \gamma \mathbf{1}$, Theorem 3.1(b) gives $0 = f(v) \geq \beta \langle q \rangle_\pi / R_0 - \gamma$. (For reversible movement, take $v = \sqrt{\pi}$ in (S2), for which $\mathcal{E}(v) = 0$.)

Strictness. Suppose $R_0 = \beta \langle q \rangle_\pi / \gamma$ and put $v = \beta \mathbf{q} / R_0 - \gamma \mathbf{1}$, so that $\langle v \rangle_\pi = 0 = f(v)$. The function $g(t) = f(tv)$ is convex with $g(0) = 0$ and $g'(0) = 0$, hence $g \geq 0$; as $g(1) = 0$, convexity gives $g \equiv 0$ on $[0, 1]$, and g is real-analytic on $\mathbb{R}$, so $g \equiv 0$. But (P2) gives $g(t) \geq \max_x (M_{xx} + t v_x)$, which is unbounded as $t \rightarrow \infty$ unless $v \leq 0$; with $\langle v \rangle_\pi = 0$ and $\boldsymbol{\pi} > 0$ this forces $v = 0$, that is, q constant. The same argument with $v = \beta(\mathbf{q} - \langle q \rangle_\pi \mathbf{1})$ treats λ_1 . $\square$

Proof of Theorem 3.3. (b) $\mathbf{K}(D_0) = \beta \mathbf{Q}(\gamma \mathbf{1} - D_0 \mathbf{M}_0)^{-1} \rightarrow \beta \mathbf{Q} / \gamma$, and the spectral radius is continuous.

(a) Let $\mathbf{E}_0 = \boldsymbol{\pi} \mathbf{1}^\top$, the spectral projector of the simple eigenvalue 0 of $\mathbf{M}_0$, and $\mathbf{Y} = (\mathbf{E}_0 - \mathbf{M}_0)^{-1} - \mathbf{E}_0$, the group inverse of $-\mathbf{M}_0$. (The inverse exists: $(\mathbf{E}_0 - \mathbf{M}_0)v = 0$ implies $\mathbf{1}^\top v = 0$, then $\mathbf{M}_0 v = 0$, then $v = 0$.) From $(\mathbf{E}_0 - \mathbf{M}_0)\mathbf{E}_0 = \mathbf{E}_0 = \mathbf{E}_0(\mathbf{E}_0 - \mathbf{M}_0)$ one gets $\mathbf{Y}\mathbf{E}_0 = \mathbf{E}_0\mathbf{Y} = 0$ and $-\mathbf{M}_0\mathbf{Y} = \mathbf{1} - \mathbf{E}_0$. A direct multiplication shows that for $\varepsilon = 1/D_0$ small

$$(\gamma \mathbf{1} - D_0 \mathbf{M}_0)^{-1} = \varepsilon(\varepsilon \gamma \mathbf{1} - \mathbf{M}_0)^{-1} = \frac{\mathbf{E}_0}{\gamma} + \varepsilon \mathbf{Y}(\mathbf{1} + \varepsilon \gamma \mathbf{Y})^{-1}, \quad (\text{S4})$$

which is analytic in ε at $\varepsilon = 0$. Hence $\mathbf{K} \rightarrow \beta \mathbf{Q} \boldsymbol{\pi} \mathbf{1}^\top / \gamma$, a matrix of rank one whose non-zero eigenvalue is $\beta \mathbf{1}^\top \mathbf{Q} \boldsymbol{\pi} / \gamma = \beta \langle q \rangle_\pi / \gamma$.

(c) Fix $v \in \mathbb{R}^n$ and let $g(t) = s(\mathbf{M}_0 + t \text{diag } v)$, which is convex and analytic by (P6), with $g(0) = 0$ and $g'(0) = \langle v \rangle_\pi$. Since s is positively homogeneous, $\varphi(D) := s(D\mathbf{M}_0 + \text{diag } v) = D g(1/D)$ for $D > 0$, and $\varphi''(D) = g''(1/D)/D^3 \geq 0$: φ is convex. Its derivative $\varphi'(D) = g(1/D) - g'(1/D)/D$ is non-decreasing and tends to $g(0) = 0$ as $D \rightarrow \infty$, so $\varphi' \leq 0$; and $\varphi(D) \rightarrow g'(0) = \langle v \rangle_\pi$. With $v = \beta \mathbf{q} - \gamma \mathbf{1}$ this is the statement for λ_1 .

For R_0 , let $H(D, R) = s(D\mathbf{M}_0 + \beta \mathbf{Q}/R - \gamma \mathbf{1})$. It is non-increasing in D (just shown), strictly decreasing in R (P3), and $R_0(D)$ is its unique zero in R (Theorem 3.1(b)). If $D_1 < D_2$, then $H(D_1, R_0(D_2)) \geq H(D_2, R_0(D_2)) = 0$, hence $R_0(D_1) \geq R_0(D_2)$. Suppose $R_0(D_1) = R_0(D_2) =: R$ and let $\varphi(D) = H(D, R)$. Then $\varphi(D_1) = \varphi(D_2) = 0$ and φ is non-increasing, so $\varphi \equiv 0$ on $[D_1, D_2]$; as φ' is non-decreasing and non-positive, $\varphi' \equiv 0$ on $[D_1, \infty)$, and $0 = \lim_{D \rightarrow \infty} \varphi(D) = \beta \langle q \rangle_\pi / R - \gamma$. This is equality in the lower bound of Theorem 3.2 for the generator $D_1 \mathbf{M}_0$, which forces q to be constant. $\square$

Proof of Proposition 3.4. (i) By (S4), $\mathbf{K} = \mathbf{K}(\varepsilon)$ is analytic at $\varepsilon = 1/D_0 = 0$, with $\mathbf{K}(0) = \beta \mathbf{Q} \boldsymbol{\pi} \mathbf{1}^\top / \gamma$ and $\mathbf{K}'(0) = \beta \mathbf{Q} \mathbf{Y}$. The eigenvalue $r_0 = \beta \langle q \rangle_\pi / \gamma > 0$ of the rank-one matrix $\mathbf{K}(0)$ is algebraically simple, with right eigenvector $\mathbf{Q} \boldsymbol{\pi}$ and left eigenvector $\mathbf{1}$; the other eigenvalues are 0. By analytic perturbation theory of a simple eigenvalue [33, Chapter II, §1], $\mathbf{K}(\varepsilon)$ has an analytic eigenvalue $r(\varepsilon)$ with $r(0) = r_0$ and $r'(0) = \mathbf{1}^\top \mathbf{K}'(0) \mathbf{Q} \boldsymbol{\pi} / (\mathbf{1}^\top \mathbf{Q} \boldsymbol{\pi}) = \beta \mathbf{q}^\top \mathbf{Y} \mathbf{Q} \boldsymbol{\pi} / \langle q \rangle_\pi$, while the other eigenvalues tend to 0; $r(\varepsilon)$ is real, being the only eigenvalue near r_0 , and hence $R_0 = r(\varepsilon)$ for small $\varepsilon > 0$. Put $\mathbf{y} = \mathbf{Y} \mathbf{Q} \boldsymbol{\pi}$. Because $\mathbf{1}^\top \mathbf{Y} = \mathbf{1}^\top \mathbf{E}_0 \mathbf{Y} = 0$ and $-\mathbf{M}_0 \mathbf{Y} = \mathbf{1} - \mathbf{E}_0$, $\mathbf{y}$ solves the stated system, whose solution is unique since the kernel of $\mathbf{M}_0$ is spanned by $\boldsymbol{\pi}$. $C \geq 0$ by Theorem 3.3(c). For symmetric rates $\boldsymbol{\pi} = \mathbf{1}/n$ and $\mathbf{Y} = (-\mathbf{M}_0)^+$.

(ii) $\mathbf{K}(D_0) = (\beta/\gamma) \mathbf{Q} (\mathbf{1} - D_0 \mathbf{M}_0 / \gamma)^{-1}$ is analytic at $D_0 = 0$, with $\mathbf{K}(0) = (\beta/\gamma) \mathbf{Q}$ and $\mathbf{K}'(0) = (\beta/\gamma^2) \mathbf{Q} \mathbf{M}_0$. The diagonal matrix $\mathbf{K}(0)$ has the semisimple eigenvalue $\varrho_0 = \beta q_{\max} / \gamma$, whose eigenprojection is the coordinate projection $\mathbf{E}_Z$ onto Z . By the reduction process for a semisimple eigenvalue [33, Chapter II, §2], the eigenvalues of $\mathbf{K}(D_0)$ near ϱ_0 are $\varrho_0 + D_0 \sigma_j + o(D_0)$, where the σ_j are the eigenvalues of $\mathbf{E}_Z \mathbf{K}'(0) \mathbf{E}_Z$ on the range of $\mathbf{E}_Z$, that is, of $(\beta q_{\max} / \gamma^2) \mathbf{M}_0[Z, Z]$. All other eigenvalues stay near values $\beta q_x / \gamma < \varrho_0$, so for small D_0 the spectral radius is the largest modulus in this group, $\varrho_0 + D_0 \max_j \operatorname{Re} \sigma_j + o(D_0)$, and the largest real part of the eigenvalues of $\mathbf{M}_0[Z, Z]$ is $s(\mathbf{M}_0[Z, Z]) = -\mu_Z^0$. $\square$

Proof of Proposition 3.5. $\mu_Z \geq 0$ by (P2). If $Z = \Omega$ the claim is $R_0 \geq \beta q_{\min} / \gamma$, which follows from $\mathbf{K} \geq \beta q_{\min} \mathbf{V}^{-1}$ and (P4). Otherwise let $Z' = \Omega \setminus Z$ and write $\mathbf{V}$ in blocks. The column sums of $\mathbf{V}_{ZZ}$ and of $\mathbf{V}_{Z'Z'}$ are at least γ , so both are non-singular M-matrices (P5). The block-inverse formula gives $(\mathbf{V}^{-1})_{ZZ} = S^{-1}$ with the Schur complement $S = \mathbf{V}_{ZZ} - \mathbf{V}_{ZZ'} \mathbf{V}_{Z'Z'}^{-1} \mathbf{V}_{Z'Z}$. Because $\mathbf{V}_{ZZ'}, \mathbf{V}_{Z'Z} \leq 0$ and $\mathbf{V}_{Z'Z'}^{-1} \geq 0$, we have $S \leq \mathbf{V}_{ZZ}$; S has non-positive off-diagonal entries and $S^{-1} = (\mathbf{V}^{-1})_{ZZ} \geq 0$, so it is a non-singular M-matrix and $S^{-1} \geq \mathbf{V}_{ZZ}^{-1} \geq 0$ by (P5). Therefore $\mathbf{K}[Z, Z] = \beta \mathbf{Q}_Z (\mathbf{V}^{-1})_{ZZ} \geq \beta q_Z \mathbf{V}_{ZZ}^{-1}$, and by (P4) $R_0 \geq \rho(\mathbf{K}[Z, Z]) \geq \beta q_Z \rho(\mathbf{V}_{ZZ}^{-1})$. Finally $\mathbf{V}_{ZZ} = \gamma \mathbf{1} - \mathbf{M}[Z, Z]$, and the Metzler matrix $\mathbf{M}[Z, Z]$ has the eigenvalue $-\mu_Z$ with a non-negative eigenvector, so $\rho(\mathbf{V}_{ZZ}^{-1}) \geq 1/(\gamma + \mu_Z)$. $\square$

Together with Theorem 3.2, $\max\{\beta \langle q \rangle_\pi / \gamma, \max_Z \beta q_Z / (\gamma + \mu_Z)\} \leq R_0 \leq \beta q_{\max} / \gamma$, and by Proposition 3.4(ii) the zone bound with $Z = \{x : q_x = q_{\max}\}$ is sharp to first order as $D_0 \rightarrow 0$.

Proof of Theorem 3.6. Symmetric rates satisfy detailed balance with uniform $\boldsymbol{\pi}$, and $\mathcal{E}(v) = \sum_{\{x,y\}} \omega_{x \rightarrow y} (v_x - v_y)^2$, summed over edges, is non-decreasing in each edge rate for every fixed v . Both expressions in (S2) are therefore maxima of functions that are non-increasing in each edge rate. The harmonic mean $2D_x D_y / (D_x + D_y)$ is increasing in D_x and D_y , and closing an edge sets its rate to zero; the proof of Theorem S2.4 does not use irreducibility when the rates are symmetric. $\square$

S2.5 Contact kernels

Proof of Proposition 3.7. (i) $\mathbf{F} \geq 0$, and $\mathbf{F} \neq 0$ because the column of $\mathbf{P}^\top \mathbf{Q}$ belonging to a cell with $q_z > 0$ is q_z times a probability vector; Theorem 3.1 was proved for such $\mathbf{F}$.

(ii) Let $\mathbf{w} \geq 0$, $\mathbf{w} \neq 0$ with $\mathbf{K} \mathbf{w} = R_0 \mathbf{w}$ (P4). Multiplying by $\mathbf{1}^\top$ gives $\sum_x n_{\text{off}}(x) w_x = R_0 \sum_x w_x$: R_0 is a weighted mean of the column sums $n_{\text{off}}(x)$, which lie between $\beta q_{\min} / \gamma$ and $\beta q_{\max} / \gamma$ by (7) and equal $\beta q_0 / \gamma$ when $q \equiv q_0$.

(iii) By (S4), $\mathbf{K} \rightarrow \beta \mathbf{P}^\top \mathbf{Q} \boldsymbol{\pi} \mathbf{1}^\top / \gamma$ as $D_0 \rightarrow \infty$, of rank one with eigenvalue $\beta \mathbf{1}^\top \mathbf{P}^\top \mathbf{Q} \boldsymbol{\pi} / \gamma = \beta \langle q \rangle_\pi / \gamma$; and $\mathbf{K} \rightarrow \beta \mathbf{P}^\top \mathbf{Q}$ as $D_0 \rightarrow 0$.

(iv) In the example $\mathbf{F} e_1 = \beta(\frac{1}{2}, \frac{1}{2}, 0)^\top$, $\mathbf{F} e_2 = 0$, $\mathbf{F} e_3 = \beta(0, \frac{1}{2}, \frac{1}{2})^\top$, and $\mathbf{K}$ commutes with the reflection that exchanges cells 1 and 3. On the antisymmetric vector $d = (1, 0, -1)^\top$, $\mathbf{M}_0 d = -d$ and $\mathbf{F} d = \frac{1}{2} \beta d$, giving the eigenvalue $\beta / (2(\gamma + D_0))$. On the symmetric subspace, spanned by $a = (1, 0, 1)^\top$ and $b = (0, 1, 0)^\top$, $\mathbf{F}$ has rank one with range spanned by $f = \frac{1}{2} a + b$, so the only

non-zero eigenvalue is β times the a -coordinate of $\mathbf{V}^{-1}f$. With $\mathbf{M}_0 a = -a + 2b$ and $\mathbf{M}_0 b = a - 2b$, solving the 2×2 system gives

$$R_0(D_0) = \frac{\beta}{\gamma} \frac{\gamma + 4D_0}{2\gamma + 6D_0},$$

which is not smaller than the antisymmetric eigenvalue, equals $\beta/(2\gamma)$ at $D_0 = 0$, increases strictly with D_0 and tends to $2\beta/(3\gamma) = \beta\langle q \rangle/\gamma$. $\square$

With $q = (1, 0.2, 1)$ the same three cells give $R_0 = 0.620 \beta/\gamma$ at $D_0 = 0$ against $\beta\langle q \rangle/\gamma = 0.733 \beta/\gamma$.

S2.6 Further numbers for the office scene and their verification

At $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$, between the two regimes of Proposition 3.4, the errors of the fast- and the slow-mixing expansion of the office scene are 0.17 and 0.09.

The statements of this section were tested on random instances of four families of generators (lattices with symmetric, departure-site and non-reversible rates, and dense matrices), and the proofs were re-derived and the computations re-implemented in the re-check (Section S11). The worked-example numbers for the canonical scenes in Sections 3.1, 4.3 and 4.4 of the main text and in Section S3 were computed after that re-check, which had checked the same quantities on another office layout (the first of the two alternative floor plans of Section 3.1: $R_0 = 5.109$, a residence time of 110.6 days in the meeting room and 97.2 % of the elasticity there, against 5.107, 114 days and 98.1 % in the canonical scene). On the canonical scenes it recomputed only R_0 (the office at $D_0 = 0.03, 1, 10$ and 100 , the other scenes at $D_0 = 1$) and $\rho(\mathbf{K}_1)$. The other worked-example numbers were cross-checked by a second script written as part of the same analysis, not by the re-check.

S3 Room size, leaks and hotspot maps: proofs and worked examples

This section proves the statements of Section 4 of the main text and gives the details of the leaky room and the worked examples on the office scene. Throughout, (P1)–(P6) are the facts of Lemma S2.1, and δ_x denotes the unit vector of cell x .

S3.1 Closed and leaky rooms

Proof of Theorem 4.1. Since $\mathbf{P}$ is row-stochastic, $\mathbf{P}\mathbf{1} = \mathbf{1}$, so $\mathbf{1}^\top \mathbf{F} = \beta q_0 \mathbf{1}^\top \mathbf{P}^\top = \beta q_0 \mathbf{1}^\top$. Since $\mathbf{1}^\top \mathbf{M} = 0$, $\mathbf{1}^\top \mathbf{V} = \gamma \mathbf{1}^\top$ and $\mathbf{1}^\top \mathbf{V}^{-1} = \gamma^{-1} \mathbf{1}^\top$. Hence $\mathbf{1}^\top \mathbf{K} = (\beta q_0/\gamma) \mathbf{1}^\top$: all column sums of $\mathbf{K}$, which are the offspring numbers $n_{\text{off}}(x)$ of (7), equal $\beta q_0/\gamma$. By (P4) there is $\mathbf{w} \geq 0$, $\mathbf{w} \neq 0$, with $\mathbf{K}\mathbf{w} = \rho(\mathbf{K})\mathbf{w}$; then $\rho(\mathbf{K}) \mathbf{1}^\top \mathbf{w} = \mathbf{1}^\top \mathbf{K}\mathbf{w} = (\beta q_0/\gamma) \mathbf{1}^\top \mathbf{w}$ and $\mathbf{1}^\top \mathbf{w} > 0$, so $R_0 = \beta q_0/\gamma$. Likewise $\mathbf{1}^\top \mathbf{J} = (\beta q_0 - \gamma) \mathbf{1}^\top$. The matrix $\mathbf{J}^\top$ is Metzler and irreducible (because $\mathbf{M}$ is, and $\mathbf{F} \geq 0$) and has the positive eigenvector $\mathbf{1}$, so by (P1) the corresponding eigenvalue is $s(\mathbf{J}^\top) = \lambda_1$. The argument uses only the irreducibility of $\mathbf{M}$ and $\mathbf{1}^\top \mathbf{M} = 0$, neither of which depends on the size or shape of Ω , on the mobilities or on the hopping rule. $\square$

Proof of Theorem 4.2. (1) $\mathbf{M}_\kappa$ has the off-diagonal entries of $\mathbf{M}$, so it is Metzler and irreducible. $\mathbf{V}_\kappa$ has non-positive off-diagonal entries and column sums $\gamma + \kappa_x > 0$, so by (P5) it is a non-singular M-matrix, $s(\mathbf{M}_\kappa - \gamma \mathbf{1}) < 0$ and $\mathbf{V}_\kappa^{-1} = \int_0^\infty e^{-\gamma t} e^{\mathbf{M}_\kappa t} dt > 0$, where $(e^{\mathbf{M}_\kappa t})_{yx}$ is the probability that a person who started in x is still in the room, in cell y , at time t . These are the only properties of $\mathbf{M}$ and $\mathbf{V}$ used in the proof of Theorem 3.1(a)–(c), which applies verbatim.

(2) By (P1), $s(\mathbf{M}) = 0$, because $\mathbf{1}^\top \mathbf{M} = 0$ with $\mathbf{1} > 0$. (P3) with $A = \mathbf{M}_\kappa$ and $E = \text{diag } \kappa$ gives $0 = s(\mathbf{M}) > s(\mathbf{M}_\kappa)$, and the same argument applied to $\kappa \leq \kappa'$ gives the monotonicity. The function $f(v) = s(\mathbf{M} + \text{diag } v)$ is convex with $f(0) = 0$ and gradient $\boldsymbol{\pi}$ at $v = 0$ (P6; the

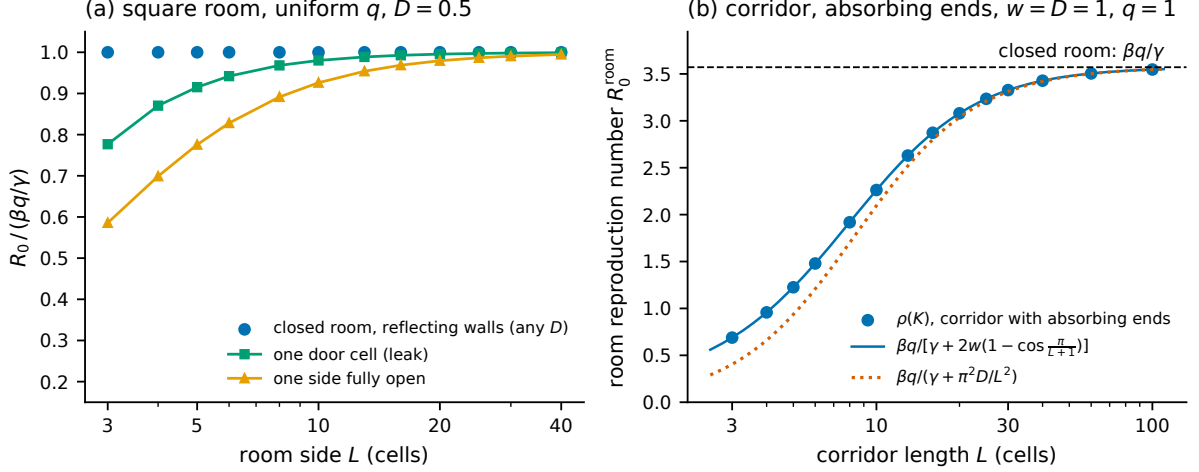


Figure S1: Room size and leaks in the mean-field model with uniform q ($\beta q = 0.5 \text{ day}^{-1}$, $\gamma = 0.14 \text{ day}^{-1}$, same-cell kernel). (a) Ratio of the reproduction number to $\beta q/\gamma$ for $L \times L$ rooms with $D = 0.5 \text{ cell}^2 \text{ day}^{-1}$: closed room with reflecting walls (circles; identical for every D), room with one door cell with exit rate $\kappa = D$ (squares) and room with one side fully open ($\kappa = D$ in each cell along that side; triangles). (b) Corridor of L cells with absorbing ends, hop rate $w = D = 1 \text{ day}^{-1}$, $q = 1$: $R_0^{\text{room}} = \rho(\mathbf{K})$ computed numerically (circles), the closed form (11) (solid) and its continuum limit $\beta q/(\gamma + \pi^2 D/L^2)$ (dotted); the dashed line is the closed-room value $\beta q/\gamma$. Numbers in Table S3.

Perron vectors of $\mathbf{M}$ are $\boldsymbol{\pi}$ and $\mathbf{1}$, so $f(v) \geq \sum_x \pi_x v_x$; $v = -\kappa$ gives $-\mu_{\text{leak}} \geq -\sum_x \pi_x \kappa_x$. For the limit write $\mu(k)$ for μ_{leak} with $\kappa = k\mathbf{1}_\Delta$ and $\Delta^c = \Omega \setminus \Delta$. The principal submatrix of $\mathbf{M}_\kappa$ on Δ^c does not depend on k , so (P2) gives $\mu(k) \leq \mu_\Delta^{\text{abs}}$; being non-decreasing, $\mu(k) \uparrow \mu^* \leq \mu_\Delta^{\text{abs}}$. Let $\mathbf{u}_k \geq 0$, $\mathbf{1}^\top \mathbf{u}_k = 1$, be the Perron vector. Row $x \in \Delta$ of $(\mathbf{M} - k \text{diag } \mathbf{1}_\Delta) \mathbf{u}_k = -\mu(k) \mathbf{u}_k$ reads $u_k(x) = \sum_{y \neq x} M_{xy} u_k(y) / (k - \mu(k) - M_{xx}) \rightarrow 0$. Any limit point $\mathbf{u}^*$ of $(\mathbf{u}_k)$ therefore vanishes on Δ , is non-negative and non-zero on Δ^c and satisfies $\mathbf{M}[\Delta^c, \Delta^c] \mathbf{u}^* = -\mu^* \mathbf{u}^*$ there; hence $-\mu^* \leq s(\mathbf{M}[\Delta^c, \Delta^c])$, that is $\mu^* \geq \mu_\Delta^{\text{abs}}$.

(3) $\mathbf{V}_\kappa \mathbf{u}_\kappa = (\gamma + \mu_{\text{leak}}) \mathbf{u}_\kappa$, so the positive matrix $\mathbf{V}_\kappa^{-1}$ has the positive eigenvector $\mathbf{u}_\kappa$ with eigenvalue $1/(\gamma + \mu_{\text{leak}})$, which is its spectral radius by (P1) and (P4); and $\beta \mathbf{Q} \mathbf{V}_\kappa^{-1} = \beta q_0 \mathbf{V}_\kappa^{-1}$.

(4) The upper bound follows from $0 \leq \beta \mathbf{Q} \mathbf{V}_\kappa^{-1} \leq \beta q_{\text{max}} \mathbf{V}_\kappa^{-1}$, (P4) and (3). For the lower bound, $f_\kappa(v) = s(\mathbf{M}_\kappa + \text{diag } v)$ is convex with $f_\kappa(0) = -\mu_{\text{leak}}$ and gradient ν at $v = 0$ (P6), so $f_\kappa(v) \geq -\mu_{\text{leak}} + \sum_x \nu_x v_x$. By (1), $f_\kappa(v) = 0$ for $v = \beta \mathbf{q}/R_0^{\text{room}} - \gamma \mathbf{1}$, which gives $0 \geq -\mu_{\text{leak}} + \beta \langle q \rangle \nu / R_0^{\text{room}} - \gamma$. $\square$

Proof of Corollary 4.3. (a) $\mathbf{M}_\kappa = \mathbf{M} - T^{-1} \mathbf{1}$ has the Perron vectors $\boldsymbol{\pi}$ and $\mathbf{1}$ of $\mathbf{M}$, so $\mu_{\text{leak}} = 1/T$, $\nu = \boldsymbol{\pi}$ and $\mathbf{V}_\kappa = (\gamma + 1/T) \mathbf{1} - \mathbf{M}$; the expected number of infections is the identity $\sum_x \pi_x n_{\text{off}}(x) = \beta \langle q \rangle \pi / \gamma$ of (7) with $\gamma + 1/T$ in place of γ . The form of $\mathbf{V}_\kappa$ does not involve the kernel, so for $\mathbf{F} = \beta \mathbf{P}^\top \mathbf{Q}$ the next-generation matrix is that of the closed room with $\gamma + 1/T$ in place of γ , and Theorem 4.1 gives the last statement. (b) $\mathbf{M}_\kappa = -w \mathbf{T}_L$, where $\mathbf{T}_L$ is the $L \times L$ tridiagonal matrix with 2 on the diagonal and -1 beside it. Its eigenvalues are $2 - 2 \cos(j\pi/(L+1))$ with eigenvectors $(\sin(j\pi x/(L+1)))_{x=1}^L$, $j = 1, \dots, L$; the eigenvector for $j = 1$ is positive, hence the Perron vector by (P1). Then apply Theorem 4.2(3). $\square$

Remark S3.1 (Other kernels and a localised door). With a localised door, parts (3) and (4) of Theorem 4.2 do not extend to other contact kernels ($R_0^{\text{room}} := \rho(\beta \mathbf{P}^\top \mathbf{Q} \mathbf{V}_\kappa^{-1})$). Take three cells in a row, hop rate 1 day^{-1} between neighbours, a door with $\kappa = 1 \text{ day}^{-1}$ at one end ($\mu_{\text{leak}} = 0.198 \text{ day}^{-1}$), $q \equiv 1$, $\beta = 0.5 \text{ day}^{-1}$ and $\gamma = 0.14 \text{ day}^{-1}$, so that the formula of part (3) gives 1.479. With $\mathbf{P}$ uniform over a cell and its neighbours, $R_0^{\text{room}} = 1.435$ (-3.0%); with the rows of $\mathbf{P}$ equal to $(0, 0, 1)$, $(0, 1, 0)$, $(0, 0, 1)$ and the door at cell 1 (the cell whose contacts land in cell 3), $R_0^{\text{room}} = 1.640$, which exceeds the upper bound of part (4) by 10.9% (with the door at

Table S3: Square rooms of side L cells with uniform q and $D = 0.5 \text{ cell}^2 \text{ day}^{-1}$ ($\gamma = 0.14 \text{ day}^{-1}$): leak rate μ_{leak} (per day) and ratio of the reproduction number to $\beta q/\gamma$, which by Theorem 4.2(3) equals $\gamma/(\gamma + \mu_{\text{leak}})$. The door cell lies in the middle of one side; ‘one side open’ means $\kappa = D$ in every cell along one side.

L	closed room	one door cell ($\kappa = D$)		one side open	
	ratio	μ_{leak}	ratio	μ_{leak}	ratio
3	1.000	0.0403	0.777	0.0990	0.586
13	1.000	0.00160	0.989	0.00676	0.954
20	1.000	0.00063	0.996	0.00293	0.979
40	1.000	0.00014	0.999	0.00075	0.995

cell 3 it is 1.127, within the bounds). The lower bound fails by continuity: as $\kappa \rightarrow 0$, $R_0^{\text{room}} \rightarrow R_0$ and the lower bound tends to $\beta \langle q \rangle_\pi / \gamma$, which the kernel of Proposition 3.7(iv) violates. In that example with hop rate 0.01 day^{-1} and a door with $\kappa = 0.01 \text{ day}^{-1}$ at one end, $R_0^{\text{room}} = 1.845$ against a lower bound of 2.292.

For heterogeneous q and a localised door the substitution $\gamma \rightarrow \gamma + \mu_{\text{leak}}$ is not exact, and only the bounds of Theorem 4.2(4) hold. In the office scene with one door cell at the end of a corridor arm, $\kappa = 1 \text{ day}^{-1}$ and $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$ give $R_0^{\text{room}} = 5.107$ where the substitution gives 5.060 ($\mu_{\text{leak}} = 0.00137 \text{ day}^{-1}$); with $\kappa = 100 \text{ day}^{-1}$ and $D_0 = 10$, $R_0^{\text{room}} = 4.220$ where the substitution gives 4.133, which is below the lower bound $\beta \langle q \rangle_\nu / (\gamma + \mu_{\text{leak}}) = 4.139$ (upper bound 4.730). A door at the end of the corridor removes people from a low- q zone, so the walk conditioned to stay is weighted towards higher q ($\langle q \rangle_\nu = 1.306$ and 1.313 in the two cases, against $\langle q \rangle = 1.300$) than under a uniform leak at the same rate, for which the substitution is exact (Corollary 4.3a).

The lattice leak rate of the corridor of Corollary 4.3(b) is 0.53, 0.86 and 0.98 of its continuum limit $\pi^2 D / L^2$ at $L = 3, 13$ and 100 (Fig. S1b).

Numerical observation S3.2 (One-door square room). For an $L \times L$ room with hop rate D and one absorbing door cell in the middle of a boundary row, the leak rate is $\mu^{\text{abs}} \approx \pi D / (L^2 \ln(L/c))$ with a fitted c that rises from 0.82 at $L = 8$ to 0.91 at $L = 128$ and has not converged. For a door cell with finite exit rate $\kappa = D$, $1/\mu_{\text{leak}} \approx 1/\mu^{\text{abs}} + 1/\langle \kappa \rangle_\pi$ (search time plus exit time), with a relative error in μ_{leak} that falls from 1.5% at $L = 8$ to 0.3% at $L = 128$.

The logarithm is that of the two-dimensional narrow-escape law [58]; we have not proved either statement.

Table S3 gives the leak rates of square rooms. They are small ($\mu_{\text{leak}} \leq 0.1 \text{ day}^{-1}$, residence times $1/\mu_{\text{leak}}$ from 10 days to 19 years) only because $D = 0.5 \text{ cell}^2 \text{ day}^{-1}$. For a fixed geometry and a door whose exit rate is itself a hop rate ($\kappa = D$, as in Table S3) μ_{leak} is proportional to the mobility scale, so at $D = 2.4 \times 10^3 \text{ cell}^2 \text{ day}^{-1}$ (Section 2.6) the same residence times would lie between 3 minutes and 1.5 days. A door with a fixed exit rate κ behaves differently: its leak rate grows with mobility but stays below $\langle \kappa \rangle_\pi$ (Theorem 4.2(2)). In the office with one door cell at the end of a corridor arm and $\kappa = 1 \text{ day}^{-1}$, $\mu_{\text{leak}} = 0.00137, 0.00356, 0.00419$ and 0.00427 day^{-1} at $D_0 = 1, 10, 100$ and $2,400$, against the cap $1/234 = 0.00427 \text{ day}^{-1}$, and $R_0^{\text{room}}/R_0 = 0.99998, 0.977, 0.971$ and 0.970 ; with κ equal to the hop rate of the door cell, $\mu_{\text{leak}} = 0.00152 D_0$ and R_0^{room}/R_0 falls to 0.916, 0.483 and 0.037 at $D_0 = 10, 100$ and $2,400$. For applications the dwell time T should therefore be an input, as in Corollary 4.3(a), not the output of a random walk towards a door.

S3.2 Hotspot maps

Proof of Theorem 4.4. (i) By (P1), λ_1 is a simple eigenvalue of $\mathbf{J}$ with spectral projector $\mathbf{u}\ell^\top / (\ell^\top \mathbf{u})$. For $c > \max_x |J_{xx}|$ the matrix $\mathbf{J} + c\mathbf{1}$ is non-negative and irreducible with a

positive diagonal, hence primitive [6], so every other eigenvalue λ of $\mathbf{J}$ satisfies $|\lambda + c| < \lambda_1 + c$ and thus $\text{Re } \lambda < \lambda_1$: $\delta > 0$. On the complementary invariant subspace $e^{\mathbf{J}t}$ is $O(e^{(\lambda_1 - \delta')t})$ for every $\delta' < \delta$, and for $\delta' = \delta$ when $\mathbf{J}$ is diagonalisable; for reversible movement $\mathbf{J}$ is similar to the symmetric matrix $\Pi^{-1/2} \mathbf{J} \Pi^{1/2}$. If $q \equiv q_0$, $\mathbf{J}\pi = (\beta q_0 - \gamma)\pi$ and $\pi > 0$, so $\mathbf{u} = \pi$ by (P1). Conversely, $\mathbf{J}\pi = \lambda_1 \pi$ reads $(\beta q_x - \gamma)\pi_x = \lambda_1 \pi_x$ for all x , so q is constant.

(ii) $\mathbf{K} = \beta \mathbf{Q} \mathbf{V}^{-1} > 0$ entrywise, so $\mathbf{K}$ is primitive and $\mathbf{K}^g / R_0^g \rightarrow \mathbf{w} \mathbf{z}^\top / (\mathbf{z}^\top \mathbf{w})$. Put $\xi = \mathbf{V}^{-1} \mathbf{w} > 0$; then $\beta \mathbf{Q} \xi = \mathbf{K} \mathbf{w} = R_0 \mathbf{w} = R_0 \mathbf{V} \xi$, so ξ solves the stated problem and $\mathbf{w} = \beta \mathbf{Q} \xi / R_0$. Any other solution ξ' gives an eigenvector $\mathbf{V} \xi'$ of $\mathbf{K}$ for the simple eigenvalue R_0 , hence $\xi' \propto \xi$. Finally $\mathbf{V} \pi = \gamma \pi$ gives $\mathbf{K} \pi = \beta \mathbf{Q} \pi / \gamma$.

(iii) For a simple eigenvalue [26], $\partial R_0 / \partial q_x = \mathbf{z}^\top (\partial \mathbf{K} / \partial q_x) \mathbf{w} / (\mathbf{z}^\top \mathbf{w})$ with $\partial \mathbf{K} / \partial q_x = \beta \delta_x \delta_x^\top \mathbf{V}^{-1}$, so that $\mathbf{z}^\top (\partial \mathbf{K} / \partial q_x) \mathbf{w} = \beta z_x \xi_x = R_0 z_x w_x / q_x$. This is the formula for e_x ; the e_x are positive and sum to one. Next, let $\mathbf{A} = \mathbf{M} + \beta \mathbf{Q} / R_0 - \gamma \mathbf{1}$. Then $\mathbf{A} \xi = 0$ and, multiplying $\mathbf{z}^\top \mathbf{K} = R_0 \mathbf{z}^\top$ by $\mathbf{V}$ from the right, $\mathbf{z}^\top \mathbf{A} = 0$: ξ and $\mathbf{z}$ are the right and left Perron vectors of $\mathbf{A}$, and

$$e_x = \frac{q_x \theta_x}{\sum_y q_y \theta_y}, \quad \theta_x = \frac{z_x \xi_x}{\mathbf{z}^\top \xi}, \quad (\text{S5})$$

where by (P6) θ is the gradient of $f(v) = s(\mathbf{M} + \text{diag } v)$ at $v_0 = \beta \mathbf{q} / R_0 - \gamma \mathbf{1}$. Detailed balance reads $\mathbf{M} \Pi = \Pi \mathbf{M}^\top$, so $\mathbf{A}^\top = \Pi^{-1} \mathbf{A} \Pi$ and $\mathbf{A}^\top \Pi^{-1} \xi = 0$: $\mathbf{z} \propto \Pi^{-1} \xi$ and $e_x \propto q_x \xi_x^2 / \pi_x$.

(iv) *Fast mixing.* $\pi \mathbf{1}^\top$ is the spectral projector of $\mathbf{M}_0$ for its simple eigenvalue 0, and $\mathbf{M}_0$ is invertible on the complementary invariant subspace, so $\mathbf{V}^{-1} = \gamma^{-1} \pi \mathbf{1}^\top + D_0^{-1} (\gamma / D_0 - \mathbf{M}_0)^{-1} (\mathbf{1} - \pi \mathbf{1}^\top) \rightarrow \gamma^{-1} \pi \mathbf{1}^\top$ and $\mathbf{K} \rightarrow (\beta / \gamma) \mathbf{Q} \pi \mathbf{1}^\top$. The limit has the simple eigenvalue $\beta \langle q \rangle_\pi / \gamma$ with right eigenvector $\mathbf{Q} \pi$ and left eigenvector $\mathbf{1}$; the eigenvectors of a simple eigenvalue depend continuously on the matrix, which gives the limits of $\mathbf{w}$ and of $e_x = z_x w_x / (\mathbf{z}^\top \mathbf{w})$. In the same way $\mathbf{J} / D_0 \rightarrow \mathbf{M}_0$, whose Perron vector is π .

Slow mixing. As $D_0 \rightarrow 0$, $\mathbf{K} \rightarrow \beta \mathbf{Q} / \gamma$, $\mathbf{J} \rightarrow \beta \mathbf{Q} - \gamma \mathbf{1}$, $R_0 \rightarrow \beta q_{\max} / \gamma$ and $\lambda_1 \rightarrow \beta q_{\max} - \gamma$. The vectors $\mathbf{u}$ lie in the unit simplex, and every limit point $\mathbf{u}^*$ satisfies $(\beta q_x - \gamma) u_x^* = (\beta q_{\max} - \gamma) u_x^*$, so it vanishes outside Z ; the same argument applies to $\mathbf{w}$. The limiting eigenvalue is degenerate when Z has several cells, so for $\mathbf{e}$ we use (S5) and convexity instead. Because $\mathbf{1}^\top (\mathbf{M} + \text{diag } v) \leq (\max_y v_y) \mathbf{1}^\top$, pairing with a non-negative right Perron vector gives $f(v) \leq \max_y v_y$; and by (P2), $f(v) \geq \max_y (v_y + M_{yy}) \geq \max_y v_y - D_0 c$ with $c = \max_y |(\mathbf{M}_0)_{yy}|$. Let $m = \max_y (v_0)_y = \beta q_{\max} / R_0 - \gamma$. Since $f(v_0) = 0$ by Theorem 3.1(b), $0 \leq m \leq D_0 c$. For $x \notin Z$ and $t = \beta (q_{\max} - q_x) / R_0 > 0$ the vector $v_0 + t \delta_x$ still has maximum m , so the tangent inequality for the convex function f gives $t \theta_x \leq f(v_0 + t \delta_x) - f(v_0) \leq m \leq D_0 c$. As $R_0 \leq \beta q_{\max} / \gamma$, $t \geq \gamma (1 - q_x / q_{\max})$, hence $\theta_x \leq D_0 c / (\gamma (1 - q_x / q_{\max}))$, and $e_x \leq (q_x / q_{\min}) \theta_x \rightarrow 0$. $\square$

Remark S3.3 (Transients). The prevalence map $\mathbf{u}$ is reached after a time of order $1/\delta$, whereas the exponential phase started by one index case among N people lasts about $\ln N / \lambda_1$. When $1/\delta$ is not small compared with $\ln N / \lambda_1$, the early pattern depends on where the index case sits and is given by the seed-specific transient $e^{\mathbf{J}t} I(0)$. The part of an outbreak that has not yet left a zone Z with uniform q evolves by $e^{\mathbf{J}[Z, Z]t}$ and grows at the zone rate $s(\mathbf{J}[Z, Z]) = \beta q_Z - \gamma - \mu_Z \leq \lambda_1$ (cf. Proposition 3.5). The threshold statement of Theorem 3.1 is unaffected.

In the office scene at $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$ (Fig. 3), the expected number of secondary cases of an index case, $n_{\text{off}}(x)$, ranges from 3.995 to 5.335 over the starting cell. The elasticity share of the meeting room falls to 20.8% at $D_0 = 10$ and 13.4% at $D_0 = 100 \text{ cell}^2 \text{ day}^{-1}$, close to its limit $q_x \pi_x / \langle q \rangle_\pi$ summed over the room, 12.8%; the prevalence share falls to 50.3% and 12.4%. The first generation of a uniformly placed index case puts 12.8% of its infections in the meeting room at every mobility (Theorem 4.4(ii)), against 79.2% asymptotically at $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$.

The transient of Remark S3.3 matters in this scene. At $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$ the gap is $\delta = 0.061 \text{ day}^{-1}$ ($1/\delta = 16$ days), while $\lambda_1 = 0.602 \text{ day}^{-1}$ and $\ln 100 / \lambda_1 = 7.6$ days. In the linearised dynamics, a point seed at a workstation far from the meeting room grows at 0.553 day^{-1} while the total rises from 10^2 to 10^4 times the seed, between the workstation-zone rate 0.531 day^{-1}

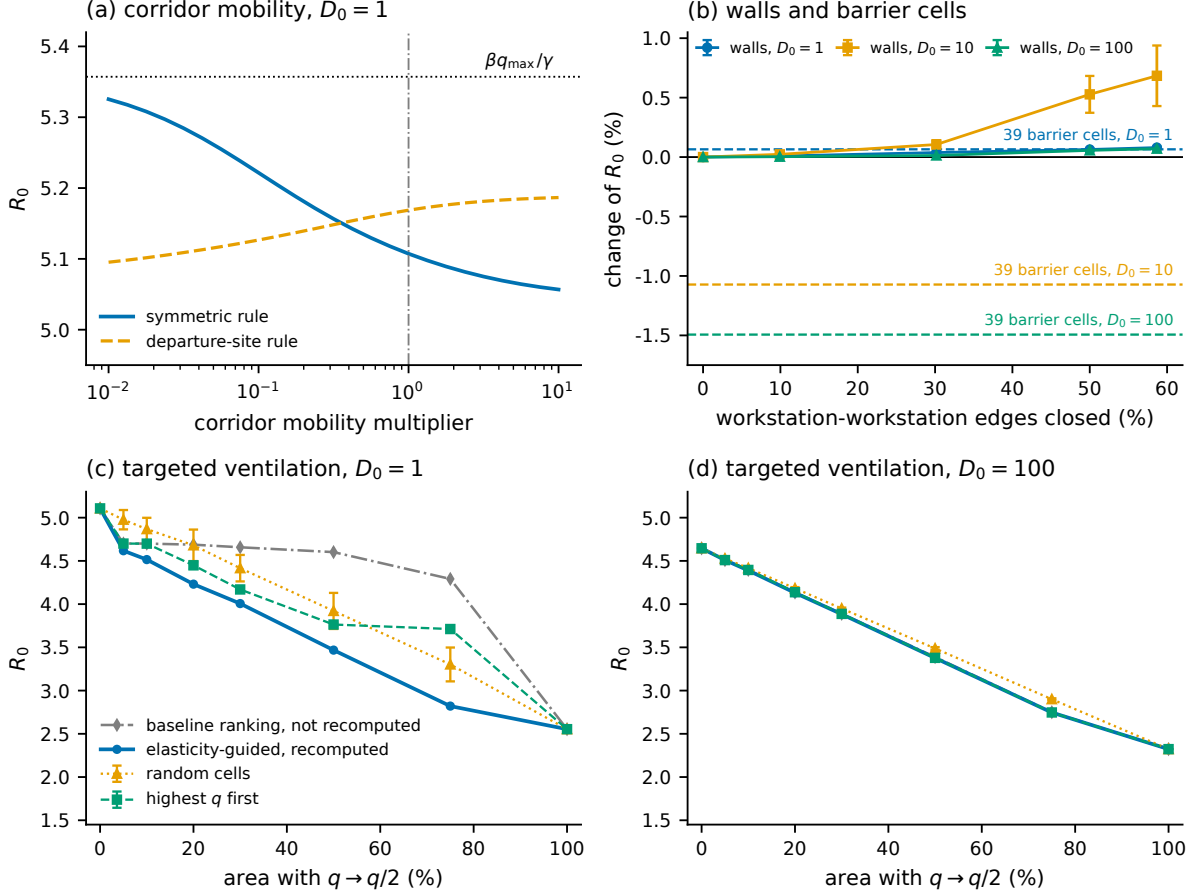


Figure S2: Mean-field R_0 of the office scene under interventions ($\beta = 0.5 \text{ day}^{-1}$, $\gamma = 0.14 \text{ day}^{-1}$, same-cell kernel). (a) Corridor mobility multiplied by a factor, for the symmetric rule (1) and for a departure-site rule; dotted line: upper bound $\beta q_{\max}/\gamma$. (b) Change of R_0 at $D_0 = 1, 10$ and $100 \text{ cell}^2 \text{ day}^{-1}$ when walls close a fraction of the 242 edges between workstation cells (symbols: mean and standard deviation over 20 random draws that keep the room connected; the last point closes as many edges as connectivity allows), and when 39 workstation cells are turned into barrier cells (dashed lines: means over 40 draws). Walls never lower R_0 (Theorem 3.6); barrier cells, which remove floor with q above the mean, lower it at $D_0 = 10$ and 100 . (c), (d) Targeted ventilation, $q \rightarrow q/2$ on a fraction of the floor, at $D_0 = 1$ and $D_0 = 100 \text{ cell}^2 \text{ day}^{-1}$: cells chosen by the elasticity map recomputed after every six cells, by the baseline elasticity ranking without recomputation, by highest q first (ties broken at random) and at random (means and standard deviations over 30 draws).

and λ_1 , and its spatial distribution is then still at total-variation distance 0.996 from $\mathbf{u}$; a seed in the meeting room grows at 0.605 day^{-1} and is at distance 0.026. At $D_0 = 100 \text{ cell}^2 \text{ day}^{-1}$ ($\delta = 0.537 \text{ day}^{-1}$) the distance is below 10^{-4} from either seed.

S3.3 Worked examples on the office scene

Table S4 and Fig. S2 give what the mean-field model says about interventions on the office scene.

Slower corridors raise R_0 . Multiplying the corridor mobility by 0.1 moves R_0 from 5.107 to 5.222, and by 0.01 to 5.325, as Theorem 3.6 requires: less movement means less mixing between high- q and low- q cells. Under a departure-site rule (hop rate D_x from cell x) the same change lowers R_0 (5.169, 5.127 and 5.095 at multipliers 1, 0.1 and 0.01), but only because people then accumulate in the slowed corridor, where q is lowest ($\langle q \rangle_\pi$ falls from 1.379 to 1.218 and 0.911). That is an occupancy effect in a different model.

Walls cannot lower R_0 ; barrier cells change the mean of q . Two kinds of partition must be

Table S4: Worked examples on the office scene (mean-field model, same-cell kernel; baseline $R_0 = 5.107$, 4.676 and 4.646 at $D_0 = 1$, 10 and $100 \text{ cell}^2 \text{ day}^{-1}$; bounds $4.643 \leq R_0 \leq 5.357$): interventions on movement (walls: mean over 20 random draws; barrier cells: mean over 40 draws, which lowers the well-mixed value $\beta\langle q \rangle/\gamma$ of the remaining floor from 4.643 to 4.571) and uniform measures. Targeted ventilation is in Table S5.

Intervention	R_0 at $D_0 = 1$	change of R_0 (%) at $D_0 =$		
		1	10	100
corridor mobility $\times 0.3$	5.159	+1.0		
corridor mobility $\times 0.1$	5.222	+2.2		
corridor mobility $\times 0.01$	5.325	+4.3		
walls on 24 of the 242 workstation–workstation edges	5.1077	+0.009	+0.023	+0.003
walls on 73 edges	5.1092	+0.038	+0.105	+0.013
walls on 121 edges	5.1105	+0.064	+0.53	+0.056
walls on 142 edges (as many as keep the room connected)	5.1114	+0.080	+0.68	+0.070
39 workstation cells made barrier cells	5.1106	+0.065	−1.07	−1.49
$q \rightarrow q/2$ in every cell	2.554	$\times 0.5$ (exact)		
$\beta \rightarrow 0.3\beta$ and $q \rightarrow q/2$ in every cell	0.766	$\times 0.15$ (exact)		

Table S5: Targeted ventilation in the office scene (mean-field model, same-cell kernel): R_0 after $q \rightarrow q/2$ on a fraction of the floor, at equal treated area for four placement rules (as in Fig. S2c,d; mean $\pm$ standard deviation over 30 draws). Baseline $R_0 = 5.107$ at $D_0 = 1$ and 4.646 at $D_0 = 100 \text{ cell}^2 \text{ day}^{-1}$.

D_0	treated area (cells)	recomputed elasticity	highest q first	random cells	baseline ranking, not recomputed
1	10 % (23)	4.515	4.699 ± 0.0002	4.867 ± 0.131	4.700
1	20 % (47)	4.231	4.449 ± 0.041	4.680 ± 0.182	4.688
1	50 % (117)	3.468	3.764 ± 0.019	3.921 ± 0.209	4.602
1	75 % (176)	2.821	3.712 ± 0.003	3.302 ± 0.196	4.291
1	100 % (234)	2.554	2.554	2.554	2.554
100	10 % (23)	4.390	4.395 ± 0.0003	4.416 ± 0.009	4.396
100	20 % (47)	4.130	4.137 ± 0.001	4.182 ± 0.014	4.135
100	50 % (117)	3.377	3.379 ± 0.001	3.487 ± 0.015	3.388
100	75 % (176)	2.746	2.747 ± 0.0003	2.900 ± 0.011	2.750
100	100 % (234)	2.323	2.323	2.323	2.323

kept apart. A wall closes edges and leaves the set of cells unchanged, so Theorem 3.6 applies: closing between 10 % of the workstation–workstation edges and as many as connectivity allows raises R_0 by 0.009 % to 0.080 % at $D_0 = 1$, by 0.02 % to 0.68 % at $D_0 = 10$ and by 0.003 % to 0.070 % at $D_0 = 100 \text{ cell}^2 \text{ day}^{-1}$, and none of the 80 draws lies below the baseline at any of the three mobilities. A barrier cell removes floor, and with it its q , which the theorem does not cover. Turning 39 workstation cells ($q = 1.4$, above the mean 1.30) into barriers lowers the well-mixed value $\beta\langle q \rangle/\gamma$ of the remaining floor from 4.643 to 4.571 (−1.5 %). At $D_0 = 1$, where R_0 is pinned by the meeting room, the net change is +0.065 % (all 40 draws above the baseline); at $D_0 = 10$ and 100 it is −1.07 % and −1.49 % (all 40 draws below). These are numerical results for this floor plan, not consequences of a theorem.

Targeted ventilation helps at low mobility, if the map is recomputed. Here ventilation is an input: q is halved by assumption on the treated cells (Fig. S2c,d; numbers in Table S5). At $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$ and equal treated area, choosing cells by the elasticity map and recomputing it after every six cells gives an R_0 that is lower than for random placement by 0.35 to 0.48 between 5 % and 75 % treated area, and lower than for ‘highest q first’ by 0.18 at 10 % and 0.30 at 50 %. The recomputation is essential, because the hotspot moves once the leading zone has been treated: a ranking by the baseline elasticity that is not recomputed is no better than random placement from 20 % treated area on (4.688 against 4.680 ± 0.182) and clearly worse at

50 % (4.602 against 3.921 ± 0.209). At $D_0 = 100 \text{ cell}^2 \text{ day}^{-1}$ the four rules differ by at most 0.11 up to 50 % treated area: in a well-mixed room $e_x = q_x \pi_x / \langle q \rangle_\pi$ and targeting reduces to treating the cells with the largest q . For discrete individuals at $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$ the placement makes no measurable difference (Section 6.4).

Uniform measures scale R_0 exactly. Because $\mathbf{K} = \beta \mathbf{QV}^{-1}$, multiplying β by s_β and every q_x by s_q multiplies R_0 by $s_\beta s_q$ for every geometry and mobility. Halving q everywhere gives 2.554; a mask factor $\beta \rightarrow 0.3\beta$ together with it gives $0.15 \times 5.107 = 0.766$. The product is plain proportionality and involves no interaction between the two measures.

S4 Pair-level theory: proofs and further results

This section proves the statements of Section 5 of the main text and gives the tables behind its figures. Movement follows the symmetric rule (1), and (P1)–(P6) are the facts of Lemma S2.1.

S4.1 Proofs

Proof of Proposition 5.1. Write $z = (x, y)$, let $r(z \rightarrow z') = \mathcal{M}_{z'z}$ ($z' \neq z$) be the hop rates of the pair chain, $r(z) = \sum_{z'} r(z \rightarrow z')$, and $u = \mathbf{1} - p$. From state z three kinds of independent exponential clocks compete: recovery (rate γ), infection (rate $h(z)$) and a hop to z' (rate $r(z \rightarrow z')$). If recovery rings first the person escapes; if infection rings first it does not; if a hop to z' rings first, the Markov property and the lack of memory of τ give the escape probability $u(z')$. Hence

$$(\gamma + h(z) + r(z)) u(z) = \gamma + \sum_{z'} r(z \rightarrow z') u(z'),$$

which is $(\gamma \mathbf{1} + \mathbf{H} - \mathcal{M}^\top)u = \gamma \mathbf{1}$, because $(\mathcal{M}^\top u)(z) = \sum_{z'} r(z \rightarrow z')(u(z') - u(z))$. The matrix $A = \gamma \mathbf{1} + \mathbf{H} - \mathcal{M}^\top$ has non-positive off-diagonal entries and row sums $\gamma + h(z) \geq \gamma > 0$; by (P5) of Lemma S2.1 applied to $A^\top$ it is non-singular with $A^{-1} \geq 0$, so $u = \gamma A^{-1} \mathbf{1}$. $\square$

The matrix $\mathbf{K}_1$ is obtained from n further solves: by the same first-step argument, the probability that the pair started at (x, y) ends with an infection while the susceptible is in cell z is $w_z = A^{-1} b_z$ with $b_z(x', y') = h(x', y') \mathbf{1}\{y' = z\}$, and $(K_1)_{zx} = \bar{\rho} \sum_y w_z(x, y)$.

Proof of Proposition 5.3. Let T_i be the infection time of the given person. Conditionally on the path of the pair and on τ , T_i has hazard $h(Z_t)$ on $[0, \tau)$, so that

$$\begin{aligned} \mathbb{P}_{(x,y)}(T_i < \tau, Y_{T_i} = z) &= \int_0^\infty e^{-\gamma t} \mathbb{E}_{(x,y)} \left[h(X_t, z) \mathbf{1}\{Y_t = z\} e^{-\int_0^t h(Z_s) ds} \right] dt \\ &\leq \int_0^\infty e^{-\gamma t} \sum_{x'} P_t(x' | x) h(x', z) P_t(z | y) dt, \end{aligned}$$

where the two walkers are independent. Because $\mathbf{M}$ is symmetric, $\sum_y P_t(z | y) = 1$. Summing over y and multiplying by $(N - 1)/n = \bar{\rho}$,

$$(K_1)_{zx} \leq \beta \sum_{x'} P_{x'z} q_{x'} \int_0^\infty e^{-\gamma t} P_t(x' | x) dt = K_{zx}$$

by Theorem 3.1(a). On the event $\{h(Z_t) > 0\}$ the pair has almost surely been in its current state for a positive time, so the discarded factor is smaller than one there and the inequality is strict wherever $K_{zx} > 0$. Column sums give $R_1(x) < n_{\text{off}}(x)$ (recall $n_{\text{off}}(x) > 0$); the uniform average and eq. (7) give $\bar{R}_1 < \beta \langle q \rangle / \gamma$; and (P4) of Lemma S2.1 gives $\rho(\mathbf{K}_1) \leq \rho(\mathbf{K})$. $\square$

Proof of Proposition 5.4. The relative position $\zeta_t = Y_t - X_t$ (modulo L_1, L_2) jumps to each neighbour at rate $2D$, so it is a Markov chain with generator $2\mathbf{M}$, and the hazard depends on

the pair only through ζ_t : it is h_0 at the origin and 0 elsewhere. Hence $p(x, y) = \varphi(y - x)$, and the argument of Proposition 5.1 applied to ζ_t gives, for $u = \mathbf{1} - \varphi$, $(\gamma \mathbf{1} - 2\mathbf{M})u + h_0 u(0) \delta_0 = \gamma \mathbf{1}$, with δ_0 the indicator of the origin. Let $\mathbf{G} = (\gamma \mathbf{1} - 2\mathbf{M})^{-1}$, so that $g(r) = G_{0r} = G_{r0}$. Since $\mathbf{M}\mathbf{1} = 0$, $\mathbf{G}\mathbf{1} = \mathbf{1}/\gamma$ and

$$u(r) = 1 - h_0 u(0) g(r).$$

Setting $r = 0$ gives $u(0) = 1/(1 + h_0 g_0)$, hence $\varphi(r) = h_0 g(r)/(1 + h_0 g_0)$. Probability conservation gives $\sum_r g(r) = 1/\gamma$, so $R_1 = \bar{\rho} \sum_r \varphi(r) = \bar{\rho} h_0 / (\gamma(1 + h_0 g_0))$, which is (13). The plane waves $e^{ik \cdot r} / \sqrt{n}$ diagonalise $\mathbf{M}$ with eigenvalues $-\mu_k$, which gives $g_0 = G_{00}$ as stated. Every μ_k with $k \neq 0$ is proportional to D , so g_0 decreases and R_1 increases with D ; g_0 does not depend on $\bar{\rho}$, so R_1 increases with $\bar{\rho}$. As $D \rightarrow \infty$ only the term $k = 0$ survives, $g_0 \rightarrow 1/(n\gamma)$, and $\bar{\rho}n = N - 1$; as $D \rightarrow 0$, $g_0 \rightarrow 1/\gamma$. $\square$

Proof of Corollary 5.5. The sum in (13) is a Riemann sum of a continuous function bounded by $1/\gamma$, so

$$g_0 \rightarrow \frac{1}{(2\pi)^2} \int_0^{2\pi} \int_0^{2\pi} \frac{dk_1 dk_2}{\gamma + 4D(2 - \cos k_1 - \cos k_2)} = \frac{1}{\gamma + 8D} \frac{1}{\pi^2} \int_0^\pi \int_0^\pi \frac{dk_1 dk_2}{1 - \frac{1}{2}k_*(\cos k_1 + \cos k_2)},$$

and the last double integral is the Green's function of the square lattice at the origin, equal to $(2/\pi)\mathcal{K}(k_*)$ [49]. Since $g_0 > 0$, $R_1 < \beta q/\gamma$. $\square$

The elliptic form agrees with a direct quadrature of the double integral to 12 digits ($g_0 = 0.241939$ at $D = 1$, $\gamma = 0.14 \text{ day}^{-1}$). For $8D \gg \gamma$ the logarithmic singularity of $\mathcal{K}$ gives $g_0 \approx \ln(64D/\gamma)/(8\pi D)$ (0.03353 against 0.03351 at $D = 10$): the correction decays with mobility only as $\ln D/D$, which reflects the recurrence of the planar walk.

With reflecting walls the pair walk is no longer translation invariant. The quantity $n^{-1} \sum_k (\gamma + 2\mu_k)^{-1} = n^{-1} \text{tr}(\gamma \mathbf{1} - 2\mathbf{M})^{-1}$ is the cell-averaged co-location Green's function of a pair started together, the mean over x of $\int_0^\infty e^{-\gamma t} \mathbb{P}(X_t = Y_t | X_0 = Y_0 = x) dt$ (on the torus, the return Green's function of the relative walk), and the observation says that replacing each cell's value by this average costs little. In the 21 rooms tested the closed form lies below the exact value, by at most 0.90 % at $D = 0.51$, 0.46 % at $D = 1$ and 0.02 % at $D = 10$.

S4.2 Finite size and mobility

The counted values test the first generation only. For an index case born according to the Perron vector of $\mathbf{K}_1$ the expected count is exactly $\rho(\mathbf{K}_1)$, and for a uniformly placed one it is $\bar{R}_1$. Of the twenty intervals in the table, nineteen contain the pair-level value. The exception is the uniform index at $D_0 = 10$ (3.697, interval 3.643–3.751, against 3.793), a batch of 20,000 runs that is reproduced bit for bit from its seed and lies 3.5 standard errors low. It is a chance fluctuation: 400,000 further runs with other seeds give 3.793 ± 0.006 (standard error), and 40,000 runs of the second simulator 3.787 ± 0.020 . The same larger samples give 2.265 ± 0.004 at $D_0 = 1$ ($\bar{R}_1 = 2.266$) and 4.340 ± 0.015 at $D_0 = 100$ (100,000 runs; $\bar{R}_1 = 4.341$); they are the counted values quoted in the main text.

Figure S3 maps the ratio of the two levels on the infinite homogeneous lattice. At $\bar{\rho} = 0.381$ and $\beta q = 0.5 \text{ day}^{-1}$ the ratio is 0.5 at $D = 0.24$, 0.9 at $D = 3.5$ and 0.99 at $D = 52 \text{ cell}^2 \text{ day}^{-1}$; for $\beta q = 1.25 \text{ day}^{-1}$ these mobilities are 0.76, 9.9 and 143. Higher occupancy moves the contours to lower mobility (ratio 0.9 at $D = 0.05$ for $\bar{\rho} = 10$, $\beta q = 0.5 \text{ day}^{-1}$). The mean-field number is therefore adequate when many people share a contact neighbourhood or when people move far enough during an infectious period to meet new ones. The scenes at $D_0 = 1$ satisfy neither condition. At $D_0 \approx 2.4 \times 10^3 \text{ cell}^2 \text{ day}^{-1}$ (Section 2.6) the ratio on the infinite lattice is 0.9997, and what remains in a finite room is the well-mixed finite-population factor.

In the office at $D_0 = 1$ the second generation of an index case and its direct cases is 1.63 times the first (Table S7). In unrestricted epidemics started from a uniformly placed index case

Table S6: Expected number of secondary cases of a lone index case in a homogeneous $L \times L$ room with reflecting walls ($q = 1$, same-cell kernel, $\beta = 0.5 \text{ day}^{-1}$, $\gamma = 0.14 \text{ day}^{-1}$, N people so that $\bar{\rho} \approx 99/260 = 0.381$ others per cell). Mean field: $R_0 = \rho(\mathbf{K})$. Closed form: eq. (13) with the reflecting-wall spectrum of Observation 5.6; last row: Corollary 5.5. Exact: $\bar{R}_1$ from eq. (12) (not computed for $L = 40$). Counted: mean number of secondary cases $\hat{R}$ in 20,000 exact simulations with a uniformly placed index case, only the index transmitting, with 95 % confidence interval.

D	L	N	mean field	closed form	exact	counted $\hat{R}$ (95 % CI)
1	4	7	3.571	1.876	1.882	1.887 (1.864–1.910)
	10	39	3.571	2.454	2.465	2.464 (2.427–2.501)
	20	153	3.571	2.588	2.596	2.599 (2.558–2.639)
	40	610	3.571	2.651	–	2.651 (2.609–2.692)
	∞	–	3.571	2.710	–	–
10	4	7	3.571	2.194	2.194	2.193 (2.168–2.218)
	10	39	3.571	3.141	3.141	3.101 (3.056–3.147)
	20	153	3.571	3.333	3.333	3.311 (3.260–3.363)
	40	610	3.571	3.385	–	3.397 (3.344–3.449)
	∞	–	3.571	3.421	–	–

Table S7: Office scene, $N = 100$, same-cell kernel (identical to the 1 m kernel at $a = 1.5$ m), as a function of the mobility scale D_0 ($\text{cell}^2 \text{ day}^{-1}$). R_0 : mean field. $\rho(\mathbf{K}_1)$ and $\bar{R}_1$: pair level (Definition 5.2). $\hat{R}$: counted secondary cases when only the index transmits, for an index case born according to the Perron vector of $\mathbf{K}_1$ (expectation $\rho(\mathbf{K}_1)$) and for a uniformly placed index case (expectation $\bar{R}_1$). Last column: total number of second-generation cases divided by the number of first-generation cases when the index (Perron-weighted) and its direct cases transmit. Intervals are 95 % confidence intervals (bootstrap for the ratio); 20,000 runs per entry up to $D_0 = 30$, 6,000 at $D_0 = 100$, and 2,000 at $D_0 = 300$ and at $D_0 = 1000$. ^a A chance-low batch (seed 301, reproduced bit for bit): 400,000 further runs give 3.793 ± 0.006 (standard error; Section 5.4).

D_0	R_0	$\rho(\mathbf{K}_1)$	$\hat{R}$, Perron-weighted index	$\bar{R}_1$	$\hat{R}$, uniform index	generation 2 / generation 1
0.03	5.348	0.710	0.719 (0.705–0.732)	0.586	0.586 (0.574–0.597)	0.357 (0.345–0.369)
0.1	5.327	1.096	1.096 (1.078–1.113)	0.895	0.898 (0.883–0.914)	0.623 (0.610–0.635)
0.3	5.271	1.568	1.571 (1.547–1.595)	1.422	1.429 (1.407–1.452)	0.986 (0.969–1.001)
1	5.107	2.321	2.338 (2.303–2.373)	2.266	2.271 (2.236–2.305)	1.631 (1.610–1.650)
3	4.813	3.111	3.125 (3.079–3.171)	3.095	3.085 (3.039–3.130)	2.403 (2.381–2.428)
10	4.676	3.803	3.847 (3.790–3.904)	3.793	3.697 (3.643–3.751) ^a	3.048 (3.022–3.074)
30	4.653	4.165	4.169 (4.107–4.231)	4.160	4.102 (4.042–4.162)	3.389 (3.362–3.413)
100	4.646	4.343	4.354 (4.235–4.473)	4.341	4.373 (4.255–4.491)	3.513 (3.470–3.560)
300	4.644	4.403	4.373 (4.176–4.570)	4.402	4.242 (4.042–4.442)	3.524 (3.441–3.608)
1000	4.643	4.425	4.502 (4.294–4.709)	4.425	4.530 (4.320–4.740)	3.605 (3.523–3.689)

the ratio is lower still: the second generation is 1.29 and 1.39 times the first in the runs of the two simulators (Section 6.2).

S4.3 A per-cell normalisation of the infection rate

Remark S4.1 (Encounter factor; an approximation). The pair hazard (5) is normalised by the mean number $\bar{\rho}$ of *other* people per cell. A rule that infects each susceptible in a cell at rate $\beta q n_I / n_{\text{tot}}$, with n_I and n_{tot} the instantaneous numbers of infectives and of all people in that cell, is a different process. An infective who shares its cell with k susceptible people then infects at total rate $\beta q k / (k + 1)$, which is zero when it is alone. If the other $N - 1$ people are taken to be independent of the infective and at stationarity at every instant (an annealed approximation, appropriate when movement is fast), k is binomial with parameters $N - 1$ and π_x , and the

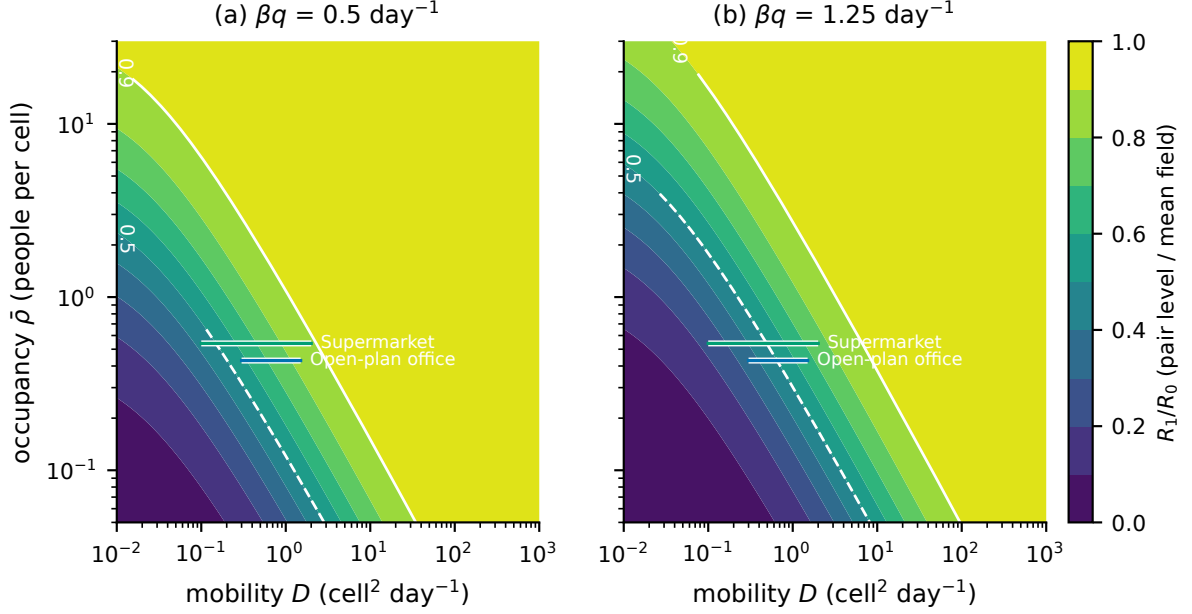


Figure S3: Where the mean-field number is a good approximation of the pair-level number. Ratio $R_1/R_0 = 1/(1 + (\beta q/\bar{\rho})g_0)$ on the infinite homogeneous lattice (Corollary 5.5; same-cell kernel, $\gamma = 0.14 \text{ day}^{-1}$) against the mobility D and the occupancy $\bar{\rho}$ (other people per cell), for (a) $\beta q = 0.5 \text{ day}^{-1}$ and (b) $\beta q = 1.25 \text{ day}^{-1}$. White contours: ratio 0.5 (dashed) and 0.9 (solid). Horizontal bars: occupancy and range of zone mobilities at $D_0 = 1$ of the office (0.43 people per cell, $D = 0.3\text{--}1.5$) and supermarket (0.54, $D = 0.1\text{--}2.0$) scenes.

identity $\binom{N-1}{k}/(k+1) = \binom{N}{k+1}/N$ gives the encounter factor

$$c_x = \mathbb{E}\left[\frac{k}{k+1}\right] = 1 - \frac{1 - (1 - \pi_x)^N}{N \pi_x},$$

so that the exposure of a case under the per-cell rule is that of the mean-field model with q_x replaced by $c_x q_x$, and its reproduction number is approximately $\rho(\beta \mathbf{Q} \text{diag}(c)(\gamma \mathbf{1} - \mathbf{M})^{-1})$. For 100 people on the 234 cells of the office $c = 0.185$ (the index is alone in its cell with probability 0.65); for 100 people on 20×20 cells $c = 0.114$. With a uniformly placed index case the annealed estimates are $c\beta\langle q \rangle/\gamma = 0.86$ and 0.41 . Exact simulation of the rule gives 0.690 ($0.676\text{--}0.704$) secondary cases in the office scene and 0.382 ($0.372\text{--}0.392$) in the homogeneous 20×20 room at $D = 1$ (20,000 runs each; the second simulator gives 0.698 ± 0.007 and 0.384 ± 0.005), and no run infected more than 37 and 23 people respectively: at these occupancies the per-cell rule is sub-critical.

The annealed formula is not exact. In a homogeneous 20×13 room with 100 people and $q = 1.4$ it predicts 0.843 ; simulation gives 0.825 ± 0.013 at $D_0 = 10$ (9,000 runs), within about 5%, but 0.754 ± 0.022 at $D_0 = 1$ and 0.545 ± 0.017 at $D_0 = 0.1$. In the two cases above it is too high by 7% (20×20 room) and by 24% (office, whose workstation cells have $D = 0.3$). The reason is again the repeated exposure of the same neighbours, which the annealed average ignores. In the four scenes at $D_0 = 1$ the per-cell rule gives $0.62\text{--}1.01$ secondary cases, a mean attack rate of $1.0\text{--}3.0\%$, and at most 4.8% of runs reach an attack rate of 10% (none of 2,000 in the metro carriage); it becomes supercritical only in the crowded carriage at high mobility (2.43 secondary cases and a mean attack rate of 52% at $D_0 = 100$).

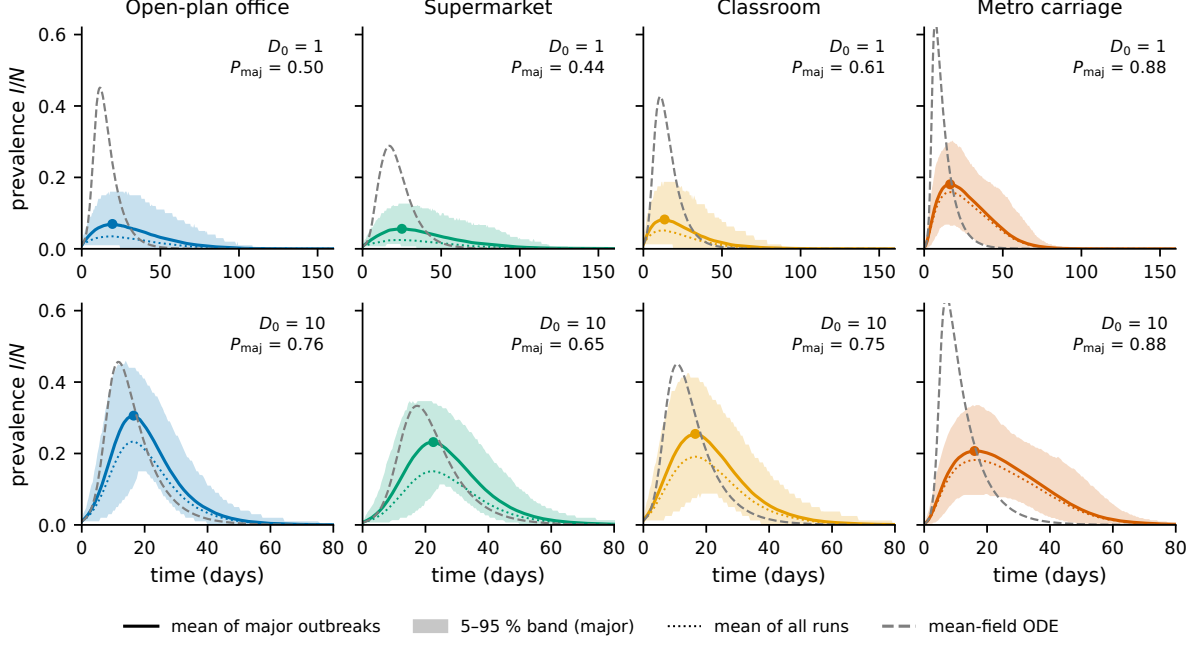


Figure S4: Prevalence I/N against time in the four scenes (columns; $N = 100, 150, 80, 310$) at $D_0 = 1$ (top) and $D_0 = 10 \text{ cell}^2 \text{ day}^{-1}$ (bottom); 1 m kernel, 2,000 exact simulations per panel, one uniformly placed index case. Solid line: mean over major outbreaks (attack rate at least 10%) at each time, with the band between the 5th and 95th percentiles; the dot marks its maximum. Dotted: mean over all runs. Dashed: mean-field equations (3) started from one infective spread uniformly over the cells. The values of P_{maj} in the panels are those of Table 4. The maximum of the mean curve (office, $D_0 = 1$: 7.0% on day 19.5) is lower than the mean of the individual peaks in Table 4 (12.7%) because outbreaks peak at different times.

S5 Simulations of the scenes and of interventions: details

This section gives the details of Section 6 of the main text. Unless stated otherwise $\beta = 0.5 \text{ day}^{-1}$, $\gamma = 0.14 \text{ day}^{-1}$, the contact kernel is the 1 m kernel, there is one uniformly placed index case, the room is occupied 24 hours a day, a ‘major’ outbreak has an attack rate of at least 10%, and $\hat{R}$ is the counted number of secondary cases of an index case that alone transmits. $\bar{R}_1$ and $\rho(\mathbf{K}_1)$ are not threshold quantities (Remark 5.7).

S5.1 The four scenes

An epidemic in the office scene at $D_0 = 1$ consists of about 5,900 events. Peak prevalence and peak day are the maximum of I/N on a grid of 0.5 day and its time, averaged over major outbreaks. The mean-field equations are started from one infective spread uniformly over the cells, and from one infective placed in a single cell (averaged over every third cell as seed); the extinction probabilities of (14) are computed by iteration from $s = 0$. Fig. S4 shows the epidemic curves of the four scenes and Fig. S5 the distribution of the attack rate over the same runs.

The values of $\hat{R}$ in Table 4 come from separate batches of 2,000 runs and are the least precise estimates of this quantity in the article; for the office, samples of 100,000 to 400,000 runs give 2.265 ± 0.004 , 3.793 ± 0.006 and 4.340 ± 0.015 at $D_0 = 1, 10$ and 100 (Section 5.4), in agreement with $\bar{R}_1$. The mean-field R_0 does not: at $D_0 = 1$ the index case of the office infects 2.26 ± 0.06 people where $R_0 = 5.11$, and the ratio $\hat{R}/R_0$ is 0.44, 0.45, 0.45 and 0.60 in the office, supermarket, classroom and metro carriage. It rises to 0.84–0.93 at $D_0 = 100$.

With the same-cell kernel the mean field is not adequate for the outbreak probability where cells are smaller than the contact range: in the metro carriage $P_{\text{br}} - P_{\text{maj}}$ is 0.54 at $D_0 = 10$

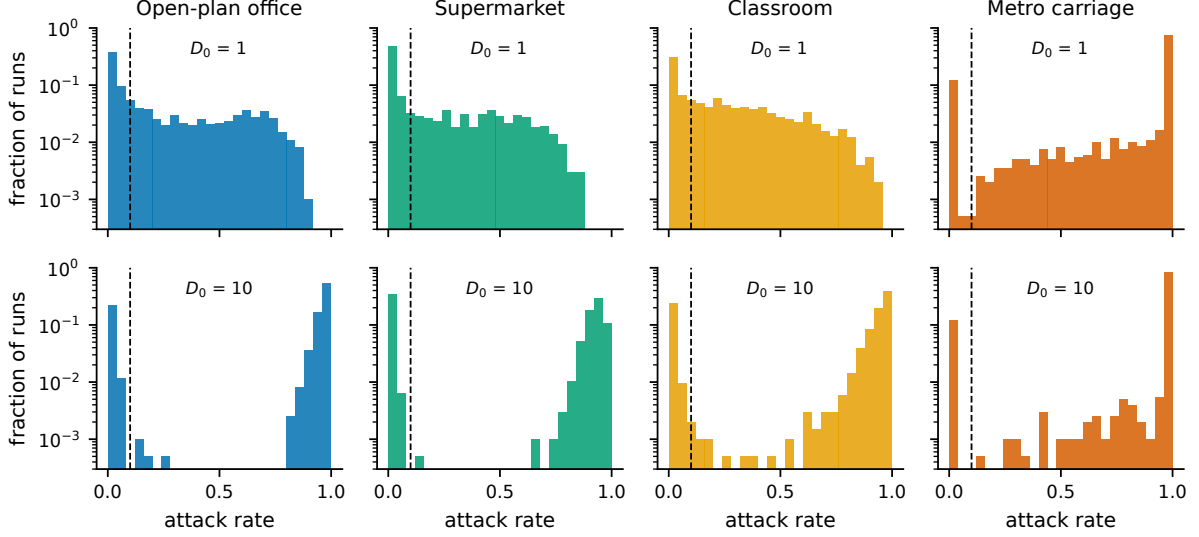


Figure S5: Distribution of the attack rate (fraction of the N people ever infected) over the 2,000 runs of each panel of Fig. S4; bins of width 0.04, logarithmic vertical axis. The dashed line is the 10 % threshold that defines a major outbreak. At $D_0 = 10 \text{ cell}^2 \text{ day}^{-1}$ the distributions are bimodal; at $D_0 = 1$ they are not, except in the metro carriage.

(0.888 against 0.349; attack rate of major outbreaks 17.3 % against 100.0 %) and still 0.031 at $D_0 = 100$ (98.6 % against 100.0 %); in the classroom it is 0.08 at $D_0 = 10$ (89.1 % against 98.8 %) and 0.014 at $D_0 = 100$ (Section S5.3).

Peak height and peak day are a different matter. At $D_0 = 10$ the simulated peaks (39.0 %, 30.6 %, 35.0 %, 27.5 %) are lower and about four to five days later than those of the mean-field equations even when these are started from a single cell (44.1 %, 33.1 %, 41.3 %, 36.3 %). At $D_0 = 100$ the two agree to within 2.5 percentage points and about one day in the office, the supermarket and the classroom. In the metro carriage they do not: its mobility is $0.028 D_0$ on area average (Table 2), the infection has to travel along the 45 cells of the carriage, and the mean-field peak depends on the seed (64.6 % on day 7.0 for a uniform seed, 42.2 % on day 13.8 for a single cell) and stays above the simulated 35.3 % on day 16.5.

Replication. The second simulator gave $P_{\text{maj}} = 0.498, 0.429, 0.603$ and 0.864 for the four scenes at $D_0 = 1$ (3,000–4,000 runs each) and 0.734 for the office at $D_0 = 10$, against $0.496, 0.436, 0.614, 0.879$ and 0.762 here. The two largest differences are 1.6 and 2.1 standard errors. The rows for $D_0 = 100$, the metro carriage at $D_0 = 10$ and the same-cell rows at $D_0 \geq 10$ were not re-run with the second simulator. Sampling prevalence every 0.5 day lowers the reported peaks by 0.3 to 1.0 percentage point relative to the exact maximum.

S5.2 Outbreak probability and early extinction

Early extinction of a chain started by one case occurs in every scene: the fraction of runs below the 10 % threshold is 0.50, 0.56, 0.39 and 0.12 at $D_0 = 1$ and 0.24, 0.35, 0.25 and 0.12 at $D_0 = 10$ (office, supermarket, classroom, metro carriage). Table S8 shows what determines it. The number of secondary cases of an index case is over-dispersed, with a variance 2.6 to 5.0 times the mean, and 10 % to 31 % of index cases infect no one. An exponential infectious period alone produces this much: a Poisson number of infections over an exponentially distributed period is geometric, with a variance-to-mean ratio of one plus the mean (3.0 to 7.4 here). The simulated ratios are lower, consistent with the limited number of people within reach. Over-dispersion lowers the outbreak probability [43]: a Poisson law with the same mean would give 0.80–1.00. A

Table S8: Offspring distribution of a lone index case and the probability of a major outbreak (1 m kernel; D_0 in $\text{cell}^2 \text{ day}^{-1}$). Columns 3–6: mean, variance, variance-to-mean ratio and probability of zero of the number of secondary cases when only the uniformly placed index case transmits (40,000 runs per row). Columns 7–9: probability that a branching process does not die out, for a Poisson offspring law with the simulated mean, for the mean-field process (14), and for a Galton–Watson process with the simulated offspring law. Last column: fraction of 2,000 full epidemics with attack rate at least 10 % (Table 4).

Scene	D_0	secondary cases of a lone index				branching processes			simulation
		mean	variance	ratio	$P(0)$	Poisson	mean field	Galton–Watson	P_{maj} (95 % CI)
Office	1	2.27	6.0	2.66	0.269	0.857	0.781	0.613	0.496 (0.474–0.518)
Supermarket	1	2.00	5.3	2.68	0.310	0.796	0.667	0.531	0.436 (0.414–0.458)
Classroom	1	2.58	6.7	2.60	0.228	0.903	0.767	0.688	0.614 (0.592–0.635)
Metro	1	5.54	21.0	3.80	0.109	0.996	0.888	0.877	0.879 (0.864–0.893)
Office	10	3.82	16.3	4.25	0.194	0.976	0.783	0.757	0.762 (0.743–0.781)
Supermarket	10	2.89	11.0	3.80	0.253	0.932	0.685	0.658	0.648 (0.627–0.669)
Classroom	10	3.73	15.5	4.15	0.197	0.973	0.777	0.753	0.750 (0.731–0.768)
Metro	10	6.35	31.9	5.02	0.104	0.998	0.888	0.883	0.878 (0.863–0.892)

Galton–Watson process fed with the simulated offspring law reproduces P_{maj} to within 0.011 wherever the attack-rate distribution is bimodal (all scenes at $D_0 = 10$, the metro carriage at $D_0 = 1$).

In the other three scenes at $D_0 = 1$ the comparison depends on the convention, because the notion of a major outbreak is not sharp there. With a threshold of 5 %, 10 % or 20 % the office gives 0.60, 0.50 or 0.40, the supermarket 0.49, 0.44 or 0.37 and the classroom 0.70, 0.61 or 0.49, whereas at $D_0 = 10$ the three thresholds change no value by more than 0.014. The Galton–Watson values (0.61, 0.53, 0.69) are 0.07–0.12 above the 10 % fractions and within 0.05 of the 5 % fractions, so no conclusion about later generations can be drawn from them. The mean-field branching values (0.78, 0.67, 0.77) are too high under either convention, by 0.15–0.29 and 0.07–0.19.

S5.3 Sensitivity to the contact kernel

The contact kernel is a modelling choice with large consequences when cells are smaller than the contact range. With the strict same-cell kernel the metro carriage (0.5 m cells) produces no outbreak at $D_0 = 1$: none of 2,000 runs reaches an attack rate of 10 % (upper 95 % limit 0.002; none of 3,000 in the second simulator) and the mean attack rate is 1.1 %, although $R_0 = 9.84$. The index case does infect: $\bar{R}_1 = 1.51$, $\hat{R} = 1.56 \pm 0.04$, and the pair-level matrix has $\rho(\mathbf{K}_1) = 2.24$. But passengers hardly move at this mobility, the people the index case can reach are those who share its cell, and the cases it makes are born in that cell with no one left to infect: over the 2,000 full epidemics the second generation is 0.61 times as numerous as the first. With the 1 m kernel the same carriage is the most explosive scene ($P_{\text{maj}} = 0.879$). In the classroom the same-cell rule gives $P_{\text{maj}} = 0.276$ against 0.614 with the 1 m kernel, with $\rho(\mathbf{K}_1) = 2.27$ and a generation ratio of 0.90. The difference between the kernels fades with mobility and is gone in the classroom at $D_0 = 100$ (0.771 against 0.779), not yet in the metro carriage (0.858 against 0.891). These cases are the evidence for Remark 5.7: $\bar{R}_1 > 1$ and $\rho(\mathbf{K}_1) > 1$ do not imply that outbreaks occur, and neither does $R_0 > 1$. Nothing computed in this article predicts the threshold of the discrete system at low mobility; it has to be simulated.

S5.4 Where infections happen

For a uniformly placed index case the cell in which a case of generation g is infected has a distribution proportional to $\mathbf{K}^g \mathbf{1}$ in the mean-field model and, in the pair-level description, to $\mathbf{K}_1^g \mathbf{1}$. Table S9(a) compares both with the infection trees of 40,000 outbreaks per scene.

Table S9: Where infections happen: prediction against simulation (1 m kernel; D_0 in cell² day⁻¹). (a) Pearson correlation r , over the walkable cells, between the predicted and the simulated distribution of the cell in which the cases of generation $g = 1, 2, 3$ are infected (40,000 outbreaks per row, uniformly placed index case, generations 0–2 infectious); mean-field prediction $\propto \mathbf{K}^g \mathbf{1}$, pair-level prediction $\propto \mathbf{K}_1^g \mathbf{1}$. ‘Index alone’: first generation in 100,000 runs in which only the index case transmits, with the value of r expected from sampling noise if the pair-level prediction is exact. Last column: cases of generation 2 per outbreak, simulated / pair level / mean field. (b) Office scene: share of the infections of generation g that occur in a class of cells divided by the share of the walkable floor it occupies (1 = uniform).

(a) Correlation between predicted and simulated infection maps

Scene	D_0	r , generations 1 / 2 / 3			r , index alone			cases per outbreak, generation 2
		mean field	pair level		mean field	pair level	if exact	
Office	1	0.09 / -0.23 / -0.37	0.88 / 0.82 / 0.71		0.284	0.948	0.958	2.8 / 5.2 / 21.8
Supermarket	1	0.37 / 0.15 / 0.22	0.91 / 0.85 / 0.85		0.550	0.981	0.977	2.6 / 4.0 / 11.6
Classroom	1	0.76 / 0.67 / 0.68	0.96 / 0.93 / 0.91		0.844	0.989	0.989	2.8 / 6.8 / 23.6
Metro	1	0.76 / 0.69 / 0.57	0.92 / 0.95 / 0.91		0.876	0.980	0.979	8.1 / 31.6 / 82.8
Office	10	0.77 / 0.52 / 0.41	0.92 / 0.85 / 0.81		0.915	0.985	0.985	8.1 / 14.4 / 21.6
Supermarket	10	0.97 / 0.97 / 0.98	0.99 / 0.99 / 0.99		0.981	0.996	0.996	6.3 / 8.3 / 10.9
Classroom	10	0.96 / 0.96 / 0.96	0.98 / 0.98 / 0.97		0.984	0.996	0.996	6.8 / 14.1 / 22.7
Metro	10	0.77 / 0.67 / 0.60	0.93 / 0.95 / 0.93		0.889	0.982	0.984	10.3 / 40.3 / 82.6

(b) Office scene: relative infection rate by class of cells, generations 1 / 2 / 3

Class of cells	D_0	cells	floor (%)	simulation	pair level	mean field
Workstations bordering a corridor	1	56	23.9	1.13 / 1.24 / 1.35	1.13 / 1.16 / 1.18	1.08 / 1.02 / 0.98
Other workstations	1	100	42.7	0.94 / 0.89 / 0.88	0.94 / 0.91 / 0.89	1.08 / 1.11 / 1.12
Corridor	1	39	16.7	0.97 / 1.09 / 1.16	0.92 / 0.93 / 0.93	0.62 / 0.55 / 0.52
Meeting room	1	26	11.1	1.00 / 0.88 / 0.68	1.06 / 1.12 / 1.16	1.15 / 1.29 / 1.44
Kitchen	1	13	5.6	0.94 / 0.76 / 0.58	1.00 / 0.98 / 0.96	0.92 / 0.83 / 0.75
Workstations bordering a corridor	10	56	23.9	1.15 / 1.19 / 1.20	1.13 / 1.12 / 1.12	1.08 / 1.05 / 1.04
Other workstations	10	100	42.7	1.06 / 1.08 / 1.09	1.06 / 1.07 / 1.07	1.08 / 1.08 / 1.08
Corridor	10	39	16.7	0.71 / 0.72 / 0.72	0.68 / 0.67 / 0.67	0.62 / 0.60 / 0.59
Meeting room	10	26	11.1	0.93 / 0.81 / 0.76	1.04 / 1.05 / 1.05	1.15 / 1.24 / 1.30
Kitchen	10	13	5.6	0.87 / 0.79 / 0.78	0.89 / 0.84 / 0.81	0.92 / 0.87 / 0.84

In the office at $D_0 = 1$ the mean-field map is uncorrelated with the simulated one in the first generation and anti-correlated in the next two ($r = 0.09, -0.23, -0.37$), whereas the pair-level map gives $r = 0.88, 0.82, 0.71$. The second simulator gives 0.11, -0.23, -0.37 and 0.88, 0.82, 0.73. The pair-level map is exact for the first generation of a lone index case, and the simulation is compatible with that: with only the index case transmitting, $r = 0.948$ where 0.958 is expected from sampling noise, and the goodness-of-fit statistic is between 0.85 and 1.20 per degree of freedom in the eight rows (smallest p -value 0.02, office at $D_0 = 1$). For later generations it is an approximation, and it over-predicts their size, because a new case starts inside the cluster of its infector: in generation 2 there are 2.8 cases per outbreak in the simulation, 5.2 in the pair-level description and 21.8 in the mean field. Over all scenes at $D_0 = 1$ the pair-level correlation lies between 0.71 and 0.96 and the mean-field one between -0.37 and 0.76. At $D_0 = 10$ the mean-field map is adequate in the supermarket and the classroom ($r = 0.96$ –0.98) but not in the office (0.77, 0.52, 0.41) or the metro carriage (0.77, 0.67, 0.60).

Which cells are hot is shown in Fig. S6 and Table S9(b). The mean-field map of the first generation is proportional to q : it marks the meeting room ($q = 1.5$) and the workstations ($q = 1.4$) and leaves the corridor ($q = 0.8$) cold, at 0.62 of the uniform rate. In the simulation at $D_0 = 1$ the corridor is close to average. With only the index case transmitting it receives 15.4% of the infections on 16.7% of the floor (pair level 15.3%, mean field 10.3%), and in full outbreaks its relative rate is 0.97, 1.09 and 1.16 in generations 1–3. So the corridors are not the hotspots. The hottest cells are workstation cells that border a corridor: their relative rate is 1.13, 1.24 and

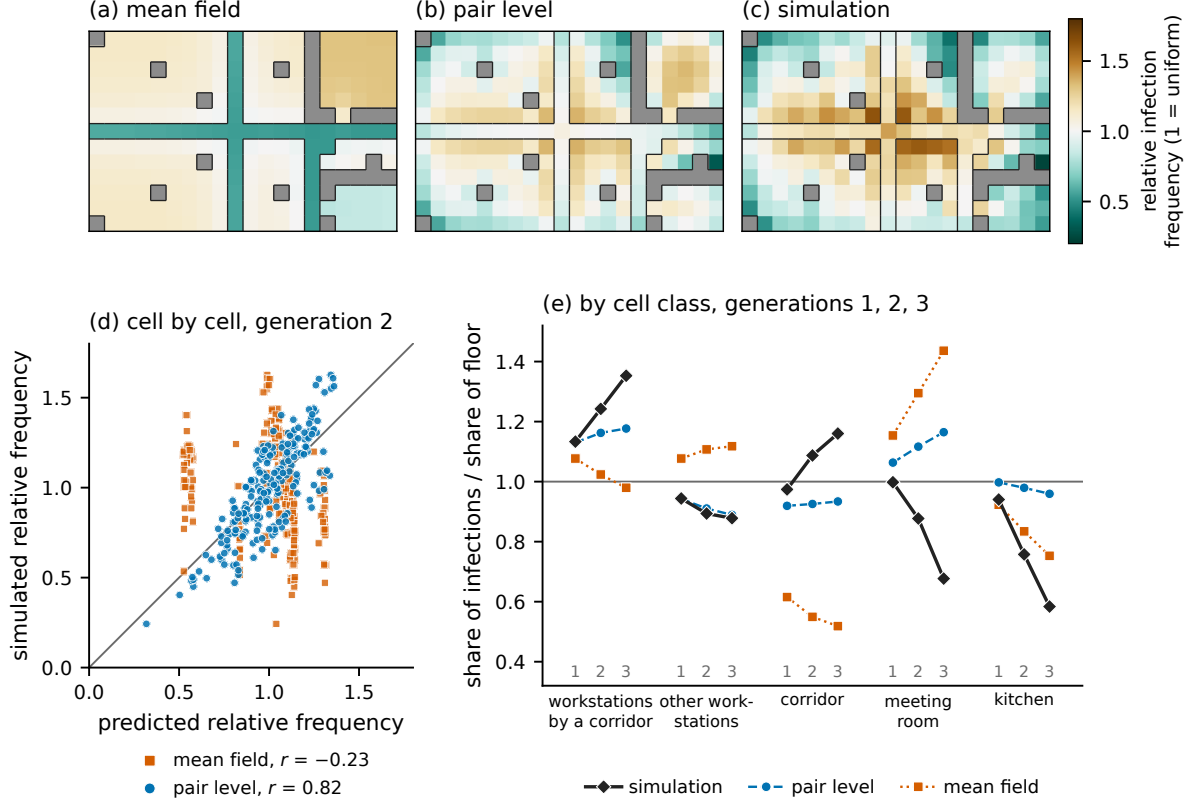


Figure S6: Where infections happen in the office scene at $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$ ($N = 100$, 234 walkable cells, 1 m kernel, equal to the same-cell kernel here; 40,000 outbreaks from a uniformly placed index case, generations 0–2 infectious). (a)–(c) Frequency with which the cases of generation 2 are infected in each cell, relative to a uniform distribution: mean-field prediction $\propto \mathbf{K}^2 \mathbf{1}$, pair-level prediction $\propto \mathbf{K}_1^2 \mathbf{1}$, and simulation. Grey cells are barriers; thin lines separate the zones of Fig. 1 (the cross is the corridor, the block at the top right the meeting room, the block at the bottom right the kitchen). (d) The same three maps cell by cell, with the correlation coefficients r of Table S9. (e) Share of infections divided by share of floor for five classes of cells and generations 1, 2, 3 (numbers in Table S9b).

1.35, against 0.94, 0.89 and 0.88 for the other workstation cells (second simulator: 1.15, 1.26, 1.36 and 0.94, 0.88, 0.87), and in each generation at least 14 of the 15 hottest cells are workstation cells, 13 to 15 of them next to a corridor. A plausible reason is that these cells combine a high q with a faster exchange of neighbours: the hop rate (1) across a workstation–corridor edge is $0.5 D_0$ against $0.3 D_0$ between two workstation cells. The meeting room, the hotspot of the mean-field analysis of Section 4, falls from the average rate to 0.68 by the third generation, where the mean field predicts 1.44. This is consistent with local depletion: only about 11 of the 100 people are in it at any time and, at this mobility, few others arrive. At $D_0 = 10$ the corridor is cold in the simulation as well (0.71–0.72; mean field 0.59–0.62) and the workstation cells are above average whether or not they border a corridor (1.15–1.20 and 1.06–1.09). A statement such as ‘infection concentrates in the meeting room and at the workstations’ is thus a mean-field statement. For discrete people at low mobility the map has to be taken from $\mathbf{K}_1$ or from simulation, and all of this concerns a room whose occupants perform a random walk without home positions.

S5.5 Ranking of the scenes

At $D_0 = 1$ the four scenes rank in the same order, metro carriage, classroom, office, supermarket, by R_0 (9.24, 5.87, 5.11, 4.39), by $\hat{R}$ (5.59, 2.62, 2.26, 1.97), by P_{maj} and by the attack rate

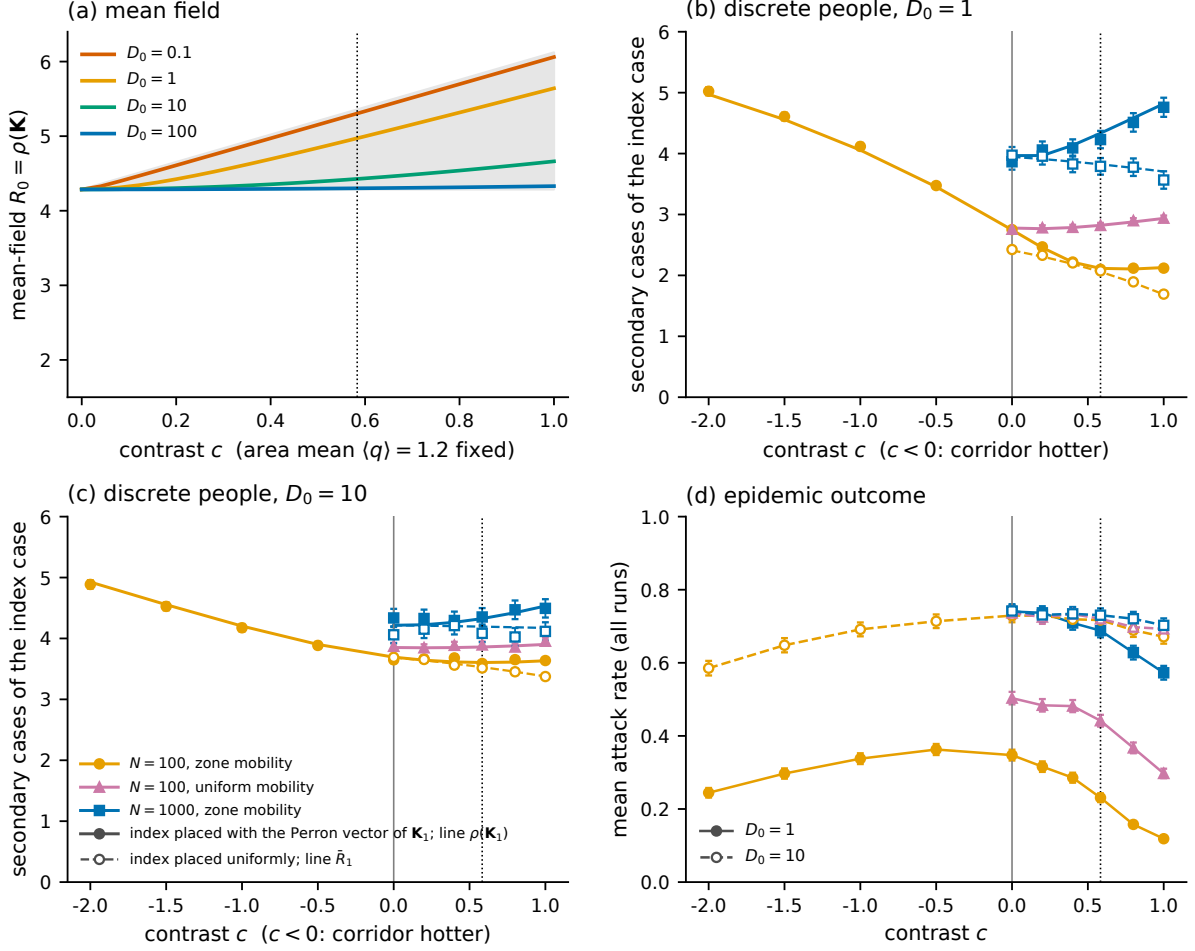


Figure S7: Two-zone room (20×13 cells; workspace 70 % of the floor with mobility $0.3D_0$, corridor 30 % with $2D_0$; same-cell kernel; FD) as a function of the contrast c of (16); the dotted vertical line marks $c = 0.583$ ($q_W = 1.5$, $q_C = 0.5$). (a) Mean-field R_0 for four mobility scales; the grey band is the interval $[\beta\langle q \rangle / \gamma, \beta q_{\max} / \gamma]$ of Theorem 3.2. (b, c) Counted secondary cases $\hat{R}$ of the index case (only the index transmits; 95 % intervals; 20,000 runs per point for $N = 100$, 4,000 for $N = 1000$) at $D_0 = 1$ and $D_0 = 10$. Filled symbols: index placed with the Perron vector of the pair-level matrix $\mathbf{K}_1$, to be compared with $\rho(\mathbf{K}_1)$ (solid lines). Open symbols: index placed uniformly, to be compared with $\bar{R}_1$ (dashed lines). ‘Uniform mobility’ is the control with $D = 0.81D_0$ in every cell (Perron-placed index only). (d) Mean attack rate over all runs with 95 % intervals (2,000 epidemics per point), $D_0 = 1$ (filled, solid) and $D_0 = 10$ (open, dashed). Numbers in Table S10.

over all runs. The order is not robust. The attack rate of major outbreaks at $D_0 = 1$ is lowest in the classroom (39.2 %), and at $D_0 = 10$ the office and the classroom cannot be told apart ($\hat{R} = 3.90 \pm 0.09$ and 3.72 ± 0.09 ; $P_{\text{maj}} = 0.762$ and 0.750). Above all, the ranking belongs to the closed-room idealisation, in which the same people share the room for the whole infectious period of $1/\gamma = 7.1$ days. Nobody spends a week in a metro carriage. Counted per visit at $D_0 = 1$, an index case causes 0.175 infections in an office day of 8 hours, 0.027 in a one-hour lecture, 0.018 in a 20-minute metro ride and 0.008 in a supermarket visit of 27 minutes (Section 6.4, Table 5), which puts the office first and the metro carriage third.

S5.6 Heterogeneity of the infection efficiency

The two-zone room of Fig. S7 has 20×13 cells without barriers, a lattice spacing of 1.5 m (so that the 1 m kernel is the same-cell kernel), FD calibration, and $N = 100$ (0.38 other people per cell) or $N = 1000$ (3.8). The contrast c of (16) is 0 in the uniform room, 0.583 for $q_W = 1.5$,

Table S10: Two-zone room of Fig. S7 (same-cell kernel, FD): mean-field R_0 , pair-level $\rho(\mathbf{K}_1)$, counted secondary cases $\hat{R}$ (95 % interval; 20,000 runs for $N = 100$, 4,000 for $N = 1000$) for an index placed with the Perron vector of $\mathbf{K}_1$ and for a uniformly placed index, probability of a major outbreak and mean attack rate over all runs ($\pm$ standard error; 2,000 epidemics per row). Mobility ‘zones’: $0.3D_0$ in the workspace and $2D_0$ in the corridor; ‘uniform’: $0.81D_0$ everywhere; $c = -2$ is the hot-corridor control. [†]From the verification (20,000 runs; standard error 0.02 to 0.03); not simulated in the main runs.

D_0	N	mobility	c	R_0	$\rho(\mathbf{K}_1)$	$\hat{R}$, Perron index	$\hat{R}$, uniform index	$P(\text{major})$	attack
1	100	zones	0.00	4.29	2.75	2.75 (2.71–2.79)	2.42 (2.39–2.46)	0.541 ± 0.011	0.347 ± 0.008
			0.58	4.97	2.12	2.10 (2.07–2.13)	2.07 (2.04–2.11)	0.460 ± 0.011	0.231 ± 0.006
			1.00	5.64	2.13	2.12 (2.09–2.15)	1.69 (1.66–1.72)	0.345 ± 0.011	0.119 ± 0.004
1	1000	zones	0.00	4.29	3.97	3.87 (3.74–3.99)	3.97 (3.84–4.11)	0.766 ± 0.009	0.740 ± 0.009
			0.58	4.97	4.33	4.23 (4.09–4.37)	3.79 (3.65–3.93)	0.727 ± 0.010	0.688 ± 0.009
			1.00	5.64	4.82	4.76 (4.60–4.92)	3.57 (3.42–3.71)	0.639 ± 0.011	0.573 ± 0.010
1	100	uniform	0.00	4.29	2.78	2.76 (2.72–2.80)	2.76 [†]	0.646 ± 0.011	0.503 ± 0.009
			0.58	4.76	2.82	2.82 (2.78–2.86)	2.61 [†]	0.620 ± 0.011	0.442 ± 0.008
1	100	zones	−2.00	9.14	4.97	5.02 (4.96–5.09)	2.50 [†]	0.381 ± 0.011	0.244 ± 0.007
10	100	zones	0.00	4.29	3.69	3.65 (3.59–3.70)	3.69 (3.64–3.75)	0.748 ± 0.010	0.729 ± 0.009
			0.58	4.43	3.60	3.59 (3.54–3.64)	3.52 (3.46–3.57)	0.744 ± 0.010	0.717 ± 0.009
			1.00	4.66	3.63	3.64 (3.58–3.69)	3.37 (3.32–3.43)	0.704 ± 0.010	0.671 ± 0.010
10	1000	zones	0.00	4.29	4.22	4.34 (4.19–4.49)	4.06 (3.92–4.20)	0.753 ± 0.010	0.742 ± 0.009
			0.58	4.43	4.33	4.35 (4.21–4.50)	4.08 (3.94–4.23)	0.745 ± 0.010	0.730 ± 0.010
10	100	uniform	0.58	4.36	3.86	3.89 (3.84–3.95)	–	0.737 ± 0.010	0.720 ± 0.009
10	100	zones	−2.00	5.70	4.93	4.89 (4.82–4.96)	–	0.614 ± 0.011	0.585 ± 0.010

$q_C = 0.5$, and negative when the higher efficiency is in the corridor. Table S10 gives the numbers. The mean-field R_0 is 4.972 at $c = 0.583$ and $D_0 = 1$, and 4.427 (+3.3 %) at $D_0 = 10$.

Discrete individuals, $D_0 = 1$, $N = 100$. Everything that is counted *falls* when the contrast is switched on (Fig. S7b,d; numbers in Table S10): $\hat{R}$ of an index case placed with the Perron vector of $\mathbf{K}_1$ goes from 2.75 to 2.10 (−24 %), $\hat{R}$ of a uniformly placed index from 2.42 to 2.07 (−15 %), the ratio of second- to first-generation cases from 2.19 to 1.58 (−28 %), the probability of a major outbreak from 0.541 to 0.460 and the mean attack rate from 0.347 to 0.231 (−33 %). (The second simulator gives $2.77 \pm 0.02 \rightarrow 2.11 \pm 0.02$ and $0.341 \rightarrow 0.239$ for the first and the last of these.) The cause is not the heterogeneity of q as such but its coincidence with low mobility: this layout puts the high efficiency where people hardly move, where an infective keeps meeting the same neighbours and its contacts fall on people who are already infected (Section 5). Two controls confirm this. With the same contrast and *uniform* mobility ($0.81 D_0$ everywhere, the area mean) $\hat{R}$ rises slightly, $2.76 \rightarrow 2.82$; with the high efficiency in the *mobile* corridor ($c = -2$) $\hat{R}$ of a Perron-placed index reaches 5.02. In both controls the mean attack rate nevertheless falls ($0.503 \rightarrow 0.442$ and $0.347 \rightarrow 0.244$): a low- q zone is a place where an index case tends to fizzle and which the epidemic reaches late.

High occupancy. With $N = 1000$ the pair-level $\rho(\mathbf{K}_1)$ rises with the contrast (3.97, 4.33, 4.82 at $c = 0, 0.58, 1$), and so does $\hat{R}$ of a Perron-placed index (3.87, 4.23, 4.76; re-check 3.92, 4.33, 4.78, each ± 0.08 to 0.09). This is the only sense in which the simulation shows the rise of R_0 . For a uniformly placed index the count falls in these runs (3.97, 3.79, 3.57; re-check 3.96, 3.86, 3.82). It is bounded above by $\beta\langle q \rangle / \gamma = 4.29$ for every contrast (Proposition 5.3), so it cannot follow the rise of R_0 ; but it is not monotone in the contrast in general: with the high efficiency in the mobile corridor ($c < 0$, $N = 100$) $\hat{R}_1$ rises from 2.41 at $c = 0$ to 2.66 at $c = -1$ before falling to 2.44 at $c = -2$ ($D_0 = 1$), and from 3.69 to 3.81 at $D_0 = 10$. The outbreak probability and the mean attack rate fall as well: in 2,000 epidemics per contrast they are 0.766 ± 0.009 , 0.727 ± 0.010 , 0.639 ± 0.011 and 0.740 ± 0.009 , 0.688 ± 0.009 , 0.573 ± 0.010 (standard errors). The first sample

of 300 epidemics per contrast (0.737, 0.690, 0.630 and 0.712, 0.654, 0.564, each ± 0.025 to 0.028) resolved only the difference between $c = 0$ and $c = 1$. R_0 is the growth factor per generation once cases are distributed as the incidence map $\mathbf{w}$, not the fate of a typical introduction.

$D_0 = 10$. For $N = 100$ the contrast $c = 0.583$ changes $\hat{R}$ of a Perron-placed index from 3.65 to 3.59, the outbreak probability from 0.748 to 0.744 and the mean attack rate from 0.729 to 0.717 (all less than 2 %, and within one standard error of 0.010 for the last two); $\hat{R}$ of a uniformly placed index falls by 5 % (3.69 $\rightarrow$ 3.52) (Fig. S7c,d). For $N = 1000$ the outbreak probability is 0.753 ± 0.010 and 0.745 ± 0.010 at $c = 0$ and 0.58, and the mean attack rate 0.742 ± 0.009 and 0.730 ± 0.010 : no change is resolved.

S5.7 Barriers with identical starts

To isolate the effect of barriers we use a uniform room (20×13 cells, $N = 100$, $q = 1.2$, so $\beta q/\gamma = 4.29$, uniform mobility D , same-cell kernel) and close randomly chosen cells at a nominal density ρ_B . The barrier sets of one layout are nested; walkable pockets cut off by barriers are closed as well, so that the realised density at nominal 0.30 is 0.35. In all arms of a layout the same people start on the same cells (drawn from the cells that are walkable in the densest arm), the index case is the same person on the centre cell, and each replicate uses the same random seed: the arms differ only by the barriers. Fig. S8 shows a scan over seven densities at $D = 1$; Table S11 gives the effects for 40 further layouts at $D = 1$ and $D = 10$, with standard errors across layouts, and for the re-check's 40 layouts.

- (i) *Mean field*. Under FD, $R_0 = \beta q/\gamma = 4.29$ at every barrier density (Theorem 4.1). Under DD it rises as the floor shrinks, by (15): 4.76 at 10 % and 6.7 at nominal 30 %.
- (ii) *Pair level*. Under FD the pair-level R_1 of the centre index falls a little, $3.02 \rightarrow 2.94 \rightarrow 2.80$ at $\rho_B = 0, 0.10, 0.30$ for $D = 1$ and $3.90 \rightarrow 3.87 \rightarrow 3.75$ for $D = 10$: barriers lengthen the time two people remain neighbours.
- (iii) *Outcomes at $D = 1$, FD*. The mean attack rate falls by 0.040 ± 0.003 at 10 % barriers and by 0.223 ± 0.011 at 30 % (re-check: 0.047 ± 0.004 and 0.235 ± 0.011), the peak prevalence of major outbreaks from 27 % to 25 % and 18 %, while the outbreak probability falls much less ($0.70 \rightarrow 0.69 \rightarrow 0.62$). Barriers act by stopping outbreaks part-way on a fragmented floor, not by preventing take-off.
- (iv) *Peak time at $D = 1$, FD*. The peak of major outbreaks is delayed by 0.81 ± 0.12 day at 10 % barriers (re-check: 0.82 ± 0.10) and by 0.93 ± 0.23 day at 30 % (re-check: 0.71 ± 0.30): a real delay of about 0.8 day. The scan of Fig. S8c cannot resolve it (its across-layout intervals are ± 1.1 day or wider).
- (v) *$D = 10$, FD*. Barriers at 10 % leave the attack rate unchanged ($+0.005 \pm 0.005$) and at 30 % lower it by 0.025 ± 0.005 ; the peak is lower by 0.8 and 5.3 percentage points and later by 0.34 ± 0.05 and 2.05 ± 0.16 days.
- (vi) *DD*. Barriers also crowd people together. At $D = 1$ the two effects cancel at 10 % (-0.001 ± 0.004) and fragmentation wins at 30 % (-0.085 ± 0.015); at $D = 10$ crowding wins and the attack rate *rises*, by 0.026 ± 0.005 and 0.077 ± 0.005 , with the peak 1.0 and 2.8 days earlier.

Two caveats. Starting everyone on cells that are walkable in the densest arm is not the stationary law of the sparser arms; the re-check measured mean attack rates of 0.600 and 0.589 at zero barriers for the two ways of starting, a difference within noise. And although the densities stay below the blocked-site percolation threshold of the square lattice, 0.407 [62], about 5 % of this small floor is already cut off as pockets at nominal 0.30.

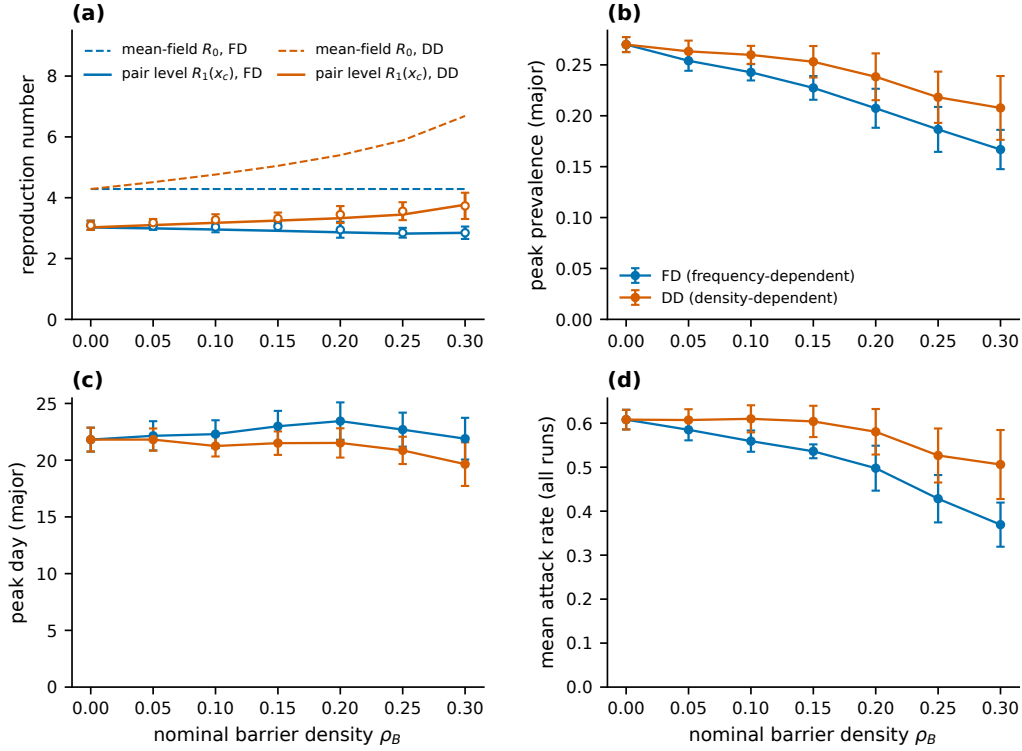


Figure S8: Barrier scan with identical starts: uniform room (20×13 cells, $N = 100$, $q = 1.2$, $D = 1 \text{ cell}^2 \text{ day}^{-1}$, same-cell kernel), random nested barriers at nominal density ρ_B ; 8 layouts with 500 epidemics per layout and density; bars are 95 % intervals across layouts. Blue: FD; orange: DD (pair hazard frozen at its barrier-free value). (a) Mean-field R_0 (dashed), pair-level $R_1(x_c)$ of the centre index (solid) and counted $\hat{R}$ of the index (open circles, 500 runs per layout with only the index transmitting). (b) Peak prevalence and (c) peak day of major outbreaks. (d) Mean attack rate over all runs. Effects with layout-level standard errors, including the peak delay that panel (c) cannot resolve, are in Table S11.

Table S11: Barriers with identical starts in the uniform room of Fig. S8: means over 40 nested layouts (1,000 epidemics per layout and arm at $D = 1$, 500 at $D = 10$). ρ_B : nominal barrier density; $R_1(x_c)$: pair-level number of the centre index; P : probability of a major outbreak; attack: mean attack rate over all runs; peak and day: peak prevalence (%) and peak day of major outbreaks. The Δ columns are differences from the barrier-free arm of the same layout, mean $\pm$ standard error across layouts (Δ_{peak} in percentage points, Δ_{day} in days). † Re-check: 40 other layouts, 1,000 epidemics each, second simulator.

D	law	ρ_B	R_0	$R_1(x_c)$	P	attack	peak	day	Δ_{attack}	Δ_{peak}	Δ_{day}
1	FD	0.00	4.29	3.02	0.698	0.609	27.2	21.5	–	–	–
1	FD	0.10	4.29	2.94	0.688	0.569	24.7	22.3	-0.040 ± 0.003	-2.5 ± 0.1	$+0.81 \pm 0.12$
1	FD	0.30	4.29	2.80	0.625	0.386	17.6	22.4	-0.223 ± 0.011	-9.6 ± 0.4	$+0.93 \pm 0.23$
1	DD	0.10	4.76	3.16	0.713	0.608	26.4	21.2	-0.001 ± 0.004	-0.8 ± 0.1	-0.33 ± 0.12
1	DD	0.30	6.68	3.69	0.727	0.524	22.0	20.3	-0.085 ± 0.015	-5.2 ± 0.5	-1.22 ± 0.25
1	FD †	0.10	4.29	–	0.683	0.564	24.6	22.3	-0.047 ± 0.004	-2.4 ± 0.1	$+0.82 \pm 0.10$
1	FD †	0.30	4.29	–	0.627	0.376	17.0	22.2	-0.235 ± 0.011	-10.0 ± 0.3	$+0.71 \pm 0.30$
10	FD	0.00	4.29	3.90	0.757	0.744	44.0	14.3	–	–	–
10	FD	0.10	4.29	3.87	0.764	0.749	43.2	14.7	$+0.005 \pm 0.005$	-0.8 ± 0.1	$+0.34 \pm 0.05$
10	FD	0.30	4.29	3.75	0.743	0.719	38.7	16.4	-0.025 ± 0.005	-5.3 ± 0.3	$+2.05 \pm 0.16$
10	DD	0.10	4.76	4.25	0.779	0.770	46.2	13.3	$+0.026 \pm 0.005$	$+2.2 \pm 0.1$	-1.05 ± 0.05
10	DD	0.30	6.68	5.47	0.826	0.821	48.8	11.5	$+0.077 \pm 0.005$	$+4.8 \pm 0.6$	-2.78 ± 0.21

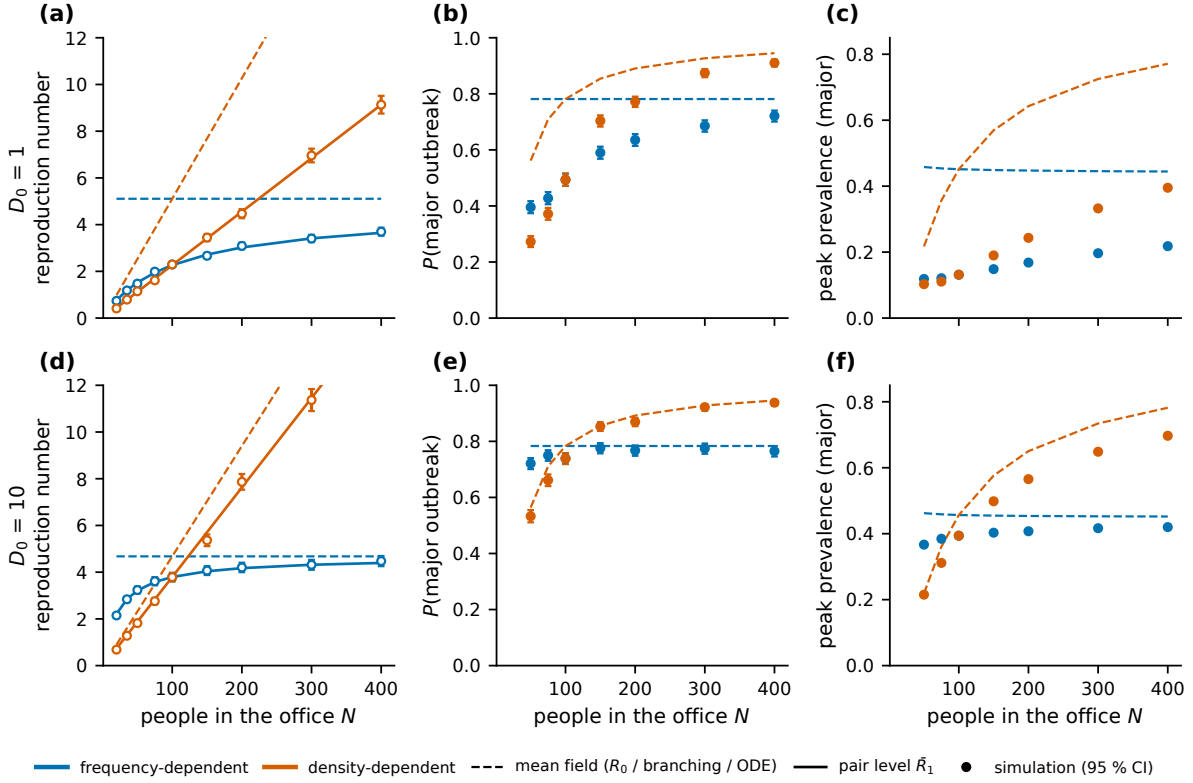


Figure S9: Occupancy dependence in the office scene (1 m kernel, 2,000 epidemics per point), $D_0 = 1$ (top) and $D_0 = 10$ (bottom). Blue: FD; orange: DD with the pair hazard frozen at its value for $N = 100$. (a, d) Mean-field R_0 (dashed), pair-level $\bar{R}_1$ (solid) and counted $\hat{R}$ (open circles, 95 % intervals). (b, e) Probability of a major outbreak: simulation (dots, 95 % intervals) and mean-field branching value (dashed); for $N < 50$ a ‘major’ outbreak means two or three cases. (c, f) Peak prevalence of major outbreaks: simulation (dots) and lattice mean-field equations (3) (dashed).

S5.8 Occupancy

In the office scene we vary the number of people from $N = 20$ to 400 (0.03 to 0.68 persons per m^2 ; 2,000 epidemics per point; Fig. S9). Under FD the mean-field R_0 does not depend on N at all: it is 5.11 ($D_0 = 1$) and 4.68 ($D_0 = 10$) for every occupancy. Discrete people do feel occupancy, through saturation: with fewer people an infective exhausts its neighbours sooner. At $D_0 = 1$ the counted $\hat{R}$ is 1.47 ± 0.04 for $N = 50$, 2.29 ± 0.05 for 100 and 3.70 ± 0.09 for 400, following $\bar{R}_1 = 1.52, 2.27, 3.65$; the attack rate of major outbreaks is 27 %, 46 % and 90 %. At $D_0 = 10$ the dependence is weaker ($\hat{R} = 3.2, 3.8, 4.5$). The elasticity of $\bar{R}_1$ with respect to N near $N = 100$ (between 75 and 150) is +0.48 at $D_0 = 1$ and +0.17 at $D_0 = 10$, against 0 for R_0 . Under DD (reference $N = 100$), R_0 is proportional to $N - 1$ by (15), from 0.98 at $N = 20$ to 20.6 at $N = 400$ for $D_0 = 1$, and the elasticity is +1.0 at both mobilities. Halving the office from 100 to 50 people then takes $\bar{R}_1$ from 2.27 to 1.12 and the outbreak probability from 0.49 to 0.27 at $D_0 = 1$ ($3.79 \rightarrow 1.88$ and $0.74 \rightarrow 0.53$ at $D_0 = 10$), whereas under FD it takes them to 1.52 and 0.40 (3.22 and 0.72).

Whether a capacity limit works is therefore an *assumption about contact behaviour* — do people keep the same number of contacts in an emptier room? — and not an output of the model.

Table S12: Interventions in the office scene (1 m kernel; 2,000 epidemics per arm; reference occupancy for DD: $N = 100$ in the unmodified scene). $\hat{R}$: counted secondary cases of the index ($\pm$ standard error); P : probability of a major outbreak; attack: mean attack rate over all runs; peak and day: peak prevalence (%) and peak day of major outbreaks. Ventilation and masks are assumptions on q and β , not model outputs. Partitions: 39 workstation cells turned into barriers. Combination: ventilation, half occupancy and partitions. A value $\bar{R}_1 < 1$ is not a control criterion (see text).

D_0	measure	law	N	R_0	$\bar{R}_1$	$\hat{R}$	P	attack	peak	day
1	baseline	FD	100	5.11	2.27	2.30 ± 0.05	0.491	0.232	12.8	24.3
1	ventilation, $q \rightarrow q/2$	FD	100	2.55	1.51	1.55 ± 0.04	0.295	0.102	9.0	23.3
1	masks, $\beta \rightarrow 0.3\beta$	FD	100	1.53	1.05	1.05 ± 0.03	0.147	0.048	7.1	21.8
1	capacity 50 %	FD	50	5.11	1.52	1.51 ± 0.04	0.393	0.126	11.7	13.0
1	capacity 50 %	DD	50	2.53	1.12	1.13 ± 0.03	0.258	0.080	10.2	13.9
1	partitions	FD	100	5.11	2.21	2.20 ± 0.05	0.459	0.166	10.7	20.3
1	partitions	DD	100	6.14	2.42	2.36 ± 0.06	0.506	0.195	11.4	19.9
1	combination	FD	50	2.56	1.10	1.09 ± 0.03	0.274	0.077	9.9	12.2
1	combination	DD	50	1.52	0.82	0.78 ± 0.02	0.165	0.055	9.0	12.3
10	baseline	FD	100	4.68	3.79	3.76 ± 0.09	0.760	0.737	39.0	16.7
10	ventilation, $q \rightarrow q/2$	FD	100	2.34	2.09	2.09 ± 0.06	0.522	0.404	21.2	27.6
10	masks, $\beta \rightarrow 0.3\beta$	FD	100	1.40	1.30	1.30 ± 0.04	0.290	0.131	10.9	27.8
10	capacity 50 %	FD	50	4.68	3.22	3.20 ± 0.07	0.697	0.629	36.2	15.8
10	capacity 50 %	DD	50	2.31	1.88	1.85 ± 0.05	0.533	0.345	21.6	19.3
10	partitions	FD	100	4.63	3.60	3.60 ± 0.09	0.730	0.693	34.2	19.6
10	partitions	DD	100	5.55	4.14	4.13 ± 0.10	0.777	0.751	38.0	17.0
10	combination	FD	50	2.31	1.79	1.74 ± 0.05	0.504	0.285	18.8	19.5
10	combination	DD	50	1.37	1.16	1.19 ± 0.04	0.329	0.123	12.6	16.7

S5.9 Interventions

Table S12 applies five measures to the office scene. Two of them are *inputs*, not results: ‘ventilation’ halves q in every cell and ‘masks’ multiply β by 0.3, by assumption.

Uniform measures scale R_0 exactly ($5.11 \rightarrow 2.55$ and 1.53 at $D_0 = 1$), but the pair-level number less than proportionally ($\bar{R}_1 = 2.27 \rightarrow 1.51$ and 1.05), because a lower hazard also wastes fewer contacts; the outbreak probability falls from 0.49 to 0.30 and 0.15. *Halving the occupancy* leaves R_0 unchanged under FD and halves it under DD (Section S5.8); in either case the peak comes earlier, on day 13 instead of day 24 at $D_0 = 1$, because a smaller group is exhausted sooner. *Partitions* (39 workstation cells turned into barriers) leave R_0 at 5.11 under FD and raise it to 6.14 under DD at $D_0 = 1$; the mean attack rate falls from 0.232 to 0.166 (FD) or 0.195 (DD). At $D_0 = 10$ it goes from 0.737 to 0.693 (FD) or 0.751 (DD); the decrease of R_0 under FD there ($4.68 \rightarrow 4.63$, -1.1%) arises because the cells removed have an above-average q (Section 4.4).

Combination (ventilation, half occupancy and partitions together). At $D_0 = 1$, $R_0 = 2.56$ (FD) or 1.52 (DD), $\bar{R}_1 = 1.10$ or 0.82 and $\hat{R} = 1.09 \pm 0.03$ or 0.78 ± 0.02 ; at $D_0 = 10$, $\bar{R}_1 = 1.79$ or 1.16 . In the classroom the combination gives $\bar{R}_1 = 1.21$ (FD) or 0.86 (DD) at $D_0 = 1$ and 1.70 or 1.10 at $D_0 = 10$. A pair-level number below one is reached only for discrete people at low mobility under the DD law, and it must not be read as a control criterion: in that arm R_0 is 1.52 and 16.5 % of the runs still reach an attack rate of 10 % (five of 50 people), and conversely a pair-level number above one does not imply outbreaks (Remark 5.7).

Targeted ventilation. Halving q on the quarter of the cells with the largest elasticity e_x (Theorem 4.4) lowers R_0 more than treating the quarter with the smallest (4.67 against 5.11 at $D_0 = 1$; 4.10 against 4.37 at $D_0 = 10$). For discrete people at $D_0 = 1$ it makes no measurable difference which quarter is treated: $\bar{R}_1 = 2.08$, 2.09 and 2.06 and mean attack rates 0.195 , 0.205 and 0.188 (standard errors about 0.005) for the hottest quarter, the coldest quarter and the hottest quarter by the pair-level elasticity; at $D_0 = 10$ the attack rates are 0.675 , 0.696 and 0.682 .

S5.10 Schedules and dwell time

Proposition S5.1 (Duty cycle). *Let $0 < f \leq 1$ and let Assumption A hold. Suppose that movement and transmission operate only during the occupied intervals $[k, k + f)$, $k = 0, 1, 2, \dots$ (time in days), and that recovery proceeds at all times: in (3), $\mathbf{M}$ and $\mathbf{F}$ are multiplied by $\chi(t)$, equal to 1 on the occupied intervals and to 0 otherwise. Put $\theta(t) = \int_0^t \chi(s) ds$ and*

$$\mathbf{J}_f = \mathbf{M} + \mathbf{F} - \frac{\gamma}{f} \mathbf{1}, \quad R_0^{(f)} = \rho(\mathbf{F}(\frac{\gamma}{f} \mathbf{1} - \mathbf{M})^{-1}). \quad (\text{S6})$$

(a) *The linearised model $\dot{I} = [\chi(t)(\mathbf{M} + \mathbf{F}) - \gamma \mathbf{1}] I$ has the solution $I(t) = \exp[\theta(t)(\mathbf{M} + \mathbf{F}) - \gamma t \mathbf{1}] I(0)$; in particular $I(k) = e^{kf \mathbf{J}_f} I(0)$ for integer k . Solutions of the nonlinear model satisfy $I(t) \leq \exp[\theta(t)(\mathbf{M} + \mathbf{F}) - \gamma t \mathbf{1}] I(0)$ componentwise.*

(b) *The growth factor per day is $e^{fs(\mathbf{J}_f)}$, and $s(\mathbf{J}_f)$ has the sign of $R_0^{(f)} - 1$.*

(c) *If $\mathbf{M} = D_0 \mathbf{M}_0$ and $R_0(D_0)$ denotes the value for continuous occupancy, then $R_0^{(f)}(D_0) = f R_0(f D_0)$. For the same-cell kernel, $f R_0(D_0) \leq R_0^{(f)}(D_0) \leq f \beta q_{\max} / \gamma$.*

Proof. (a) The matrices $A(t) = \chi(t)(\mathbf{M} + \mathbf{F}) - \gamma \mathbf{1}$ are constant on each occupied and each empty interval and commute with one another for all times, so the product of the exponentials over successive intervals is the exponential of $\int_0^t A(s) ds = \theta(t)(\mathbf{M} + \mathbf{F}) - \gamma t \mathbf{1}$; at $t = k$, $\theta = kf$. In the nonlinear model $S_y \leq N_y^*$ gives $\dot{I} \leq A(t)I$ componentwise, and since $A(t)$ is Metzler the comparison argument of Theorem 3.1(d) applies on each interval. (b) $\mathbf{M} + \mathbf{F}$ is Metzler and irreducible, so $e^{fs(\mathbf{J}_f)}$ is a positive matrix with spectral radius $e^{fs(\mathbf{J}_f)}$ (Lemma S2.1, P1); the sign statement is Theorem 3.1(c) with γ replaced by γ/f . (c) $\mathbf{F}(\frac{\gamma}{f} \mathbf{1} - D_0 \mathbf{M}_0)^{-1} = f \mathbf{F}(\gamma \mathbf{1} - f D_0 \mathbf{M}_0)^{-1}$. The inequalities follow from Theorem 3.3(c), because $f D_0 \leq D_0$, and from Theorem 3.2. $\square$

Remark S5.2 (What is exact and what is approximate). $R_0^{(f)}$ is an exact threshold parameter of the periodic mean-field model, for any cycle length, because one and the same operator is switched on and off; in general periodic models the threshold is given by a next-generation operator on periodic functions and not by time-averaged parameters [4, 73]. By part (c) a duty cycle does more than scale R_0 by f : it also lowers the effective mobility to $f D_0$. As a *number of secondary cases*, however, the substitution $\gamma \rightarrow \gamma/f$ is an approximation. It replaces the expected infectious time spent in the room by f/γ (2.381 days for $f = 1/3$), whereas a case that becomes infectious at the start of an occupied interval spends $(1 - e^{-\gamma f})/[\gamma(1 - e^{-\gamma})] = 2.493$ days there (+4.7%), and one that becomes infectious at a uniformly distributed time within an occupied interval, as secondary cases approximately do (their infection times within an interval follow the incidence, which changes during the interval), 2.383 days (+0.07%). For discrete individuals the same substitution in the pair-level formulas of Section 5 is likewise an approximation. The exact pair-level value follows from the argument of Proposition 5.1. Let $A = \gamma \mathbf{1} + \mathbf{H} - \mathcal{M}^\top$ be the pair operator of that proposition and $u_0(z)$ the probability that the given person escapes infection when the index case becomes infectious at the start of an occupied interval with the pair in state z . During the occupied interval of length f the pair moves and the hazard acts, so $(e^{-At} \mathbf{1})(z)$ is the probability that neither recovery nor infection has occurred by time $t \leq f$; during the rest of the day nobody moves, the index recovers with probability $c_f = 1 - e^{-\gamma(1-f)}$, and otherwise the next day starts from the same state. Hence

$$u_0 = \gamma A^{-1}(\mathbf{1} - e^{-Af} \mathbf{1}) + e^{-Af}(c_f \mathbf{1} + (1 - c_f) u_0), \quad (\text{S7})$$

a linear fixed-point equation whose map is a contraction in the maximum norm (factor at most $e^{-\gamma} < 1$, because e^{-Af} is non-negative with row sums at most $e^{-\gamma f}$), solved by iteration; the expected number of secondary cases is $(N - 1)$ times the mean of $1 - u_0$ over the n^2 pair states. For an index case that becomes infectious at a time uniformly distributed over the occupied interval, the first day is shortened to its remaining part r and the same expression is averaged over $r \in (0, f)$. With $f = 1$ the iteration reproduces the solution of (12) to 5×10^{-10} .

Office, eight hours a day ($f = 1/3$). The mean-field number drops from 5.11 to $R_0^{(f)} = 1.75$ at $D_0 = 1$ and from 4.68 to 1.60 at $D_0 = 10$; the pair-level number with $\gamma \rightarrow \gamma/f$ is $\bar{R}_1 = 0.889$ and 1.335 (Table 5a). The exact pair-level values of (S7) are 0.930 at $D_0 = 1$ and 1.397 at $D_0 = 10$ for an index case that becomes infectious at the start of a working day, 4.6 % and 4.7 % above the rule, as its longer time in the room leads one to expect, and 0.889 and 1.336 for one that becomes infectious at a uniformly distributed time of the working day, 0.07 % above the rule. Simulation with the schedule imposed explicitly agrees: 0.9309 ± 0.0009 and 0.8889 ± 0.0009 at $D_0 = 1$ (2,000,000 runs each) and 1.400 ± 0.002 and 1.337 ± 0.002 at $D_0 = 10$ (800,000 runs each), all within 1.6 standard errors of the exact values. In full epidemics with the schedule, 6.1 % of the runs (95 % interval 5.1–7.2 %) reach a 10 % attack rate at $D_0 = 1$ and 30.2 % (28.2–32.2 %) at $D_0 = 10$. The index case of an office occupied eight hours a day thus infects about 0.9 people on average at $D_0 = 1$ and 1.3 to 1.4 at $D_0 = 10$, against 2.3 and 3.8 for a room occupied around the clock; this is not a threshold statement (Remark 5.7). For a class that meets one hour a day ($f = 1/24$) the numbers are $R_0^{(f)} = 0.27$, $\bar{R}_1 = 0.18$ with the rule and 0.194 exactly (index infectious from the start of the lecture; counted 0.189 ± 0.003) at $D_0 = 1$, and no run in 2,000 reaches a 10 % attack rate: with $\beta = 0.5 \text{ day}^{-1}$ one daily lecture cannot sustain transmission in this model.

Further simulation estimates of the duty-cycle counts. Before the exact values of (S7) were computed, the number of secondary cases of an index case in the office occupied eight hours a day was estimated by simulation several times. For an index case infectious from the start of a working day at $D_0 = 1$: 0.918 ± 0.009 (one batch of 20,000 runs), 0.944 ± 0.006 (the second simulator), 0.928 ± 0.003 (pooled, in the re-check’s report) and 0.935 ± 0.003 (200,000 runs), against the exact 0.930; at $D_0 = 10$, 1.408 ± 0.012 and 1.395 ± 0.004 against 1.397. For an index case infectious from a uniformly distributed time: 0.896 ± 0.003 and 1.331 ± 0.004 (200,000 runs each) against 0.889 and 1.336. The 200,000-run samples at $D_0 = 1$ lie 2.0 and 2.5 standard errors above the exact values; the samples of 2,000,000 runs quoted in Section S5.10 do not (+1.5 and −0.3).

One visit. Table 5b gives the expected number of people that one infective infects during a single stay of length T . For a uniformly placed infective the mean-field value is $\beta \langle q \rangle (1 - e^{-\gamma T}) / \gamma$ whatever the mobility and the kernel, because the walk is stationary; the counted exposure agrees with it within 0.3 %, and the counted infections are lower only where pairs saturate within the visit. The values are 0.21 for an office day (0.175 counted at $D_0 = 1$, 0.198 at $D_0 = 10$), 0.027 for a lecture, 0.018 for a 20-minute metro ride and 0.0086 for a 27-minute supermarket visit. A per-visit number is not the reproduction number of a setting that is visited repeatedly: an infective who rides the metro twice a day for the whole infectious period infects $f \beta \langle q \rangle / \gamma = 0.25$ fellow passengers in expectation, and one who shops once a day 0.06 fellow customers. Weighting each setting by the fraction of time spent in it is the residence-time (Lagrangian) formulation of multi-setting epidemic models [7]; here it is applied to one setting at a time. These numbers, and not the closed-room values, are what the settings contribute, and they undercut the ranking of Section 6: as closed rooms the scenes rank metro, classroom, office, supermarket ($R_0 = 9.2, 5.9, 5.1, 4.4$ at $D_0 = 1$); with the dwell times above the mean-field numbers are 1.75 for the office and 0.27 for the lecture ($R_0^{(f)}$, closed groups), 0.25 for the metro carriage and 0.06 for the supermarket ($f \beta \langle q \rangle / \gamma$).

A daily schedule of the parameters. In the metro scene we let the mobility field and a transmission multiplier follow a daily signal with two rush hours (24 hourly segments; the multiplier is 1 in the rush hours and 0.8 otherwise, mean 0.833), and compare three arms: constant rush-hour values (the scene of Section 6), the hourly schedule, and constants equal to

Table S13: Daily schedule in the metro scene ($N = 310$ in every arm, 1 m kernel, FD, 2,000 epidemics per arm). Arms: constant rush-hour values (‘peak’), hourly schedule of mobility and transmission multiplier (‘schedule’), constants equal to the time averages (‘average’). R_0 is not defined by a single matrix for the schedule arm. 95 % intervals in parentheses.

D_0	arm	R_0	$P(\text{major})$	attack, major (%)	peak prevalence, major (%)	peak day, major
1	peak	9.24	0.871 (0.855–0.885)	93.6	25.0 (24.7–25.2)	22.2 (21.6–22.7)
1	schedule	–	0.837 (0.820–0.853)	93.4	23.0 (22.7–23.2)	25.3 (24.6–26.0)
1	average	7.66	0.844 (0.827–0.859)	93.7	22.9 (22.7–23.1)	25.2 (24.5–25.9)
10	peak	9.16	0.887 (0.873–0.901)	97.9	27.8 (27.5–28.1)	21.0 (20.5–21.5)
10	schedule	–	0.869 (0.854–0.883)	99.3	28.6 (28.3–28.9)	21.0 (20.5–21.4)
10	average	7.58	0.862 (0.846–0.876)	99.3	28.6 (28.4–28.9)	20.9 (20.5–21.4)
100	peak	9.08	0.880 (0.866–0.894)	99.9	35.4 (35.1–35.8)	16.5 (16.2–16.8)
100	schedule	–	0.853 (0.836–0.867)	99.8	44.2 (43.9–44.6)	13.9 (13.7–14.1)
100	average	7.55	0.855 (0.839–0.870)	99.8	44.4 (44.0–44.7)	13.9 (13.8–14.1)

the time averages of the mobility field and of the multiplier. The number of people is $N = 310$ in all arms: only mobility and the multiplier vary, not the occupancy.

Numerical observation S5.3 (Time average). In the metro scene the hourly schedule and its time average give the same outbreak probability, attack rate, peak prevalence and peak day within Monte Carlo error at $D_0 = 1, 10$ and 100 (Table S13); for example the peak prevalence of major outbreaks is 23.0 % against 22.9 % on day 25.3 against 25.2 at $D_0 = 1$.

A test in the re-check, run with the production simulator, in the office scene (half of each day with hop rates multiplied by 1.8 and transmission by 1.3, the other half by 0.2 and 0.7, against constants; 6,000 epidemics each) agrees: outbreak probability 0.491 against 0.498, peak prevalence 12.8 % against 12.8 %, peak day 23.7 against 23.3 (standard error 0.3). This is not a theorem: the hourly generators do not commute and the hop rates are not small compared with one cycle per day. Relative to constant rush-hour values the schedule lowers the multiplier but raises the mobility; the first effect wins at $D_0 = 1$ (peak 25.0 % $\rightarrow$ 23.0 %, three days later), the second at $D_0 = 100$ (35.4 % $\rightarrow$ 44.2 %, 2.6 days earlier), and at $D_0 = 10$ they nearly cancel.

S5.11 What remains at higher mobility

The experiments above were run at $D_0 = 1$ and 10 . The pair-level number $\bar{R}_1$ is exact for the first generation at any mobility (Definition 5.2) and costs one linear solve, so each effect can be followed up to $D_0 = 2.4 \times 10^3 \text{ cell}^2 \text{ day}^{-1}$, the estimate of Section 2.6 for people who walk 5 % of the time (Table S14). The effects of the contrast in q and of barriers under FD shrink below 1 % at $D_0 = 100$ and below 0.05 % at $D_0 = 2.4 \times 10^3$. Simulation at $D_0 = 100$ agrees: the contrast $c = 0.583$ changes the mean attack rate from 0.744 to 0.750 (2,000 epidemics each, standard error 0.009), and barriers at 10 % and 30 % change it by $+0.007 \pm 0.005$ and $+0.013 \pm 0.007$ (20 layouts with 250 epidemics each, FD). Three things survive. Uniform reductions of q or β act on $\bar{R}_1$ almost in proportion, as they do on R_0 . Occupancy matters in full under DD; under FD a residue of about 5 % remains for halving the office, which is a finite-population effect (the limit $D \rightarrow \infty$ in Proposition 5.4) and not a spatial one. And the time spent in the room enters through Proposition S5.1 at any mobility. Epidemics at $D_0 = 2.4 \times 10^3$ were not simulated.

S5.12 Computational cost

Table S15 compares, for the office scene, two ways of computing *the same quantity*, charging Monte Carlo only for the runs needed to reach a stated accuracy (its cost scales as accuracy⁻²); times are single-core CPU times on one laptop.

Table S14: Relative change (%) of deterministic reproduction numbers caused by the measures of this section, as a function of the mobility scale D_0 (cell² day⁻¹). Two-zone room and barrier room: $N = 100$, same-cell kernel (barriers: mean over the 40 layouts of Table S11); office scene: 1 m kernel; DD reference: the unmodified room. Last row: the ratio $\bar{R}_1/R_0$ in the office scene.

measure	quantity	$D_0 = 1$	10	100	2.4×10^3
two-zone contrast $c = 0.58$	R_0	+16.0	+3.3	+0.4	+0.01
	$\bar{R}_1$	-14.6	-4.3	-0.6	-0.03
barriers, $\rho_B = 0.30$, FD	$\bar{R}_1$	-7.6	-4.7	-0.9	-0.04
barriers, $\rho_B = 0.30$, DD	$\bar{R}_1$	+20	+38	+50	+52
capacity 50 %, FD (office)	$\bar{R}_1$	-32.9	-15.2	-6.2	-4.5
capacity 50 %, DD (office)	$\bar{R}_1$	-50.5	-50.5	-50.5	-50.5
ventilation, $q \rightarrow q/2$ (office)	$\bar{R}_1$	-33.4	-45.0	-48.3	-48.8
office scene, baseline	$\bar{R}_1/R_0$	0.44	0.81	0.93	0.95

Table S15: Cost of deterministic evaluation against Monte Carlo for the same quantity (office scene, $N = 100$; single-core CPU). $n_{\text{off}}(x)$: offspring number (7) of a case introduced in one cell; CG: conjugate gradients. Speed-up: Monte Carlo CPU time needed for a relative standard error of 1 % (an absolute standard error of 0.01 for the outbreak probability) divided by the deterministic CPU time; the three values for R_0 are the first run, the re-check’s re-run of the same script, and the re-check’s compiled estimator; the two values for the other rows are the first run and the re-check’s re-run; no speed-up is quoted where the deterministic value is biased. Bias: deterministic value minus simulated value of the individual-based model.

quantity	deterministic method	deterministic CPU		speed-up at equal accuracy		bias
		$D_0 = 1$	$D_0 = 10$	$D_0 = 1$	$D_0 = 10$	$D_0 = 1; 10$
$R_0 = \rho(\mathbf{K})$	sparse LU, power iteration	4.6 ms	1.4 ms	8; 9; 3	133; 113; 86	none
$n_{\text{off}}(x)$, one cell	one sparse solve	0.27 ms	0.29 ms	35; 22	331; 351	none
$\bar{R}_1$	CG on the pair lattice	68 ms	137 ms	18; 22	85; 97	none
$P(\text{major})$	mean-field branching	9.4 ms	9.7 ms	–	–	+0.28; +0.03
attack rate, major	lattice equations (3)	0.40 s	1.54 s	–	–	+0.55; +0.03

One sparse factorisation of $\gamma \mathbf{1} - \mathbf{M}$ followed by power iteration gives R_0 in 1.4 to 4.6 ms (within 10^{-8} of a dense eigen-solve). Estimating the same number to 1 % by a particle power method costs 8 to 9 times more at $D_0 = 1$ and 113 to 133 times more at $D_0 = 10$ with a vectorised NumPy estimator, and 3 and 86 times more with a compiled one; at 0.1 % all factors are 100 times larger. The equal-accuracy speed-up is thus a few to about 10^2 at 1 % accuracy and a few hundred to about 10^4 at 0.1 %; it grows with mobility (more hops per infectious period) and depends on the implementation by a factor of two to three. The pair-lattice solve for $\bar{R}_1$ (conjugate gradients on $n^2 = 54,756$ unknowns, 0.07 to 0.14 s) is about 20 ($D_0 = 1$) to 90 ($D_0 = 10$) times faster than counting secondary cases to 1 % and agrees with the count within 0.1 standard error. For whole-epidemic outputs the mean-field route is not a substitute for simulation at $D_0 = 1$: the branching value of the outbreak probability is too high by 0.28 and the attack rate of the lattice equations by 0.55; no speed-up is meaningful for a biased answer. At $D_0 = 10$ both biases are 0.03, and the lattice equations (1.5 s) are slower than simulating to 1 % (0.19 s, 33 runs). The exact simulator takes 0.5 ms for one office epidemic at $D_0 = 1$ (5,900 events) and 5.9 ms at $D_0 = 10$. In short, the spectral formula and the pair solve are the inexpensive ways to obtain R_0 and $\bar{R}_1$, and simulation is cheap enough to be the method of choice for outbreak probabilities and sizes.

S6 The outbreak records

This section documents the two datasets of outbreak records used in Section 7 of the main text. For the first dataset, 27 records of outbreaks and exposure cohorts (Section S6.1), it says how they were found, what was recorded, how reliable each record is, and which numbers were digitised or reconstructed. The second dataset, eleven records of ten outbreaks with positions, follows in Section S6.2.

S6.1 The first dataset: 27 records of outbreaks and exposure cohorts

Search and inclusion. Candidate outbreaks were located by web search, with one set of queries per scene (office and call centre; supermarket and retail; classroom and school; metro, bus, train and aircraft; restaurant; choir), and by following the citations of the papers found. This is not a systematic review and the coverage is not exhaustive. A record was included if (a) it is described in a peer-reviewed article or an official surveillance report, (b) the DOI of that source is registered, and (c) the source gives at least one of: the number of exposed people, positions at seat or zone level, the geometry of the room, the duration of exposure. The result is 27 records from 31 sources: 5 offices and workplaces, 4 retail settings, 5 classrooms and class-like rooms, 7 vehicle cabins, 2 restaurants, 2 choir or church events, and 2 records of large settings that lie outside the scope of the model (a cruise ship and worker dormitories). These two are recorded for reference only, as examples of large shared settings with long exposure that a model of one closed room does not cover, and take part in no comparison. Three records without a denominator (S3, C4, T7) and one with a reported attack rate but no transcribed counts (T6: 54 persons aboard, 72 % of the passengers ill, from the abstract) were kept as context for their settings; like the two out-of-scope records, they take no part in any quantitative comparison. No documented outbreak in a metro carriage with an exposed denominator was found; buses, a coach, a minibus, a train cohort and one flight stand in for that scene. The supermarket has one record with denominators for staff and customers (S1) and none with geometry, positions or dwell times.

What was recorded. Each record has 51 fields: setting, place, dates and pathogen; the sources and how each was read; type and duration of exposure; room geometry, occupancy, ventilation and mask use; the index case; the numbers exposed and infected, with an alternative count where sources or case definitions differ; the attack rate with its exact (Clopper–Pearson) 95 % interval; the spatial pattern reported; a quality grade; a list of the fields that were derived and not transcribed; and the mapping to the model. Table S16 lists the records and Fig. S10 shows the attack rates. Three records concern other pathogens (O5 tuberculosis, C4 measles, T6 influenza).

Table S16: The 27 records of the outbreak dataset. Counts exclude the index case unless marked †. Attack rates and exact (Clopper–Pearson) 95 % intervals are computed from the counts in the third column; ‘(or ...)’ gives the count of a second source or case definition. Exposure is the duration of the event, or of each exposure where it was repeated. Grades are those assigned when the dataset was built (criteria in the text). Record T4 pools 2,334 index patients.

ID	Setting (place)	Cases / exposed	Attack rate, % (95 % interval)	Exposure	Ventilation	Gr.	Sources
<i>Offices and workplaces</i>							
O1	Call centre, 11th floor, open-plan desks (Seoul)	94/216†	43.5 (36.8–50.4)	working days over about 2 weeks	not reported	B	[56]

continued on the next page

Table S16 (continued)

ID	Setting (place)	Cases / exposed	Attack rate, % (95 % interval)	Exposure	Ventilation	Gr.	Sources
O2	Team of 13 in an open-plan office (Zurich)	8/12 (or 10/12)	66.7 (34.9–90.1)	11 h	mechanical, about 1 air change per hour	B	[74]
O3	Public-facing office, three open-plan floors (England)	22/40 [†]	55.0 (38.5–70.7)	weeks	CO ₂ typically ≤1000 ppm	C	[53] ^a
O4	Beef-processing line, fixed work stations (Germany)	20/78	25.6 (16.4–36.8)	3 shifts	<1 air change per hour, recirculating cooling fans	B	[20]
O5	Office building, tuberculosis (Massachusetts)	27/67	40.3 (28.5–53.0)	160 h (4 weeks)	about 7 l/s outdoor air per occupant	C	[52] ^a
<i>Supermarket and retail</i>							
S1	Supermarket, staff (Liaocheng)	11/120 [†]	9.2 (4.7–15.8)	not reported	not reported	B	[67]
S2	Department store, sales staff (Tianjin)	6/194 [†] (or 7/194)	3.1 (1.1–6.6)	not reported	described as poorly ventilated	C	[10, 41], [30] ^a
S3	Shopping mall, 8 floors (Wenzhou)	no denominator	–	not reported	not reported	C	[9]
S4	Grocery store, staff, PCR screen (Massachusetts)	21/104 [†]	20.2 (13.0–29.2)	not reported	not reported	C	[40] ^a
<i>Classrooms and class-like rooms</i>							
C1	Primary-school classroom, 24 pupils in 5 rows (Marin County)	12/24	50.0 (29.1–70.9)	school days; hours not reported	portable HEPA filter; doors and windows open	A*	[39]
C2	High school, 35–38 pupils per classroom of 39–49 m ² (Jerusalem)	153/1,161 [†]	13.2 (11.3–15.3)	8–9 school days of 6.3–6.7 h	split air conditioning, recirculating	B	[64]
C3	20 primary schools, masks worn, seats 0.9 m apart (Utah)	5/728	0.69 (0.22–1.60)	school days; not quantified	not quantified	B	[23]
C4	Primary school, 14 classrooms on one recirculating system, measles (New York State)	28 cases; no denominator	–	not in abstract	largely recirculated air	C	[57] ^a
C5	Fitness dance classes in a 60 m ² room (Cheonan)	57/217 [§]	26.3 (20.5–32.7)	50 min per class	not reported	B	[29]
<i>Vehicle cabins (stand-ins for the metro carriage)</i>							
T1	Bus, 15 rows of seats (Zhejiang)	23/67	34.3 (23.2–46.9)	50 min each way	air conditioning on recirculation	A	[59] ^e
T2	Coach, 49 seats, windows closed (Hunan)	7/48 (or 7/46; 45 with seat positions)	14.6 (6.1–27.8)	2.5 h (200 min in the second report)	1.72 l/s per person, measured	A	[45, 55]
T3	Minibus, 18 seats (Hunan)	2/12 (or 2/17)	16.7 (2.1–48.4)	60 min	3.22 l/s per person, measured	B	[45, 55]
T4	High-speed trains, pooled contact-tracing cohort (China)	234/72,093	0.32 (0.28–0.37)	mean 2.1 h	not reported	A*	[27] ^e
T5	Long-haul flight, business cabin (London–Hanoi)	12/20	60.0 (36.1–80.9)	10 h	aircraft cabin; not quantified	A	[35]

continued on the next page

Table S16 (continued)

ID	Setting (place)	Cases / exposed	Attack rate, % (95 % interval)	Exposure	Ventilation	Gr.	Sources
T6	Airliner held on the ground, influenza (Alaska)	72 % of passengers; 54 aboard (counts not transcribed)	–	3 h	none (system inoperative)	C	[51] ^a
T7	National cluster surveillance; commuter rail as a venue (Japan)	no denominator	–	–	–	C	[14]
<i>Restaurants</i>							
R1	Restaurant, 18 tables, 89 patrons (Guangzhou)	5/79 [‡]	6.3 (2.1–14.2)	82 min (index table)	0.56–0.77 air changes per hour, measured	A	[42, 44]
R2	Restaurant, 9.2 m × 10.5 m (Jeonju)	2/13	15.4 (1.9–45.4)	overlaps of 5 and 21 min	no ventilation system; air-conditioner jet	A	[38]
<i>Choir and church</i>							
H1	Choir rehearsal in a church hall (Skagit County)	52/60 (32/60 confirmed)	86.7 (75.4–94.1)	2.5 h	0.3–1.0 h ^{−1} , estimated	A*	[47] ^m , [21] ^e
H2	Church services, singer in a choir loft (Sydney)	12/508	2.4 (1.2–4.1)	1 h per service	minimal: fans off, doors and windows largely closed	B	[31]
<i>Outside the scope of the model (recorded for reference)</i>							
X1	Cruise ship under quarantine (Yokohama)	634/3,711 [‡]	17.1 (15.9–18.3)	weeks	–	X	[48]
X2	Migrant-worker dormitories (Singapore)	17,758/295,000 [‡]	6.0 (5.9–6.1)	weeks to months	–	X	[37]

[‡] numerator and denominator include the index or primary case(s). [‡] upper bound on the number infected in the restaurant (see text). * graded A, but not all grade-A criteria are met (see text). § numerator derived from the reported attack rate (26.3 % of 217); the record pools the classes of eight infected instructors in 12 facilities and is not a single-index event. Reading access to the source: ^a abstract only; ^e publisher's page available only as an automated text extraction; ^m accepted manuscript; unmarked, full text.

Grades. Grade A (8 records) was defined as: a single index case, an enumerated exposed group, positions at seat or zone level, a reported duration, and a geometry that is reported or measurable. Grade B (9 records) requires a denominator and a duration with partial spatial information; grade C (8 records) covers aggregate, multi-generation, abstract-only or incompletely enumerated events; the 2 out-of-scope records are graded X. The re-check found that only 5 of the 8 grade-A records (T1, T2, T5, R1, R2) meet all the grade-A criteria. The classroom C1 has no exposure duration in hours and no room size; the train cohort T4 pools 2,334 index patients and its source prints no cell denominators; and no seat or zone positions were published for the choir H1. These three are marked A*. They remain in use (T4 is the only matrix of attack rates by seat offset, H1 the benchmark for the level of the attack rate), but the dataset contains five, not eight, single-index events with positions, a duration and a geometry.

Reading access. All 31 DOIs were confirmed in the DOI registry, and the bibliographic fields were taken from Crossref; the re-check repeated both checks. The numbers were transcribed from the full text for 21 sources and from the accepted manuscript for one [47]. Three sources [21, 27, 59] were available only as an automated text extraction of the publisher's web page, not as the PDF. Their numbers were compared with the abstracts and extracted a second time in the re-check, with the same values, and two internal checks hold (the risk ratios recomputed

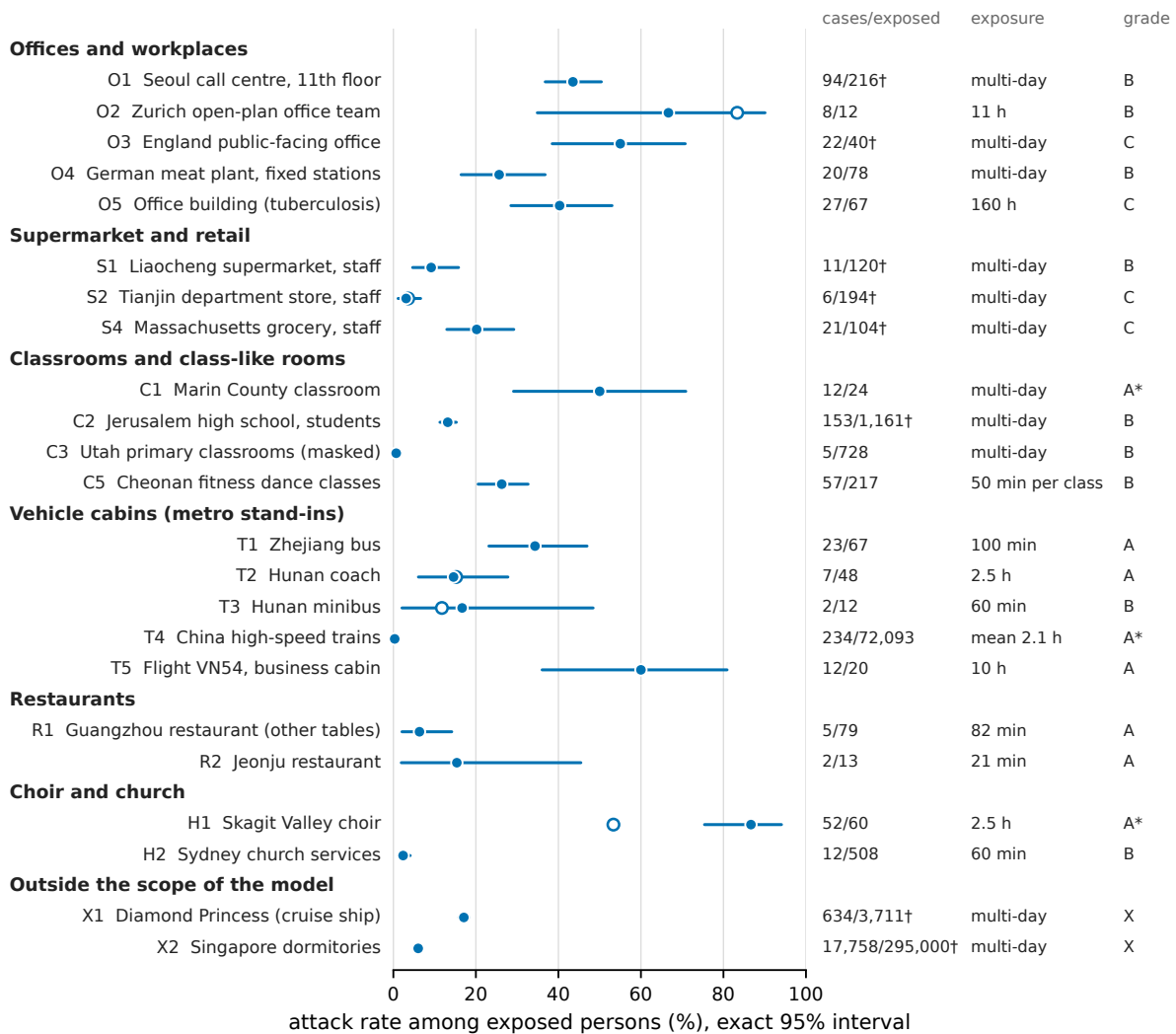


Figure S10: Attack rates of the 23 records that have an exposed denominator, with exact (Clopper–Pearson) 95 % intervals, in the order of Table S16. Filled circle: the count used in this article; open circle: the alternative count (O2: 10 instead of 8 secondary cases; S2: 7 instead of 6; T2: 7/46; T3: 2/17; H1: the 32 laboratory-confirmed cases only). †: counts include the index or primary case(s). ‘Multi-day’ means that no duration in hours is recorded. A*: graded A without meeting every grade-A criterion. The events differ in pathogen, variant, mask use, testing and number of generations, so the attack rates are not comparable as levels of risk.

from the zone counts of the bus, Table S17, equal those printed by Shen et al. [59]; the train table is discussed below). These three sources have not been checked against the PDF files. Six sources were read as abstracts only [30, 40, 51–53, 57]; the five records that rest on abstracts alone (O3, O5, S4, C4, T6) are graded C.

Conflicting counts and an upper bound. Where sources disagree, both versions are recorded. The two investigations of the Hunan coach give 7 cases among 48 passengers over 2.5 h [45] and 7 among 46 over 200 min [55], and they place the infected passengers differently; the minibus of the same outbreak has 2/12 against 2/17; the department store has 6 or 7 staff cases; the Zurich team has 8 or 10 first-generation cases among 12; and 20 of the 52 choir cases are probable, not laboratory-confirmed. For the Guangzhou restaurant the five secondary cases at the two neighbouring tables are an upper bound on the number infected at the lunch. The first report of that outbreak states that at least one member of each of the two families was infected

Table S17: Near/far contrasts available in the sources. Risk ratios with log-normal (Katz) 95 % intervals and Fisher’s exact test, computed from the counts shown. The two rows of T1 and the two rows of R1 are alternative splits of the same people. ^c: one cell is zero and 0.5 was added to every cell. [‡]: upper bound (see text). ^e: pupils enrolled in grades 7–9 and 10–12 (581 and 583); 1,161 were tested, 580 of them in grades 10–12. For T5 the source defines the near stratum as ‘ ≤ 2 seats away’. O1 and C2 are multi-generation outbreaks, and the position of the index case in O1 is unknown.

ID	Contrast (near against far)	Near: cases/ <i>n</i> (%)	Far: cases/ <i>n</i> (%)	Risk ratio (95 % CI)	<i>p</i>
O1	north wing against south wing (mapped seats)	79/137 (57.7)	4/62 (6.5)	8.9 (3.4–23.3)	5.2×10^{-13}
C1	front two rows against back three rows	8/10 (80.0)	4/14 (28.6)	2.8 (1.2–6.8)	0.036
T1	rows 5–11 (within about 2 m of the index) against rows 1–4 and 12–15	14/33 (42.4)	9/34 (26.5)	1.6 (0.8–3.2)	0.20
T1	rows 6–10 against rows 1–5 and 11–15 (alternative split)	11/23 (47.8)	12/44 (27.3)	1.8 (0.9–3.3)	0.11
T2	rear rows 8–13 (index zone) against front rows 1–7	3/19 (15.8)	4/26 (15.4)	1.0 (0.3–4.1)	1.0
T5	within 2 seats of the index against farther seats, business cabin	11/12 (91.7)	1/8 (12.5)	7.3 (1.2–46.2)	7.7×10^{-4}
R1	neighbouring tables against remote tables	5/16 [‡] (31.2)	0/63 (0.0)	41.4 ^c (2.4–713)	1.9×10^{-4}
R1	same air-conditioning zone against other zones (alternative split)	5/11 [‡] (45.5)	0/68 (0.0)	63.2 ^c (3.7–1,071)	2.0×10^{-5}
C2	grades 7–9 against grades 10–12 (separate wings)	135/581 (23.2)	18/583 ^e (3.1)	7.5 (4.7–12.1)	1.3×10^{-26}
S1	staff against customers	11/120 (9.2)	0/8,224 (0.0)	1,563 ^c (92.7–26,381)	3.4×10^{-21}

Table S18: Negative controls: groups with a documented opportunity for exposure and almost no infection. The last numerical column is the one-sided exact 95 % upper bound on the attack rate. The two restaurant rows are alternative groupings of the same patrons. Access marks as in Table S16.

Group	Record	Cases/exposed	Upper bound, %	Source
Supermarket customers	S1	0/8,224	0.036	[67]
Other bus travelling to the same event	T1	0/60	4.9	[59] ^e
Restaurant, remote tables	R1	0/63	4.6	[42]
Restaurant, other air-conditioning zones	R1	0/68	4.3	[42]
Flight, premium-economy cabin	T5	0/35	8.2	[35]
Tested school contacts, masked classrooms	C3	5/728	1.44	[23]
Call-centre building, floors 7–9	O1	1/595	0.79	[56]

in the restaurant and that the further cases in those families may be due to transmission within the family [44]; the number infected at the lunch outside the index table is therefore between 2 and 5. The counts marked [‡] in Tables S16 and S17 carry this caveat.

Spatial contrasts and negative controls. Table S17 gives every near/far contrast that the sources allow, and Table S18 the groups used as negative controls. Three cautions apply. First, the wing contrast of the call centre refers to the 84 case seats shown in the published plan; if the ten cases without a seat had all worked in the south wing the ratio would be 2.6 (95 % interval 1.6–4.1), not 8.9. Second, the four single-index events with one split each (C1, T1, T2, T5) are statistically compatible with a common risk ratio (Cochran’s $Q = 3.77$ on 3 degrees of freedom, $p = 0.29$), so sorting events into ‘near-field’ and ‘well-mixed’ ones is a reading of point estimates; the coach, with no front-to-rear gradient, has a side-to-side contrast of 6/24 against 1/20 (risk ratio 5.0, 95 % interval 0.7–38). Third, the levels are heterogeneous: over the nine SARS-CoV-2

events with a single index case and a single exposure period, the cumulative hazard per hour, $-\ln(1 - \text{attack rate})/\text{duration}$, spans a factor 34 when the 20 probable choir cases are counted and a factor 20 with confirmed cases only.

Digitisation of the call-centre floor plan. The seat map of record O1 was obtained from the floor plan published as Fig. 2 of Park et al. [56]; the coordinates are derived from that figure and are not original data of this article. Desk rectangles were segmented by colour and shape. The two open-plan wings contain 137 desks (north) and 62 desks (south). The plan marks 84 case seats: 79 in the north wing, 4 in the south wing and 1 in the offices on the east side, whose other desks were not digitised. The text of the source reports 94 cases among the 216 employees of the floor, so 10 cases have no seat in the plan. The plan has no scale bar: coordinates are in units of the desk pitch, their conversion to metres is unknown, and one employee per drawn desk is assumed. The re-check typed the map again desk by desk from enlargements of the figure: the two readings match one to one, no desk has a different case flag, and the largest difference in position is 3.8 pixels at a desk pitch of 32.5 pixels. Figure S11 is drawn from this file.

Reconstruction of the train cell counts. Hu et al. [27] print, for 23 seat offsets from the index passenger, an attack rate with a 95 % interval but no counts. Two reconstructions were needed. On the publisher’s page the five same-row values appear one column to the left of where they belong (a contact cannot sit zero rows and zero columns away); with the values moved to columns 1–5, the column means recomputed from the cells differ from the printed ones by 0.02 percentage points in total, against 1.17 for the layout as printed. Then, for each cell, the re-check searched for the integer pair (cases, contacts) whose rate and Wilson interval reproduce the printed values; we checked that every recovered pair does so to within 0.012 percentage points. The recovered cells sum to 230 cases among 72,692 contacts, against the 234 among 72,093 reported for the cohort; cell denominators range from 1,033 to 5,420, and the 11 cells with at most three cases are the least certain. The calibration of Section 7 uses these reconstructed counts. The source itself warns that relatives and friends travelling together may have infected each other before or after the journey, which inflates the adjacent-seat cell (3.53 %) most.

Limitations and release of the first dataset. Investigated outbreaks are mostly large ones, so the attack rates over-represent highly infectious index cases; records C3 and T4 are closer to ordinary exposures. Variants, mask use, vaccination and testing differ between records and are not harmonised. Positions are zonal in most records: only T4 gives a matrix of offsets and only O1 is resolved to seats. The released repository contains the dataset (`outbreaks.csv`, `outbreaks.json`), the list of sources with the access level of each, the auxiliary tables (strata, risk ratios, the train matrix and its reconstruction, the digitised seat coordinates) and the scripts that build and check them. Copies of the source texts and figures used for transcription are not redistributed.

S6.2 The second dataset: eleven records of ten outbreaks with positions

Search and inclusion. The second dataset was assembled after the first test, by queries to Europe PMC for outbreaks with a seat map or with counts by distance for the whole exposed group, and from the references of Hertzberg and Weiss [24]. Eleven events were kept (Table S19); the meat-processing line is record O4 of the first dataset, now with its counts by distance. The events found and not used, with the reason, were fixed in the protocol before any model was run: no identified source on board, one or two cases only, exposure before the flight, denominators missing by seat, or positions not published (among them the aircraft outbreak of Moser et al. [51] and the measles outbreak analysed by Riley et al. [57]). No outbreak in a courtroom, a fitness class or a restaurant with positions for the whole exposed group was found in openly accessible form. The search is not a systematic review, and the re-check did not repeat it.

Table S19: The eleven records of the second dataset (Section 7.5); W1 and W2 are the same ward with the same index patient. Near: within two rows of a source (aircraft), in the index patient’s cubicle or bay (ward), within 8 m (processing line). Risk ratio near/far with its 95 % interval. Grade A: identified source or sources, enumerated exposed group, positions for the whole group; B: more than one generation of cases. Role: fixed by the pathogen before any model was run.

	Event	Pathogen	Setting, duration	Position data	Near vs far	Risk ratio	Gr.	Role
F1	Hong Kong–Beijing, CA112 [54]	SARS-CoV-1	B737-300, 3 h	full seat map (redrawn in [24])	9/29 vs 9/82	2.83 (1.24–6.43)	A	calibration
F2	Chicago–Honolulu [34]	<i>M. tuberculosis</i>	wide-body, 8.75 h	two strata of the abstract; no seat map	4/13 vs 2/55	8.46 (1.73–41.3)	A	calibration
F3	Los Angeles–Auckland [5]	influenza A(H1N1) pdm09	B747-400, rear section, 13 h	seat map; 9 sources	3/67 vs 0/40	4.20 ^a (0.22–79.2)	A	calibration
F4	Cancun–Birmingham [77]	influenza A(H1N1) pdm09	B767, 9.5 h	seat map; 6 sources	5/126 vs 4/90	0.89 (0.25–3.23)	A	calibration
W1	Ward 8A, students [75]	SARS-CoV-1	hospital ward, 40 min	three strata by bed; floor plan	7/11 vs 0/8	11.09 ^a (0.73–168.6)	A	calibration
W2	Ward 8A, inpatients [78]	SARS-CoV-1	same ward, days	three strata by bay; floor plan	13/20 vs 17/54	2.06 (1.24–3.43)	B	calibration
F5	Flight to Naha [69]	SARS-CoV-2	B737-800, 2 h	seat map; index in 23C	7/23 vs 7/117	5.09 (1.97–13.1)	A	hold-out
F6	Sydney–Perth [60]	SARS-CoV-2	A330-200, mid cabin, 5 h	seat map; 4 sources	8/46 vs 3/52	3.01 (0.85–10.7)	A	hold-out
F7	Dubai–Auckland, EK448 [65]	SARS-CoV-2	B777-300ER, 18 h	seats within rows 23–30; 2 sources	4/13 vs 0/71	47.67 ^a (2.72–835.1)	A	hold-out
F8	Tel Aviv–Frankfurt [25]	SARS-CoV-2	B737-900, 4.7 h	seat map; 7 sources	2/20 vs 0/51	12.56 ^a (0.63–250.5)	A	hold-out
P1	Meat-processing line [20]	SARS-CoV-2	32 × 8.5 m hall, 3 shifts	counts by 1-m distance band	17/26 vs 3/52	11.33 (3.65–35.2)	A	hold-out

^a With a continuity correction of 0.5 (a stratum without a case).

Transcription and its checks. Seat maps were transcribed by eye from the published figures, and each map was checked against the totals printed in its source before any model was run. The totals agree exactly for the flights from Hong Kong, from Los Angeles and from Tel Aviv. They differ by one to four persons, in far strata, for three flights: 216 passengers at risk with a seat against 220 in the text for the flight from Cancun; 141 persons on the map against 142 for the flight to Naha; 110 passengers of the mid cabin against 111 or 112 for the flight to Perth. Two full texts are not openly accessible: the flight from Hong Kong uses the seat map as redrawn by Hertzberg and Weiss [24], and the tuberculosis flight only the two strata of the abstract, with seats placed at random inside them. The counts of the ward and of the processing line were typed from the text and from an appendix table of the sources. The re-check re-read every seat symbol of the four held-out flights and of the flight from Hong Kong against the stored figures, re-typed the counts of the processing line, and fetched the abstract and full-text numbers of the other events again; all stratum counts match. Two calibration maps (the flights from Los Angeles and from Cancun) were not re-read seat by seat. Three limits of the data are not in the counts: about 22 passengers of the flight to Naha who were lost to follow-up are counted as non-cases, mostly in far rows; most non-cases of the flight from Tel Aviv were never tested; and the published figure of the ward gives each student’s bed, which the analysis replaced by random beds within the three strata (using the exact beds changes the pooled score against the well-mixed model by +0.13 and nothing else). The near zone of the flight to Naha is not the one

its source analysed: the source’s own group ‘within two rows’ has 14 persons, against the 23 in rows 21–25 used here. And one flight (AF171) was excluded as a secondary tabulation, although two calibration events are themselves secondary in that sense (the flight from Hong Kong, from a redrawn map, and the tuberculosis flight, from an abstract); with two cases it would change nothing. The released tables give, for every event, the strata, the persons at risk and the cases, and for the seat-map events the transcribed position of every person, with the reference and the figure or table of origin.

S7 The first outbreak test in detail

This section gives the details of the first outbreak test of Section 7 of the main text; its protocol, sensitivity analyses and post-hoc log are in Section S8.

S7.1 Structure in the first dataset

In a cohort of contacts of 2,334 index cases on Chinese high-speed trains [27] the attack rate is 3.53 % in the adjacent seat and 0.03–0.25 % one to three rows away; the source notes that companions travelling together inflate the adjacent-seat value. The Seoul call centre is the only multi-generation record with seat positions (Fig. S11). Its two wings differ: 79 of 137 desks in the north wing are case seats against 4 of 62 in the south wing, a ratio of 8.9 (3.4–23) for the 84 mapped seats, and of 2.6 (1.6–4.1) if the ten unmapped cases all sat in the south wing. Inside the affected wing there is no clustering at desk scale. Of the 286 pairs of adjacent desks, 91 are pairs of cases; random relabelling of the 79 cases among the 137 desks gives 94.6 ± 5.6 ($z = -0.64$, one-sided $p = 0.76$; join-count statistic of Moran [50]). In the re-check the same seat map was obtained by reading enlarged crops of the published figure desk by desk, without the colour segmentation, the same result for four other definitions of adjacency, and a power of at least 0.94 against neighbour-driven spread whose local weight is at least equal to the background.

S7.2 Calibration

Vehicle cabins are calibrated on the train cohort: binomial likelihood on 22 seat-offset cells, with the adjacent-seat cell excluded because of the companions. Seated rooms are calibrated on the Guangzhou restaurant [42, 44], with interval-censored counts at the two tables next to the index table, because the first report states only that at least one member of each of those families was infected at the lunch. The distribution of the mixing coefficient used for prediction is a pseudo-posterior: proportional to the profile likelihood (the infectiousness parameter ε_k maximised, not integrated out) times a flat prior on $\log_{10} D$ between 10^{-2} and $10^4 \text{ m}^2 \text{ h}^{-1}$. The second test uses the same construction. For brevity it is called the posterior below and in Section 7 of the main text.

Hold-out. Four single events (bus, coach, flight, classroom) are tested on the split of their cases between a near and a far stratum, and the call centre on the join-count z of its north wing. For each single event ε_k is set from the event total, never from the strata; the prediction is the exact distribution of the number of near cases given the total, averaged over 300 draws of the posterior, of the ventilation prior and of the unknown seating inside each stratum. The logarithm of its value at the observed split is the score S , and $\Delta = S(\text{model}) - S(\text{M0})$.

On the train cohort both spatial models gain 26.6–26.7 units of log-likelihood over M0 for one more parameter and cannot be told apart (Table S20, Fig. S12a,b). M2 gives $D_{\text{air}} = 25 \text{ m}^2 \text{ h}^{-1}$ at the maximum of the likelihood (90 % posterior interval 9.7–43, with its mode at $20 \text{ m}^2 \text{ h}^{-1}$ once the ventilation prior is integrated out; 8–96 if the likelihood is tempered for its over-dispersion). Both are nevertheless rejected by the data they were calibrated on (deviance 72.6–72.9 on 20 degrees of freedom). The isotropic kernel cannot follow the anisotropy of the train table: two seats away in the same row the observed attack rate is 1.65 % and the fit 0.47 %; one

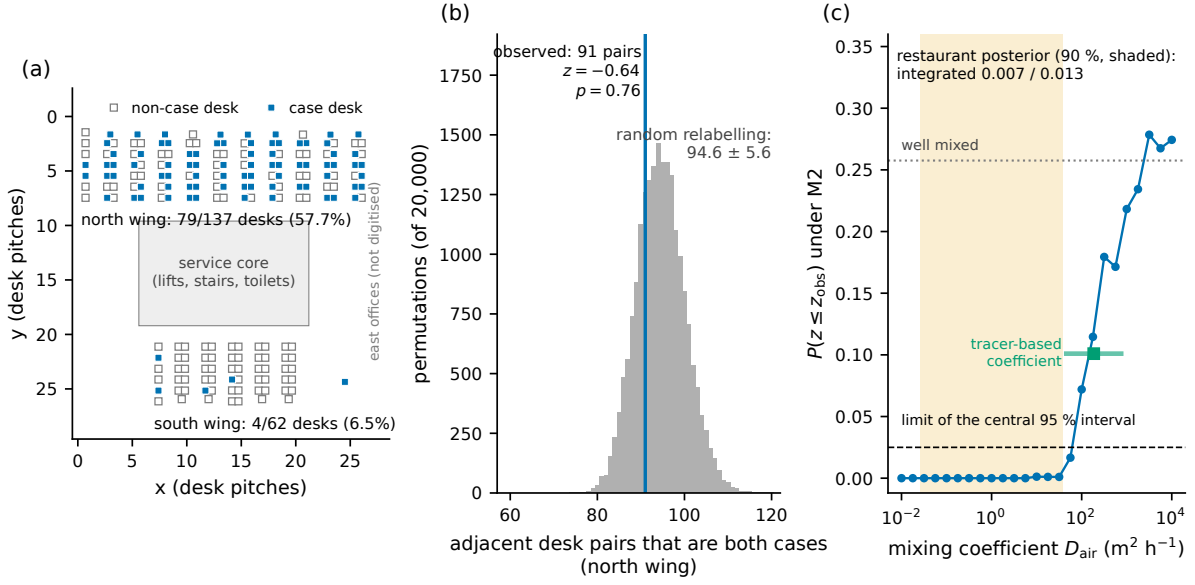


Figure S11: Seoul call centre, 11th floor. (a) Desks with (filled) and without (open) a case. The positions were digitised by us from Fig. 2 of Park et al. [56]; the published plan has no scale bar, so coordinates are in desk pitches, and it shows 84 of the 94 reported case seats (one of them in the east offices, which were not digitised otherwise). (b) Number of adjacent desk pairs in the north wing (centres within 1.25 desk pitches; 286 pairs) in which both desks are case seats: observed value (line) and its distribution under 20,000 random relabellings of the 79 cases among the 137 desks; z is the standardised difference and p the one-sided probability of a count at least as large as observed. (c) Probability, under M2, of a joint-count z as small as observed, as a function of the mixing coefficient (25 grid values, computed in the re-check with a mode-sum propagator that shares the numerical defect of Section 7.4; the probability is 0 at every grid value below $5.6 \text{ m}^2 \text{h}^{-1}$, so the defect can act only through the outbreak-size weights of the second integral below). Shaded: 90 % posterior interval of the coefficient calibrated on the restaurant; integrated over that posterior the probability is 0.007 (posterior draws weighted equally) or 0.013 (weighted by their probability of producing an outbreak of the observed size), below the limit 0.025 of the central 95 % interval (dashed). Dotted line: well-mixed model. Square with bar: coefficient taken from tracer measurements (Section 7.8). With the 40 posterior draws of the Monte Carlo estimate of the protocol the probabilities were 0.017 and 0.031 (Table S30, entry P5).

Table S20: Calibration fits of the first test (in-sample). Mixing coefficient: D_p for M1, D_{air} for M2; value at the maximum of the likelihood and 90 % posterior interval. The deviance is relative to the saturated binomial model; for the restaurant the counts are interval-censored and no deviance is defined.

Record	Model	Mixing coefficient ($\text{m}^2 \text{h}^{-1}$)	Log-likelihood	Deviance (d.f.); p
Trains, 22 cells	M0	—	−99.70	126.1 (21); 5×10^{-17}
	M1	0.50 (0.31–0.79)	−73.12	72.9 (20); 6×10^{-8}
	M2	25 (9.7–43)	−72.97	72.6 (20); 7×10^{-8}
Restaurant, 16 tables	M0	—	−6.08	—
	M1	0.20 (0.016–14)	−2.47	—
	M2	0.50 (0.026–39)	−2.48	—

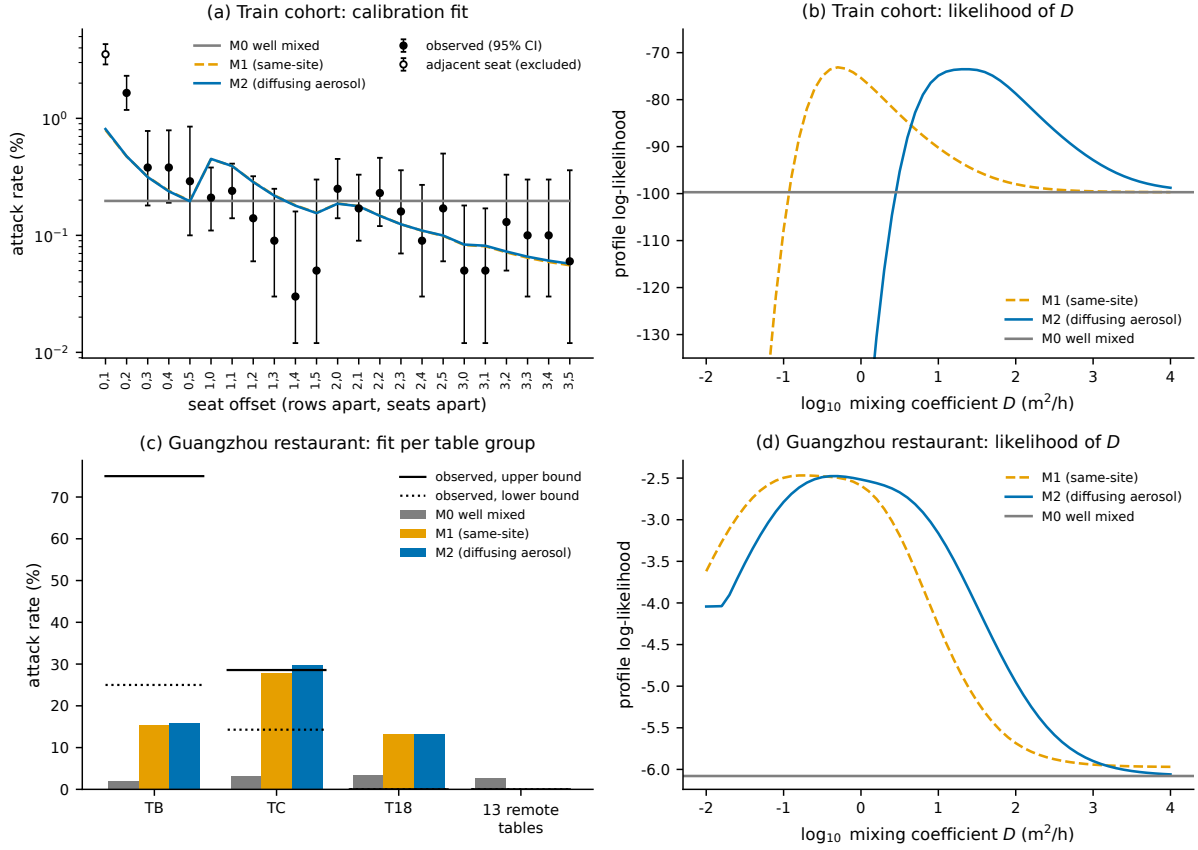


Figure S12: Calibration of the first test. (a) Train cohort of Hu et al. [27]: attack rate by seat offset (rows apart, seats apart) with 95 % intervals, and the fitted models; the open symbol is the adjacent seat, excluded from the likelihood. (b) Profile log-likelihood of the mixing coefficient D (D_p for M1, D_{air} for M2). (c) Guangzhou restaurant: fitted attack rates at the two tables next to the index table (TB, TC), at the table across from it (T18) and at the 13 remote tables, with the bounds of the interval-censored counts; no case was observed at T18 or at the remote tables. (d) As (b) for the restaurant. The models are those of Table 6.

row away in the same column they are 0.21 % and 0.45 %. For the restaurant, with two to five cases acquired at the lunch, D is identified only within three decades, and the fitted M2 conflicts with the tracer-gas measurement of Li et al. [42]: the measured concentration at the two neighbouring tables was 87–98 % of that at the index table and 55–86 % at five tables outside their air-conditioning zone, whereas the fitted exposure is at most 0.3 % of the index-table value at all seven (Fig. S12c,d). The infections were confined to one air-conditioning zone although the air was not; an isotropic kernel can imitate this only with a coefficient that does not describe the air. A re-implementation with separately written code and a different propagator method reproduces the train log-likelihoods to three decimals.

S7.3 Hold-out events

Fig. S13 and Table 7 give the results. A re-implementation with separately written code and its own event builders reproduced the results of the frozen code to Monte Carlo accuracy, including, in the classroom, the numerical defect described below. We take the events in turn for M2.

Flight VN54: reproduced. Eleven of the 12 passengers seated at most two seats from the index case (≤ 2 seats away in the source) and one of the 8 farther away were infected. M0 is rejected ($p = 0.0008$) and M2 is not ($p = 0.61$). The result is robust to the calibration ($p = 0.60$ at the mode of the train posterior, $20 m^2 h^{-1}$) and to the ventilation prior ($p = 0.19$ and 1.0

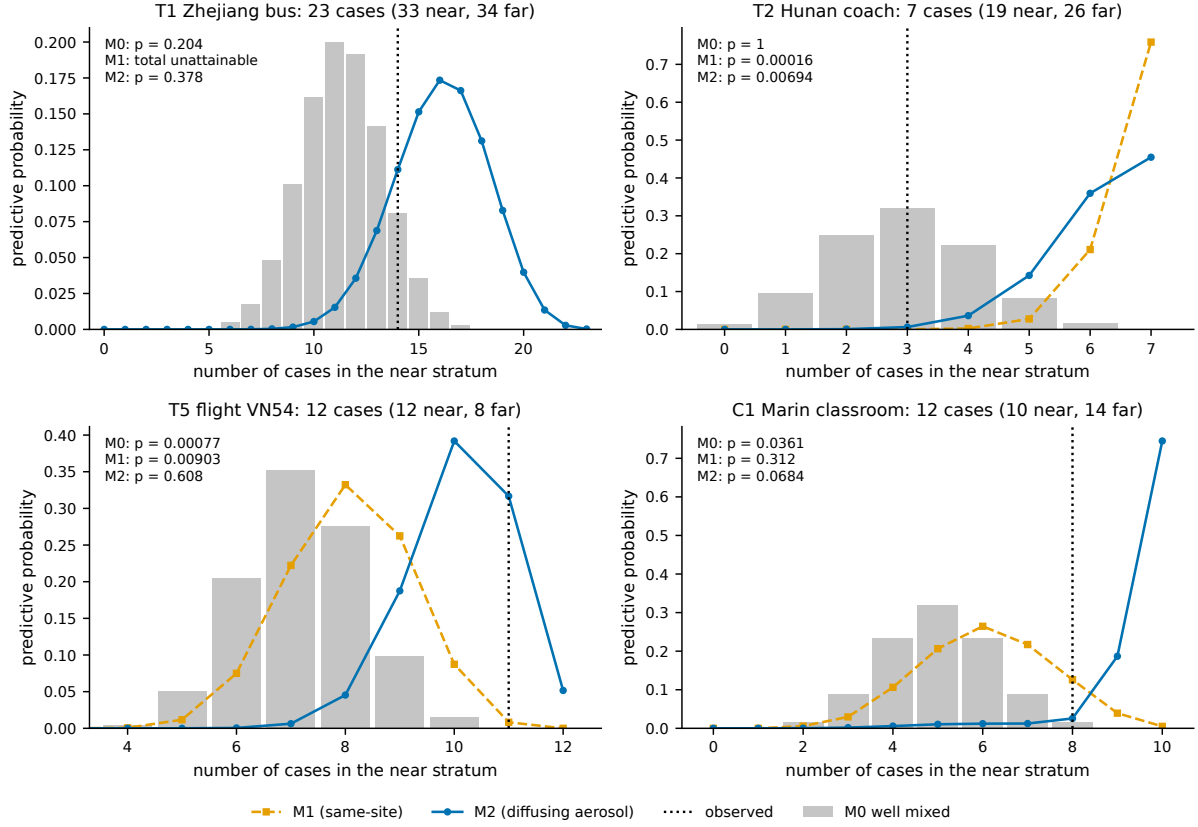


Figure S13: Hold-out events of the first test. Predictive distribution of the number of cases in the near stratum, given the total of the event, under M0 (bars; hypergeometric), M1 (dashed; exact simulator for the bus, the flight and the classroom, closed form for the coach) and M2 (solid); dotted line: observed number. The distributions average over 300 draws of the calibrated posterior, of the ventilation prior and of the seating inside each stratum; p is the two-sided predictive p -value. For the bus no M1 curve is drawn because the model reaches the observed total only with probability 0.0002 (Table S21). Exposures are computed with the round-off-safe propagator (see the classroom below); only the classroom curve of M2 differs from that obtained with the frozen propagator.

when it is scaled by $1/3$ and by 3). This is the one robust positive result of the first test, and it is thin: 20 exposed passengers, among them four companion couples seated side by side, three of which are pairs of cases. No correction for companions was applied, so the p -values against M0 are too small.

Hunan coach: failure. Infection was uniform along the coach (3/19 rear, 4/26 front). With the train-calibrated coefficient and the measured ventilation of the coach, M2 puts six of the seven cases in the rear half (predicted ratio 11, $p = 0.007$); M0 fits exactly. The failure persists in 10 of the 12 pre-specified sensitivity variants (Table S28; the pooled test passes in 11), under the tempered posterior ($p = 0.026$) and with the ventilation rate cut to a third ($p = 0.040$). The model gives all 45 passengers the same 200 minutes of exposure, whereas two of them boarded 40 minutes late and four left early.

Marin classroom: a pass that is not a prediction. The predictive p -value is 0.068 (0.050 to 0.097 over five other Monte Carlo seeds), and it comes from the width of the restaurant posterior, not from the calibrated kernel. At the posterior mode ($D_{\text{air}} = 0.5 \text{ m}^2 \text{ h}^{-1}$) the model infects all ten front desks and the observed 8/10 against 4/14 is rejected ($p = 0.0009$). The 63 % of the posterior mass between 0.1 and $5.6 \text{ m}^2 \text{ h}^{-1}$ contributes 10 % of the predictive probability of the observation, and the 18 % below $0.1 \text{ m}^2 \text{ h}^{-1}$ contributes 0.1 %; the rest, 90 %, comes from the diffuse upper tail above $5.6 \text{ m}^2 \text{ h}^{-1}$ (19 % of the mass), where the kernel is nearly flat across the room. In 83 % of the posterior draws the emission rate needed to produce the 12 cases

exceeds 10^4 quanta h^{-1} (median 8×10^7), which is not physical. With the frozen propagator the predictive p was 0.11, and more than half of the predictive probability came from coefficients below $0.1 \text{ m}^2 \text{ h}^{-1}$; that half was a numerical artefact. The frozen propagator is a sum over cosine modes whose absolute error reaches about 10^{-10} of the largest exposure; at coefficients of $0.1 \text{ m}^2 \text{ h}^{-1}$ and below the true exposures of the farthest desks are 10^{-33} to 10^{-16} of that largest value, and the mode sum returns round-off noise of either sign in their place, which made far infections look possible. The same defect was found in the second test (Section 7.5). The values above use a propagator that keeps the mode sum where it is resolved and replaces it elsewhere by a positive sum of image terms; on classroom layouts this agrees with a third method (uniformisation) to better than 10^{-8} relative, receiver by receiver. Only the classroom is affected: for the bus, the coach and the flight no exposure in the posterior range is unresolved and the predictive distributions are unchanged to the last digit; the restaurant calibration changes by about 10^{-7} in log-likelihood at most.

Seoul call centre: failure. For this multi-generation outbreak the final infection pattern was simulated as a chain of generations with pair probabilities (18), scaled to the observed size. Desks are mapped to the nearest lattice site, so that the 137 desks occupy 113 distinct sites (24 sites hold two desks, which then have the largest possible mutual exposure); the re-check found the result unchanged on lattices two and three times finer. With the restaurant-calibrated kernel M2 predicts strong desk-scale clustering (median $z \approx 11$), against $z = -0.64$ observed. Integrated over the posterior by quadrature, the probability of a z as small as observed is 0.007 when posterior draws are weighted equally and 0.013 when each draw is weighted by its probability of producing an outbreak of the observed size; it depends on the arbitrary upper limit of the prior on $\log D$ (0.004 and 0.007 for $10^3 \text{ m}^2 \text{ h}^{-1}$; 0.013 and 0.024 for 10^6) and stays below 0.025 throughout. The observation is outside the central 95 % interval under either weighting (Fig. S11c). These probabilities were computed in the re-check with its own propagator, a mode sum that shares the numerical defect described above, and were not recomputed with the corrected propagator. The probability of a z as small as observed is 0 at every coefficient below $5.6 \text{ m}^2 \text{ h}^{-1}$, including those at or below $0.1 \text{ m}^2 \text{ h}^{-1}$ (18 % of the restaurant posterior) where the defect acts, so the equally weighted value is unaffected. The defect can act only through the outbreak-size weights of the second value, in either direction; with zero weight at coefficients up to $0.1 \text{ m}^2 \text{ h}^{-1}$ that value would be 0.015 (0.014 with half the weight), still below 0.025, and the failure stands. The Monte Carlo estimate of the protocol, from 40 posterior draws, had given 0.017 and 0.031. M0 is compatible ($P = 0.26$). The contrast between the wings is not counted for or against any model: the wings are separate rooms, and a lattice with a wall reproduces any contrast if the coupling through the door is free.

Pooled test and what it shows. The pooled score of M2 is $\Delta = +2.83$ (exact one-sided $p = 2.5 \times 10^{-4}$ under M0), so A1 is passed (+3.87 with the frozen propagator). It rests on the flight: without it $\Delta = -3.19$ and A1 fails; without the classroom $\Delta = +2.41$ ($p = 0.0024$).

S7.4 The model with same-site transmission

Calibrated on the trains, M1 has $D_p = 0.50 \text{ m}^2 \text{ h}^{-1}$ (0.31–0.79): a seated passenger changes site a few times per hour; the classroom takes its mobility from the restaurant posterior, which is wide (draws from 0.02 to $15 \text{ m}^2 \text{ h}^{-1}$, median 0.2). At these mobilities the mean number of cases stays below the observed number in three of the four held-out events, however large the transmission rate, because the attack rate saturates at the probability that a susceptible ever shares a site with the index case (Table S21). The mean number of cases is at most about 9 on the bus, where 23 were observed, about 6 in the business cabin (12 observed) and about 1 in the classroom at the posterior mode (12 observed). The observed total itself is reached with probability 0.0002 on the bus, 0.025 on the flight and 0.081 in the classroom (averaged over the posterior draws; on the flight the largest simulated total is 19, in the classroom six draws give the observed total a positive probability, three of them 0.55 to 0.61). Under the rule adopted when this was found

Table S21: Reach ceiling of M1 in the exact simulator of Section 6 (same-site rule, only the index case infectious, 24 draws of D_p from the calibrated posterior). Third column: number of draws in which the mean number of cases can reach the observed number for some transmission rate. Fourth column: for the other draws, mean number of cases as the transmission rate tends to infinity (median and range over draws). Last column: probability that the number of cases reaches the observed number, averaged over the 24 draws, each at the largest transmission rate of its search (or at the rate at which its mean reaches the observed number).

Event	Cases / exposed	Draws reaching	Largest mean total	$\mathbb{P}(\text{total} \geq \text{observed})$
Zhejiang bus	23/67	0 of 24	8.9 (7.5–12.2)	0.0002
Hunan coach	7/45	21 of 24	6.8 (6.8–6.9)	0.57
Flight VN54	12/20	0 of 24	6.3 (3.9–8.2)	0.025
Marin classroom	12/24	2 of 24	1.1 (0.0–11.9)	0.081

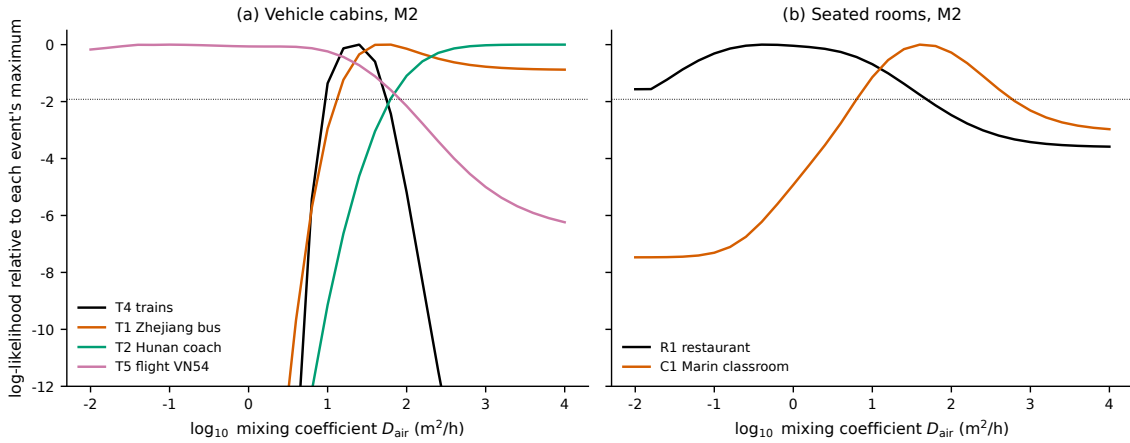


Figure S14: Log-likelihood of the mixing coefficient D_{air} of M2 for each event taken alone, relative to its maximum: (a) vehicle cabins, (b) seated rooms. The horizontal line at -1.92 bounds the 95 % likelihood interval of each event. For the hold-out events the likelihood is that of the near/far split given the total.

(post-hoc log, item P4: a total reached with probability below 0.01 counts as a failure), only the bus total is unattainable. A separately written simulator gives 9.1 for the bus, with no replicate reaching 23. Where the total can be reached, the pattern is wrong: on the coach M1 predicts a ratio of 34 ($p = 0.0002$), and on the flight a pattern that is too flat ($p = 0.009$). Its pooled score is -2.94 (Table 7).

S7.5 Mixing coefficients and levels

Fitted alone, the vehicle events of the first test ask for mixing coefficients that differ by orders of magnitude (Fig. S14): $0.06\text{--}0.1\text{ m}^2\text{ h}^{-1}$ for the flight, 25 for the trains, 63 for the bus, and at least $6,300\text{ m}^2\text{ h}^{-1}$ for the coach, whose likelihood is flat up to the end of the grid ($10^4\text{ m}^2\text{ h}^{-1}$). A common value is rejected: the likelihood ratio is 9.52 on 3 degrees of freedom (9.56 in the re-check), with its maximum at $40\text{ m}^2\text{ h}^{-1}$, and $p = 0.023$ from the chi-square reference. That p is approximate: the coach's maximum lies on the edge of the grid and each event has few cases, so the regularity conditions of the reference distribution are not met. This analysis was added after the primary results were seen. The protocol expected the measured ventilation rate to explain which cabins show a gradient; it does not explain the coach. The coefficient calibrated in one setting therefore cannot be assumed to transfer to another.

One infectiousness parameter is fitted per event, in both tests, so no attack rate in this section is a prediction. The second test conditions on the event totals and has no level check; the following numbers are those of the first. The implied emission rates span a factor of 76

Table S22: Emission rate implied by each event, in quanta per hour at a breathing rate of $0.5 \text{ m}^3 \text{ h}^{-1}$. Upper block: well-mixed scaling (shared by M0 and M2 for a group spread over the room), with the exact binomial 95 % interval. Lower block: emission that M2 needs in the hold-out events (median and 5–95 % range over the calibrated posterior and the ventilation prior; for the classroom with the round-off-safe propagator). Λ : prior median (upper block).

Event	Cases / exposed	Duration (h)	$\Lambda \text{ (h}^{-1}\text{)}$	Implied emission
Skagit Valley choir [21]	52/60 ^a	2.5	1.6	2,734 (1,904–3,832)
Jeonju restaurant [38]	2/13	0.35	1.2	1,921 (223–6,969)
Zurich office [74]	8/12	11	1.9	768 (300–1,614)
Cheonan fitness classes [29]	57/217 ^d	1.7	2.5	283 (214–367)
Hunan minibuss [55]	2/17 ^e	1.0	10.5	63 (7–229)
Hunan coach [55]	7/46 ^e	3.3	5.7	36 (14–74)
Trains (mean per index case)	230/72,692	2.1	6–21	1.3 (0.8–2.3) ^b
Zhejiang bus (M2)	23/67	1.7		132 (103–231)
Hunan coach (M2)	7/45 ^e	3.3		46 (34–67)
Flight VN54 (M2)	12/20	10		1,594 (435–11,823)
Marin classroom (M2)	12/24	13 ^c		8×10^7 ($514-9 \times 10^{17}$)

^a Includes 20 probable cases; 1,034 quanta per hour with the 32 confirmed cases only. At a breathing rate of $1.0 \text{ m}^3 \text{ h}^{-1}$, the middle of the range 0.65–1.38 sampled by Miller et al. [47], the value is about 1,370 quanta per hour; they report 970 ± 390 for the same event, with lower loss rates. ^b 90 % posterior interval. ^c Assumed (two school days of 6.5 hours); the source does not report hours. ^d Not a single-index event: eight infected instructors taught classes in 12 facilities; 57 is derived from the reported attack rate (26.3 % of 217), and the emission is computed as if for one index case in one room of 180 m^3 . ^e Counts of Ou et al. [55]: 46 persons on the coach including the driver in their dose table, 45 passengers with a seat in their stratum table (Luo et al. [45] report 7/48); 17 persons in the minibuss (Luo et al. [45]: 2/12).

across five single-index events and one cluster with several instructors, from 36 quanta per hour on the coach to 2,734 at the choir rehearsal (a factor of 53 if only the confirmed choir cases are counted), and the train cohort, which is not selected for large outbreaks, has a mean of 1.3 (0.8–2.3) (Table S22). Published outbreaks are selected for being large; their level cannot be predicted from a typical index case. For the one event that M2 reproduces, the flight, the emission it needs is about 1,600 quanta per hour (435–11,800); for the classroom the median is 8×10^7 . Two level checks have a defined reference. The same index case rode the Hunan coach and then a minibuss: fitted on the coach, M2 predicts 1.3 cases among the 17 people in the minibuss (90 % interval 0–4); 2 were observed. Anchored on the train mean, M2 predicts 2.1 cases (0–5) among 728 contacts in masked Utah classrooms [23] with an assumed mask factor of 0.35; 5 were observed. Both checks pass, and neither is sharp: the second fails at a mask factor of 0.2 (predicted 1.2, interval 0–3), inside the pre-specified range 0.2–0.6. No negative control fails (Table S18), but the criterion has almost no power: the premium-economy cabin of the flight (0 of 35 infected) passes although the 95th percentile of the predicted attack rate, 18.6 %, is more than twice the one-sided 95 % upper bound of the observation, 8.2 %.

S7.6 Outcome of the first test

Table S23 applies the rule of the first test. The outcome is W4 for both models: M1 fails the pooled test and three single events; M2 passes the pooled test and fails two adequacy tests, the coach and the call centre. In the re-check the calibration, the hold-out predictions and the call-centre test were re-implemented and the counts retyped from the sources; the call-centre probabilities by quadrature and the comparison with a constant ratio come from it. Its propagator shared the numerical defect of the classroom, whose correction changes no verdict. The second test has the outcome V5 for M2 and for M1 (Section 7.5 of the main text); in its re-check every number that decides the outcome was reproduced, the held-out seat maps were re-read against the published figures, and the comparisons that qualify it were added. Fig. S21 shows all events

Table S23: First test: decision rule applied to both models, with the corrections of the re-check and the numerical correction of the classroom (Section 7.4). The call-centre probabilities are given for the two weightings of posterior draws (by outbreak probability; equal); those for M1 are from the 40 posterior draws of the protocol and were not recomputed by quadrature. Outcome W4: two or more adequacy failures; no general statement of agreement is licensed.

Criterion	M1 (same-site)	M2 (diffusing aerosol)
A1 pooled test against M0	$\Delta = -2.94$: fail	$\Delta = +2.83$, $p = 2.5 \times 10^{-4}$: pass (-3.19 without the flight)
A2 single events	fail: bus (total unattainable), coach, flight	fail: coach
A2 call-centre join count	inside ($P = 0.056$; 0.040)	outside ($P = 0.013$; 0.007)
A3 level checks	minibus inside; classrooms outside	minibus inside; classrooms inside at mask factor 0.35, outside at 0.2
A4 negative controls	none fails	none fails (criterion nearly powerless)
Outcome	W4	W4

together.

What these data cannot decide. (i) Whether the same-row excess in the train table is air or companions. The choice decides two verdicts of the first test: with the adjacent-seat cell included in the calibration the bus is rejected as well and the pooled score is -4.9 ; with the whole index row excluded nothing fails and it is $+4.9$. (ii) The kernel in rooms: the restaurant has two to five informative cases, and the two ward datasets of the second dataset contradict each other. (iii) Anything about level. (iv) The definition of near: on aircraft the comparison with the constant ratio changes sign with the near zone and the strata (Section 7.5), and the two-row zone is itself a rule derived from earlier outbreaks, one of them a calibration event here. (v) The models ignore documented features of the events: the bus passengers attended a 150-minute event together, coach passengers boarded and left at different stops, households sat together on the flights, and most passengers of one flight were never tested. Seat maps were transcribed by eye from published figures; those of the held-out events were re-read in the re-check and found to match, two calibration maps were not re-read seat by seat.

S8 Protocol of the first outbreak test, sensitivity analyses and post-hoc log

This section documents the first test of Section 7 of the main text (the second test and the further analyses are documented in Section S10): what was fixed before any model was run, the decision rule and its power, the sensitivity analyses, and every choice made after results had been seen. The protocol, its power addendum, the freeze record and the post-hoc log are released with the code as redacted copies: passages that do not concern the analyses of this article are omitted or reworded, and identifiers are renamed as in the rest of the repository; the repository lists every changed line (`code/REDACTIONS.md`). The SHA-256 hashes of the freeze records refer to the unredacted originals, which are kept by the author and can be provided to the editor and referees; the timing evidence, self-attested in any case, can therefore not be checked from the repository alone. The protocol file is named `PREREGISTRATION.md` and describes itself as a pre-registration; it was not lodged with any registry, and in this article it is called the protocol and its contents pre-specified. ‘The re-check’ denotes the second pass of Section 1: a re-implementation of the whole analysis with separately written code, within the same project and on the same machine; where it corrected a result, the corrected result is the one reported. Where the frozen protocol texts speak of a ‘verifier’, they mean this same-project re-check, not an independent party, and

by ‘the analyst’ they mean the first analysis (both passes were made by the author).

S8.1 What was frozen, and the evidence for when

The protocol was frozen on 1 October 2026 at 05:12:03 (British Summer Time, as are all times below). The freeze record holds the SHA-256 hashes of six files: the protocol text (hash beginning 632db9b7), the modules that define the event geometries, the lattice kernels, the statistics and the interface to the exact simulator, and the file of hold-out stratum counts. Recomputed for this article on the unredacted originals, all six hashes match. These files fix the observation model of every event (Table S25), the three models (Table 6), the split into calibration and hold-out records, the estimation procedure, the metrics, the decision rule (Table S26) and twelve sensitivity analyses. Table S24 lists how they depart from the protocol first proposed with the dataset, which is released as a redacted copy (`code/bsc_outbreaks/PROTOCOL_PROPOSAL.md`).

The protocol text was last modified at 05:04:36. The first calibration run started at 05:13:30. The predictive distributions of the hold-out events and the power of the rule were stored at 05:15:43 and 05:15:46, by a script that does not read the hold-out counts, and the first comparison with those counts was stored at 05:19:04. The intervals are minutes because the protocol, the code and the runs were prepared in one continuous stretch of work; the re-check (Section 1) was a separate, later pass.

Four limits apply. (i) The protocol is *data-aware* and only *model-blind*: the outbreak counts had been read before the protocol was written, and the strata, the ventilation priors and the exclusion of the adjacent-seat cell were chosen with that knowledge. (ii) The evidence for the order of events is file-system metadata alone. There is no version-control history, and the freeze record was itself rewritten at 05:29:32, when the hashes were re-checked, so its time of freezing is self-reported. (iii) The code that computes the hold-out predictions and applies the rule was written after the calibration results had been seen and is not among the hashed files; one module was edited at 05:21:59, after the hold-out counts had been read. The re-check re-ran the current code and recovered the stored predictive distributions exactly. (iv) Two details differ from the protocol text: the p -value of A1 is computed by exact enumeration, not from 20,000 Monte Carlo replicates, and the predictive distributions average 300 draws where the text says 400.

S8.2 Decision rule and its power

The rule of Table S26 replaced a criterion that any constant near/far ratio between 1.16 and 3.19 would have passed (amendment V4). Its power is in Table S27(a). If the calibrated M2 were true it would pass A1 with probability 0.997, and A1 with no adequacy failure with probability 0.87; if the events were well mixed, a spatial model would pass A1 with probability at most 0.002. The re-check reproduced these three figures from its own predictive distributions (0.998, 0.877, 0.0019). With the corrected classroom distribution they are 0.998, 0.874 and 0.0017.

This power is against one alternative only, and the rule keeps the weakness it was meant to remove. Table S27(b) applies it to a generic model with a constant near/far risk ratio. With a ratio of 2, 2.5 or 3 that model passes A1 with $\Delta = +7.1$ to $+7.7$, more than the $+2.83$ of the calibrated M2 ($+3.87$ with the frozen code and $+3.76$ in the re-check’s own implementation, both with the numerical defect of the classroom), and has no single-event failure. Passing A1 therefore shows that a proximity gradient is present in these data; it does not show that the lattice kernel or its calibration predicts it. This comparison was made after unblinding and is not part of the protocol.

S8.3 Sensitivity analyses

The protocol lists twelve sensitivity analyses, never to be used to upgrade the primary outcome. Eleven bear on the pattern tests and, with the primary analysis, make the 12 variants of Table S28;

Table S24: Changes to the protocol first proposed with the dataset, all made before any model was run. V1–V7 were required by the re-check of the dataset (Section S6); D1–D6 were forced by gaps in the data. The analyses S-... are those of Table S28.

	Objection or gap	Amendment or consequence
V1	Restaurant counts are an upper bound: the first report states only that at least one member of each of two neighbouring families was infected at the lunch [44]	interval-censored likelihood (1–3 of 4 at one table, 1–2 of 7 at the other); the index case’s own table is excluded; S-R1-upper, S-R1-lower
V2	Train cohort: companions inflate the risk, most of all in the adjacent seat [27]	adjacent-seat cell excluded from the likelihood; no near-field factor transferred to hold-out events; S-adj, S-row, S-nu
V3	Three ‘grade A’ records do not meet the grade criteria (classroom: no hours, no room size; trains: pooled index cases, no denominators; choir: no positions)	classroom size, hours and ventilation enter as explicit priors, and its test conditions on the class total; trains used as a pooled cohort; choir used for level only
V4	The pattern criterion first proposed ($ z < 1.96$ on the log risk ratio) is passed by any constant near/far ratio between 1.16 and 3.19, and the level criterion by a constant hazard	rule replaced by A1–A4 (Table S26); power computed before unblinding (Table S27)
V5	The train table prints percentages and intervals but no cell denominators	cell counts reconstructed (230/72,692 against the reported 234/72,093), binomial likelihood; S-logit
V6	‘Three regimes’ of proximity gradient are not established (heterogeneity $p = 0.29$)	no analysis assumes them
V7	The contrast between the wings of the call centre depends on ten unmapped cases	not used as a test; the join count inside the north wing is used
D1	Seat maps of the bus, the coach, the minibus, the aircraft cabin, the classroom and the restaurant were not digitised	published stratum sizes and index seat are used; the unknown occupied seats inside each stratum are drawn at random and averaged over (300 configurations)
D2	Meat-processing plant [20]: appendix counts not transcribed	not evaluated
D3	Jerusalem school [64]: needs coupled rooms; no per-class denominators	not evaluated
D4	Sydney church [31]: no room dimensions	excluded from the level checks
D5	A square lattice cannot place one seat per site in 3+2 seating	rectangular lattice with one site per seat position and hop rates D/a_x^2 , D/a_y^2
D6	Index seat on the flight not in the stored text of the source	uniform over the 28 business-class seats; S-5K

the twelfth concerns the choir (Section S8.5). The pooled test A1 is passed in 11 variants. It fails when the adjacent-seat cell of the train table is put back into the calibration (S-adj): the kernel steepens, the bus is rejected as well and Δ turns negative. The coach is rejected in 10 variants and passes in the two that flatten the train kernel, S-row and S-pitch. The classroom is rejected if the restaurant counts are taken at their upper bound; the flight is never rejected. The verdict thus turns on how the same-row cells of the train table are used, which is what the data cannot settle: companions and air produce the same excess. M1 in its closed form (unchanged by the correction) fails A1 in every variant except S-row (Δ from -48.2 to -0.25 ; $+2.63$ in S-row, where it fails on the flight). Three checks added during the re-check, not pre-specified, change no verdict: the coach stays rejected when the train likelihood is tempered for its over-dispersion ($p = 0.026$) or its ventilation rate is cut to a third ($p = 0.040$), and the flight stays compatible when the ventilation priors are scaled by $1/3$ or 3 . With the corrected propagator the classroom stays compatible when its ventilation prior is scaled by $1/3$ ($p = 0.11$) and is rejected when it is scaled by 3 ($p = 0.045$); the pooled Δ is $+2.79$ and $+1.98$. Monte Carlo error is small except in the classroom: over five other seeds the predictive p of M2 ranges over 0.37 – 0.40 (bus), 0.0054 – 0.0089 (coach), 0.60 – 0.61 (flight) and 0.050 – 0.097 (classroom, close to the limit 0.05), and Δ over $+2.64$ to $+3.50$.

Table S25: Observation model of each event, as frozen. Lattices have reflecting walls and one site per seat position; pitches are across $\times$ along the cabin. Strata are those of the sources, with their sizes in brackets. The row pitches of the trains and the flight, the width of the bus, the classroom plan and hours, and three ceiling heights are assumptions of the protocol, not data.

Event and role	Lattice	Index case	Statistic	Duration
Trains [27]; calibration, vehicles	17×6 sites (3 seats, aisle, 2 seats); 0.5×1.0 m	every seat in turn	attack rate in 23 seat-offset cells, model averaged over all seat pairs; 22 cells in the likelihood	gamma, mean 2.1 h, SD 1.8 h
Restaurant [42, 44]; calibration, rooms	17×8 sites, 1 m; height 3.14 m; 18 tables on a 6×3 grid	index table	censored counts at the two neighbouring tables, zero at the others; remote layout averaged over	23–82 min, per table
Bus [59]; hold-out	15×6 sites (3 seats, aisle, 2 seats); 0.42×0.75 m; height 2.0 m	row 8, middle of the 3-seat side	rows 5–11 (33) against other rows (34)	2×50 min
Coach [55]; hold-out	13×5 sites; 0.5×0.877 m; 60.42 m^3	seat 12D	rear rows 8–13 (19) against front rows 1–7 (26)	200 min
Flight [35]; hold-out	business cabin 7×6 sites, 0.9×1.1 m, inside a 46-row cabin	uniform over 28 seats	the 12 occupied seats nearest the index against the other 8	10 h
Classroom [39]; hold-out	11×13 sites, 1 m; 25 desks at 2-site pitch	teacher, front centre	front two rows (10) against back three (14)	13 h
Call centre [56]; hold-out	137 digitised desks of the north wing, each mapped to the nearest lattice site (113 distinct sites); desk pitch uniform on 1.0–1.6 m	uniform over desks	join-count z of the final pattern of a chain of generations scaled to a mean of 79 cases; outbreaks of 69–89 cases kept	steady state
Minibus [55]; level, same index as the coach	6×4 sites; 21.69 m^3	row 4	cases among 17 exposed	60 min
Choir, office, restaurant, fitness classes [21, 29, 38, 74]; level	volume only: 810; 1,890; 309; 180 m^3	– (fitness classes: eight instructors, treated as one index)	implied emission from 52/60, 8/12, 2/13, 57/217 (57 derived from 26.3%)	2.5 h; 11 h; 21 min; 100 min
Utah classrooms [23]; level and control	volume log-uniform on $180\text{--}350 \text{ m}^3$	51 index cases	cases among 728 contacts, emission anchored on the train mean	6.5–13 h
Supermarket customers [67]; premium economy of the flight; one metro ride [14]; controls	$6,000 \text{ m}^3$; 30 sites of the cabin lattice; 148.5 m^3	–	predicted attack rate against the 95% upper bound of 0/8,224 and of 0/35; probability of five or more cases among 309 riders	27 min; 10 h; 20 min

Table S26: Decision rule, evaluated separately for M1 and M2. Δ is the sum over the four single events of the log-score difference to M0. An event is ‘reproduced’ if its two-sided predictive p is at least 0.05, and ‘discriminating’ if M0 has $p < 0.05$. Whatever the outcome, the sentence may name only settings for which a comparison was run, and no level agreement may be claimed beyond A3. The outcome reached is W4 for both models (Table S23).

<i>Criteria</i>		
A1	discrimination	$\Delta > 0$ with exact one-sided $p < 0.05$ under M0
A2	adequacy	$p \geq 0.05$ for each of bus, coach, flight and classroom, and the call-centre z inside the central 95 % predictive interval
A3	level checks	counts of the minibus and of the Utah classrooms inside their 90 % predictive intervals
A4	negative controls	no control fails; a control fails if the probability that the predicted attack rate exceeds the one-sided 95 % upper bound of the observation is above 0.5
<i>Outcomes (condition; sentence licensed)</i>		
W1	A1, A2 without failure, A3, A4, and power of A1 at least 0.8	‘good out-of-sample agreement with the spatial patterns of documented outbreaks’
W2	A1 and A4, at most one failure in A2	‘after calibration, with one infectiousness parameter per event, the model reproduces out-of-sample the near/far contrasts of [events]; it does not capture [event]’
W3	A2 with at most one failure, A1 fails	‘consistent with the documented spatial contrasts, but not distinguishable from a well-mixed model on these data’
W4	two or more failures in A2	‘reproduces [events] and fails [events]’; no general statement of agreement
W5	A4 fails, or every discriminating event is not reproduced	no statement of agreement

In the pre-specified leave-one-event-out analysis, D_{air} is refitted on the other events of the class and the left-out event is scored against M0. The difference is +23.9 for the trains, +5.50 for the flight, +0.76 for the bus and −3.86 for the coach. Taken alone the events prefer 0.1, 25, 63 and 6,300 m² h^{−1} (flight, trains, bus, coach; the coach likelihood is flat up to the end of the grid at 10⁴, and the re-check’s finer search puts the flight at 0.06). A common coefficient for the vehicles is rejected: the likelihood ratio is 9.52 on 3 degrees of freedom, $p = 0.023$ (9.56 and the same p in the re-check), with the common maximum at 40 m² h^{−1}. The p -values of this paragraph are approximate: they refer the statistic to a chi-square distribution, whose regularity conditions fail when an event’s maximum lies on the edge of the grid, as the coach’s does, and each event has few cases. This test was added after the primary results had been seen (entry P10 below). For the two rooms it gives 3.07 on 1 degree of freedom ($p = 0.080$; 3.12 with the frozen propagator); taken alone, the classroom prefers about 40 m² h^{−1}.

S8.4 Level checks and negative controls

Both A3 and A4 are formally met by M2 (Table S29), and neither carries much weight. The Utah check passes at the assumed mask factor of 0.35 with the observation at the edge of the interval, and fails at 0.2, inside the pre-specified range; the factor was chosen knowing the observed 5/728. A negative control fails only if more than half of the predictive mass lies above the 95 % bound of the observation, which almost nothing can do: the premium-economy cabin passes although the 95th percentile of its predicted attack rate, 18.6 %, is more than twice the bound of 8.2 %. The metro control rests on the absence of reported rail clusters, although exposure on public transport is rarely traced. The pre-specified dispersion test asks whether scaling by duration, volume and removal rate narrows the spread of implied infectiousness across four selected outbreaks (three single-index events and the Cheonan fitness classes, a cluster with eight infected instructors that

Table S27: (a) Power of the rule as computed before unblinding (frozen code), after calibration and before any script read the hold-out counts: probability of each result when the data are generated by the model in the column heading at its calibrated posterior (exact enumeration over the joint outcomes of the four single-event tests, event totals fixed; the call centre is not included). (b) The rule applied to the observed data for the calibrated M2 and, by the re-check after unblinding, for a model with no lattice and no calibration that assigns the same near/far risk ratio to every event: pooled score Δ , its one-sided p under M0, the two-sided predictive p of each event, and the events failing A2. Recomputed with the corrected propagator of Section 7.4, the entries of (a) change by at most 0.011: for M1, ‘A1 passes’ under a true M2 is 0.914; for M2, ‘A1 passes’ is 0.0017 under M0 and 0.998 under M2, and ‘A1 and no A2 failure’ under M0 is 0.0001.

(a) Judged model	Result	truth: M0	truth: M1	truth: M2
M1	A1 passes	$< 10^{-4}$	1.000	0.903
	no A2 failure	$< 10^{-4}$	0.906	0.043
	at most one A2 failure	0.0081	0.997	0.778
	A1 and no A2 failure	$< 10^{-4}$	0.906	0.043
M2	A1 passes	0.0019	1.000	0.997
	no A2 failure	0.0003	0.069	0.874
	at most one A2 failure	0.022	0.758	0.994
	A1 and no A2 failure	0.0002	0.069	0.874

(b) Model	Δ	p under M0	Bus	Coach	Flight	Classroom	Fails A2
M2 (calibrated)	+2.83	2.5×10^{-4}	0.38	0.0069	0.61	0.068	coach
constant ratio 1.5	+5.86	4×10^{-5}	1.00	0.69	0.021	0.21	flight
constant ratio 2	+7.63	4×10^{-5}	0.59	0.41	0.14	0.66	none
constant ratio 2.5	+7.74	4×10^{-5}	0.27	0.22	0.30	1.00	none
constant ratio 3	+7.09	4×10^{-5}	0.088	0.18	0.58	1.00	none
constant ratio 4	+4.75	4×10^{-5}	0.014	0.056	1.00	0.24	bus
constant ratio 5	+1.77	4×10^{-5}	0.003	0.030	1.00	0.073	bus, coach
constant ratio 8	−39.47	6×10^{-4}	< 0.001	0.006	1.00	< 0.001	bus, coach, classroom

is treated as one index case). The standard deviation of its logarithm is 0.88 with that scaling and 0.81 with a constant hazard per hour. The power of the test, stated before it was run, is 0.29: the result is uninformative.

S8.5 Post-hoc log, a superseded analysis and a further sensitivity analysis

Table S30 lists the log. The re-check overturned two of its entries; one further pre-specified sensitivity analysis is reported at the end of this section.

The 40-draw call-centre analysis. The Monte Carlo estimate of the protocol drew 40 values of D_{air} from the restaurant posterior and obtained $\mathbb{P}(z \leq z_{\text{obs}}) = 0.031$ or 0.017 according to the weighting (P5); the result was reported as borderline and the outcome for M2 as W2 ‘as run’ (P11). This was an artefact of the small number of draws: three of the 40 had $D_{\text{air}} > 100 \text{ m}^2 \text{ h}^{-1}$, where the posterior expects 1.2. Integrating $\mathbb{P}(z \leq z_{\text{obs}} | D_{\text{air}})$ over the same posterior on a grid of 25 values gives 0.007 with equal weight and 0.013 with pooled weight. The observation is outside the central 95% interval under both conventions, the call centre is a second adequacy failure of M2, and the outcome is W4 without any judgement. The value is sensitive to the arbitrary upper limit of the prior on $\log_{10} D_{\text{air}}$, because the restaurant likelihood is flat towards the well-mixed limit: 0.004 and 0.007 (equal and pooled weight) for a limit of $10^3 \text{ m}^2 \text{ h}^{-1}$, 0.013 and 0.024 for 10^6 , below 0.025 in each case. The probabilities for M1 (0.056 and 0.040) are those of the 40-draw estimate and were not recomputed; M1 fails on three single events in any case.

The choir with confirmed cases only. The twelfth pre-specified sensitivity analysis counts the 32 laboratory-confirmed cases of the choir rehearsal in place of the 52 that include 20 probable cases. It gives an implied emission of $1,034 \text{ quanta h}^{-1}$ against 2,734. With it the spread of

Table S28: The primary analysis and the eleven pre-specified sensitivity analyses of the pattern tests, for M2, computed with the round-off-safe propagator (Section 7.4; only the classroom and Δ differ from the frozen code). For each hold-out event: predicted risk ratio (ratio of the expected attack rates of the two strata) and two-sided predictive p . D_{air} is the mode of the recalibrated mixing coefficient in $\text{m}^2 \text{h}^{-1}$. Bold: the event is rejected ($p < 0.05$), an A2 failure. Δ : pooled score against M0; its one-sided p under M0 lies between 7×10^{-5} and 1.0×10^{-3} in every row, so A1 fails only where $\Delta < 0$.

Variant	Change	Bus	Coach	Flight	Classroom	Δ
primary	protocol as frozen; $D_{\text{air}} = 25$ (vehicles), 0.5 (rooms)	2.4; 0.38	11; 0.0069	3.6; 0.61	5.5; 0.068	+2.83
S-adj	adjacent-seat cell put back (23 cells); $D_{\text{air}} = 10$	5.4; 0.0084	51; 3.2×10^{-5}	5.9; 1.00	5.5; 0.068	-4.86
S-row	whole index row excluded (18 cells); $D_{\text{air}} = 100$	1.3; 0.65	2.6; 0.31	1.9; 0.12	5.5; 0.068	+4.91
S-logit	logit-normal likelihood on printed intervals (19 cells); $D_{\text{air}} = 16$	2.8; 0.25	15; 0.0024	4.1; 0.62	5.5; 0.068	+1.58
S-pitch	train row pitch 0.4 m (source) instead of 1.0 m; $D_{\text{air}} = 40$	1.6; 1.00	4.5; 0.075	2.5; 0.36	5.5; 0.068	+4.79
S-aniso	fitted seat-back factor on hops between rows ($\theta = 0.32$); $D_{\text{air}} = 40$	2.6; 0.28	13; 0.0051	3.0; 0.67	5.5; 0.068	+2.12
S-R1-upper	restaurant counts at their upper bound (3, 2); $D_{\text{air}} = 0.25$	2.4; 0.38	11; 0.0069	3.6; 0.61	6.4; 0.0051	+1.08
S-R1-lower	restaurant counts at their lower bound (1, 1); $D_{\text{air}} = 2.0$	2.4; 0.38	11; 0.0069	3.6; 0.61	5.1; 0.11	+3.34
S-5K	flight: index case in seat 5K (recalled, not verified)	2.4; 0.38	11; 0.0069	3.3; 0.60	5.5; 0.068	+2.70
S-Luo	coach: 150 min and 7/48 [45]	2.4; 0.38	11; 0.0065	3.6; 0.61	5.5; 0.068	+2.77
S-corner	classroom: teacher's desk in a front corner	2.4; 0.38	11; 0.0069	3.6; 0.61	2.6; 1.00	+6.11
S-nu	train adjacent-seat excess (factor 4.4) applied to adjacent seats of hold-out events	2.5; 0.36	12; 0.0046	3.6; 0.61	5.5; 0.068	+2.39

Table S29: Level checks (A3) and negative controls (A4). Prediction: mean number of cases with the 90% predictive interval (A3), or median predicted attack rate (A4). The mask factor multiplies the dose in the masked Utah classrooms; 0.35 is assumed, with a pre-specified range of 0.2–0.6. Last column: for A3, the probability of at least the five observed cases in the re-check's re-run; for A4, the probability that the prediction exceeds the bound, which must be above 0.5 for a failure.

Check	Model (mask factor)	Prediction	Verdict	
A3: coach $\rightarrow$ minibus, 17 exposed, 2 observed	M0	1.18 (0–3)	inside	
	M1	2.05 (0–5)	inside	
	M2	1.31 (0–4)	inside	
A3: trains $\rightarrow$ Utah classrooms, 728 contacts, 5 observed	M2, factor 0.35	2.10 (0–5)	inside	$\mathbb{P}(X \geq 5) = 0.092$
	M2, 0.2	1.17 (0–3)	outside	$\mathbb{P}(X \geq 5) = 0.016$
	M2, 0.6	3.51 (0–8)	inside	$\mathbb{P}(X \geq 5) = 0.312$
	M2, no masks	5.82 (1–12)	inside	$\mathbb{P}(X \geq 5) = 0.633$
	M1, factor 0.35	1.47 (0–4)	outside	
A4: supermarket customers, 0/8,224 (bound 0.0364%)	M0/M2	0.0011% (95th percentile 0.0059%)	pass	$\mathbb{P}(\text{exceed}) = 0.0002$
A4: masked Utah classrooms, 5/728 (bound 1.44%)	M0/M2, 0.35	0.29%	pass	$\mathbb{P}(\text{exceed}) = 0.0003$
A4: premium economy of the flight, 0/35 (bound 8.2%)	M2	0.31% (95th percentile 18.6%)	pass	$\mathbb{P}(\text{exceed}) = 0.16$
A4: one 20-minute metro ride, 309 other riders	M0/M2	0.032 expected cases	pass	0 of 20,000 rides with ≥ 5 cases

Table S30: The eleven entries of the post-hoc log: every choice made after a model had been run against data, with the time recorded in the log (self-reported). None was used to upgrade a conclusion. Last column: the effect stated in the log and, in italics, the status after the re-check.

	Time	Item	Effect
P1	05:12	The header of the protocol gives a writing window of 05:00–05:40; the file was frozen at 05:12:03 and not edited afterwards.	none
P2	05:17	Geometry of three negative controls, the premium-economy average and the prior of the minibus check were missing from the protocol; fixed in the power addendum before the corresponding scripts existed.	none (not data-driven)
P3	05:18	M1 in the exact simulator: the conditional distribution of near cases was first averaged with equal weight per posterior draw; corrected to weight each draw by its probability of producing the observed total. Found before the hold-out comparison was run.	M1 rows that use the simulator
P4	05:18	Unplanned outcome of a pre-specified check: at the calibrated mobility M1 cannot reach the observed totals of the bus, the flight and the classroom (Table S21). Rule added: an event whose total is reached with probability below 0.01 counts as an A2 failure; the pooled score is reported with the simulator and with the closed form.	stricter for M1
P5	05:26	Call centre: the protocol did not say how posterior draws are combined. Pooling the retained outbreaks gave $\mathbb{P}(z \leq z_{\text{obs}}) = 0.031$ for M2 (inside the central 95 % interval), equal weight per draw 0.017 (outside); both reported.	conservative reading adopted. <i>Superseded: 0.013 and 0.007 by quadrature, outside under both</i>
P6	05:22	The call-centre test uses the closed-form kernel of M1, which overstates its reach; the exact rule is more local and would predict at least as much clustering.	M1 only
P7	05:20	‘Model risk ratio’ is the ratio of expected stratum attack rates; its interval is that of the sample ratio.	presentation
P8	05:20	Simplifications to stay within the run-time budget, fixed before the outputs were inspected (coarser grids for S-aniso and for leave-one-out; 4,000 simulations for the Utah check; no occupancy factor in the minibus check of M1).	negligible
P9	05:21	The coach has 7 cases among 45 passengers in the pattern test and among 46 persons, with the driver, in the level check: two tables of the same source [55].	none
P10	05:24	Added after the primary results: five further seeds, a brute-force cross-check, the coefficient preferred by each event and the test of a common coefficient.	descriptive only
P11	05:28	Reading of the outcome for M2: W2 as run, W4 under the conservative reading of P5; the W4 sentence was recommended.	wording. <i>Superseded: W4 under either reading</i>

implied emission across the six events of the upper block of Table S22 (five single-index events and the Cheonan cluster) falls from 76-fold to 53-fold.

S8.6 Software verification

Before the freeze the code was tested on synthetic inputs only. The closed-form exposures of M1 and M2 agree with numerical quadrature of the matrix exponential of the lattice generator to 5×10^{-15} and 9×10^{-15} ; M2 with $D_{\text{air}} \rightarrow \infty$ equals M0 to 3×10^{-10} ; and the conditional distribution of near cases agrees with enumeration to 2×10^{-16} . The closed form of M1 was compared with the exact simulator for two walkers: it gives an infection probability of 0.0215 against 0.0209 ± 0.0006 at a small hazard, and 0.886 against 0.556 ± 0.002 at a hazard 100 times larger. It is exact only to first order, which is why the protocol required the simulator check that led to entry P4. After unblinding, the exposures of M2 on the actual configurations of the coach,

the flight and the classroom were recomputed by brute force (relative error at most 2×10^{-14}) and the conditional distributions by rejection sampling (largest difference 0.0012–0.0025, within Monte Carlo error). Apart from the re-run of the analysis’s own scripts (Section S8.1, item (iii)), the re-check shares no code with the analysis. It obtains the propagator by numerical eigendecomposition of the generator and reproduces the train log-likelihoods to three decimals, the hold-out p -values to Monte Carlo accuracy (pooled Δ between +3.76 and +3.79 against +3.87), the power table and, with its own simulator, the ceiling of M1 on the bus (9.1 cases against 8.9). Both implementations computed the classroom with a mode sum and shared its numerical defect, which neither the brute-force check above (errors relative to the largest exposure) nor this comparison could reveal; the corrected values are those of Section 7.4.

S9 The second outbreak test and the further analyses in detail

This section gives the details of the second outbreak test and of the analyses of contact records, tracer measurements and zone structure (Section 7 of the main text). Their protocols, decision rules, power and data are in Section S10.

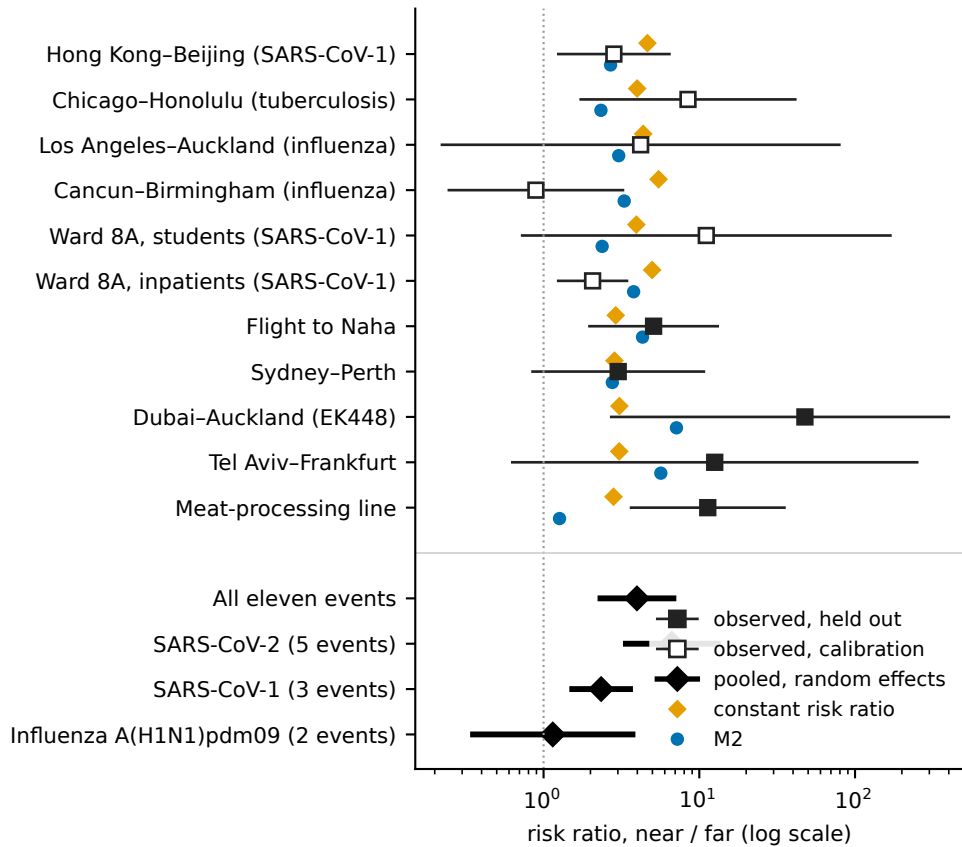


Figure S15: Near/far risk ratios of the eleven outbreaks of the second dataset, with 95 % intervals (a continuity correction of 0.5 where a stratum has no case), and random-effects pooled estimates. Near: within two rows of a source (aircraft), the index patient’s bay or cubicle (ward), within 8 m (processing line). Small symbols: ratio of the expected stratum attack rates under the constant-ratio model and under M2 (Section 7.2; for the six calibration events, from a fit to the other calibration events of the class).

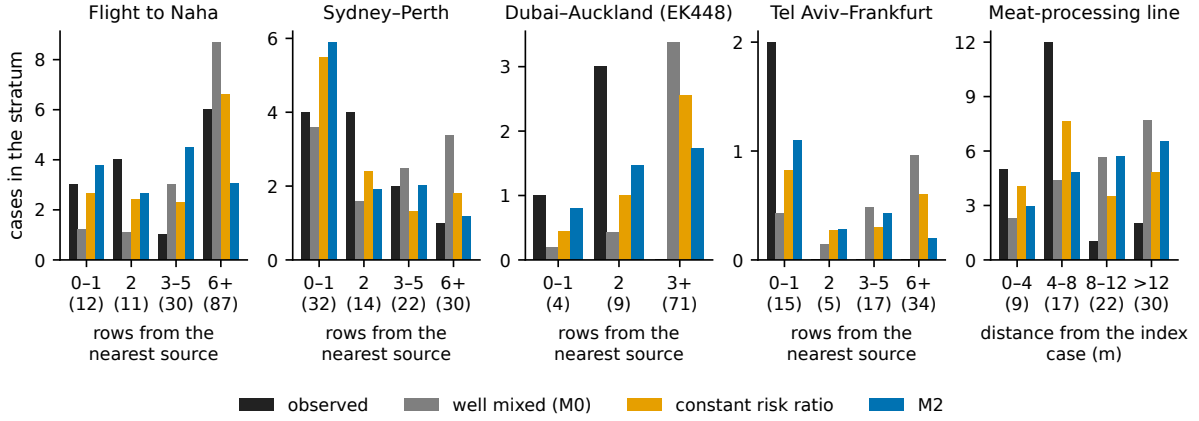


Figure S16: Second test: the five held-out SARS-CoV-2 events. Observed cases by distance stratum and the numbers expected, given the event total, under the well-mixed model, the constant-ratio model and M2, all calibrated on the six pre-2020 events. In brackets under each stratum: number of people at risk. On the four flights M2 and the constant ratio cannot be told apart; on the processing line the constant ratio uses a near zone of 8 m, which the authors of the source selected on these same data.

S9.1 Second outbreak test

Fig. S16 shows the observed and expected cases by distance stratum for the five held-out events. The two ward datasets contradict each other, and no model fits the inpatients in-sample. The room value is a correction made after the held-out events had been scored. The lattice propagator of the first test, copied unchanged, loses precision at receivers whose exposure is below its round-off error, which reaches about 10^{-10} of the largest exposure; the room posterior first computed (mode $0.02 \text{ m}^2 \text{ h}^{-1}$) came from two such noise values. A propagator written as a positive sum of image terms removes the defect. The correction changes no number for the flights, moves the pooled scores from +13.37 and -10.13 to the values reported below, and leaves the outcome of the rule unchanged; the re-check confirmed it with a third propagator and found the restaurant calibration of the first test unaffected (Section S10.2). The hold-out prediction for the classroom of the first test was affected and is reported corrected (Section 7.4).

It is positive in each of the 15 pre-specified sensitivity variants (+10.3 to +18.1). These variants were computed with the room kernel as first frozen, before the numerical correction (primary values +13.37 and -10.13 with that kernel); the flight parts are unaffected, and the pooled values would move by about +1.

It is significantly in favour of the constant ratio in the three variants that change the outcome definition of the flights: -2.52 (mirror $p = 0.013$) with the confirmed and probable cases of the flight to Naha, -1.90 ($p = 0.026$) with flight-associated cases only on the flight from Sydney to Perth, and -3.42 ($p = 0.005$) with all four changes of outcome definition. It is significantly in favour of M2, +1.21 ($p = 0.044$), when the members of two households seated together are counted once. With the published ratio 2.4 [24] in place of the calibrated one M2 is ahead on the flights (+1.18, $p = 0.028$) and behind over the five events (-4.44, mirror $p = 0.002$, with the room kernel as first frozen; -4.21, mirror $p = 0.002$, with the re-check's propagator). Comparisons added during the re-check give +2.4 ($p = 0.003$) with two strata in place of four, and +7.8 and +2.2 with a near zone of one and of five rows. On aircraft M2 is therefore not shown to be better than the two-row rule, nor shown to be worse; the sign is not determined.

On the processing line, $p = 1.4 \times 10^{-5}$: the attack rate is flat out to 8 m (5 of 9, 12 of 17) and then drops (1 of 22, 2 of 30), whereas the ward-calibrated kernel is almost well mixed and expects 2.9, 4.8, 5.7 and 6.5 cases. This is mostly a failure of the transfer from the ward to the line; at the coefficient that the line itself prefers ($40 \text{ m}^2 \text{ h}^{-1}$) the log score of M2 is -9.5, still below the -6.86 of the constant ratio.

Secondary analyses of the protocol. None changes the outcome, and two bear on what follows. (i) The kernel of the first test, calibrated on the train cohort ($D_{\text{air}} \approx 20 \text{ m}^2 \text{ h}^{-1}$), was applied to the eight flights with no parameter fitted on any of them. It beats the well-mixed model ($\Delta_0 = +7.56$) and is rejected on four flights ($p = 0.020, 0.004, 0.0002$ and 0.015): it is too steep for aircraft, although it reproduces the two flights whose cases sat next to the sources ($p = 0.72$ and 1.0). (ii) With the roles exchanged, calibrating on the SARS-CoV-2 events and predicting the earlier ones, M2 beats the constant ratio on the four pre-2020 flights ($\Delta_C = +3.58, p = 0.004$), because the ratio learned on the 2020 flights, about 7, over-predicts the earlier gradients, above all on the influenza flight with none; in the ward it loses (-3.64) and fails on the inpatients ($p = 2 \times 10^{-6}$). Pooled, $\Delta_C = -0.06$. (iii) In the 15 sensitivity variants (outcome definitions, sets of sources, row pitch, cabin ventilation, households, ward calibration) Δ_C lies between -12.8 and -7.7 and the outcome is V5 in 12 and V3 ('better than well mixed, but a constant ratio predicts better') in three (kernel as first frozen, see above). (iv) Leaving one event out within each class, M2 and the constant ratio are not separated on the flights: $\Delta_C = -0.06$, negligible in size but improbable under either model (one-sided $p = 0.046$ under the constant ratio, mirror $p = 0.044$ under M2); formally the mirror criterion is met, but the value is about as improbable under the constant ratio, and the re-check gives $+0.02$; in the rooms M2 is no better than the well-mixed model ($\Delta_0 = -0.02$) and the constant ratio predicts better ($\Delta_C = -14.0$). (v) Scored seat by seat on the three held-out flights with seat maps, the constant ratio does best (-76.4), ahead of M2 (-78.6) and the well-mixed model (-83.5). (vi) The train kernel of the first test, applied to the rooms, is rejected on the ward inpatients ($p = 2 \times 10^{-5}$) and on the processing line ($p = 4 \times 10^{-4}$). (vii) The exploratory two-scale variant of the protocol (M2 plus a well-mixed fraction), evaluated by the same rule, has $\Delta_0 = +17.5$ and $\Delta_C = -5.95$ (mirror $p = 2 \times 10^{-4}$) and fails on the processing line: outcome V3.

In the second test M1 was evaluated in its closed form only, as a further column of the same rule. Its protocol fixed the reading in advance: the first test had shown with the exact simulator that the closed form overstates the reach of M1 (on the bus its log score differs from that of the simulated model by 9.9 units, -9.58 against $+0.30$ relative to M0; pooled -11.11 against -2.94), so a pass of the closed form would not rehabilitate M1 and a failure would confirm the first test. Against the well-mixed model it passes with $\Delta_0 = +15.45$ over the five held-out events, more than M2 ($+14.35$), largely through the processing line ($+9.11$ against $+4.16$); against the constant ratio its pooled score is $\Delta_C = -8.06$ (mirror $p = 1.5 \times 10^{-5}$), as for M2. It fails on three of the five held-out events (flight to Naha $p = 0.008$, flight EK448 $p = 0.012$, processing line $p = 0.004$), one more than M2, and on the four flights the constant ratio predicts better ($\Delta_C = -4.63$; a value this low has probability 0.002 under M1), where M2 ties. Its formal outcome is the same as that of M2, V5. Its gain over the well-mixed model is that of an approximation whose error on comparable seated events of the first test is of the same size, and by the protocol it does not bear on the same-site model; its failures are consistent with the first test. The same-site model is not supported by the first test, nor, as Section 7.7 shows, by recorded contacts.

The second test confirms this with a test that its protocol lists. For the three room events a common coefficient is rejected (likelihood ratio 17.4 on 2 degrees of freedom, approximate $p = 1.6 \times 10^{-4}$): the students in the ward prefer the lower end of the grid, the inpatients about $2,000 \text{ m}^2 \text{ h}^{-1}$ and the processing line $40 \text{ m}^2 \text{ h}^{-1}$ (16–126). For the eight flights the heterogeneity is weaker (15.2 on 7 degrees of freedom, approximate $p = 0.034$; common value $158 \text{ m}^2 \text{ h}^{-1}$, 79–316). These p -values, too, use the chi-square reference although one maximum lies on the edge of the grid and the events have 2 to 20 cases each; the per-event profiles (Table S36) stand regardless. By pathogen the flights prefer $158 \text{ m}^2 \text{ h}^{-1}$ for SARS-CoV-2 (63–501), 126 for SARS-CoV-1 (40–794), 16 for tuberculosis (no lower limit) and about 500 for influenza, where the gain over the largest coefficient of the grid ($10^4 \text{ m}^2 \text{ h}^{-1}$, close to well mixed) is 0.26 units of log-likelihood, that is, no detectable gradient (Table S36, Fig. S22). Together with the transfer of the train kernel, which fails on four of eight flights (Section 7.5), this leaves no setting class with a single coefficient,

Table S31: Contacts generated by each model against RFID contact records in six indoor settings (calibrated on the first half of the days, scored on the second half). Families: number of the seven families of contact statistics passed, per dataset, in the order office 2013, primary school, hospital ward, office 2015, high school, conference (six families apply at the conference, which has no groups). Scenarios: SIR scenarios passed, of 24 (four per dataset). Ratio: secondary cases of the index case on model contacts divided by those on the recorded contacts, range over datasets of the mean over scenarios. Last column: mean absolute error of the attack rate over the 24 scenarios. The first four rows are the pre-specified test; the others are not.

Model	Parameters	Families passed (of 7)	Scenarios	Ratio	Error
Well mixed	2	1, 2, 2, 1, 1, 2	0	1.25–1.79	0.195
Constant within/between-group ratio	3	2, 4, 3, 2, 2, 2	0	1.20–1.63	0.158
Lattice walk, homogeneous	2	1, 2, 2, 1, 1, 1	0	1.25–1.72	0.167
Lattice walk, two zones	4	2, 2, 2, 2, 2, 2	0	1.24–1.64	0.144
Static network of training-day pair rates ^a	–	–	12	–	–
Walk anchored to clustered home sites ^b	4	4, 3, 3, 4, 2, 1	2	1.13–1.31	0.041
with a slow seating zone ^b	5	5, 3, 3, 5, 2, 2	3	1.11–1.30	0.038
with a slow own seat ^b	5	5, 4, 3, 4, 4, 2	5	0.97–1.26	0.050

^a Added post hoc during the re-check: contact rate of each pair taken from the training days, no space; exploratory. ^b Developed on the first three datasets after the walk had failed on them; description after the fact (the confirmatory test on the last three datasets was uninformative).

and no transfer between classes. Section 7.8 compares the fitted coefficients with tracer fields, which correspond, summarised by a diffusion coefficient, to values of order 100 to 1,000 m² h^{−1} in vehicles: far above the vehicle value calibrated in the first test (25 m² h^{−1}); its room value, 0.5 m² h^{−1}, is a few hundred times below the 170 to 185 m² h^{−1} that a regression on two houses gives for rooms, for which no tracer measurement exists. The values of the second test are not far below them: the aircraft value, 126, lies inside the range measured in cabins (112 to 708), and the room value, 794, lies above the regression values. That does not make an outbreak-fitted coefficient a measurement of air mixing, since within the same test no common coefficient holds.

S9.2 Recorded contacts

Two things are scored on the test days. *Structure*: seven families of contact statistics (total contact time; contact durations; gaps between contacts and recurrence of pairs; number of partners; heterogeneity between people; repetition of pairs from one day to the next; share of contact within groups), each passed if the model lies within a factor 0.8 to 1.25 of the data, or within a stated absolute margin. *Epidemic consequence*: an SIR epidemic is run on the recorded temporal network of the test days and on the model-generated contacts, at the same total contact time, in four scenarios (nominal reproduction number 1.5 or 3; infectious period 1 or 4 days); a scenario is passed if the secondary cases of the index case agree within 10 % and the probability of a major outbreak and the mean attack rate within 0.05, and the epidemic criterion is met with three of four scenarios. The rule: the movement component is validated on a dataset if it meets the epidemic criterion and at least six of the seven families, and overall if it does so on two-thirds of the datasets; the outcome is called A0 if it neither does this nor meets the epidemic criterion where the well-mixed model fails. On synthetic data a true model passed in 20 of 20 replicates and a wrong one reached the per-dataset criterion in none of 44. The protocol recorded the expectation A0.

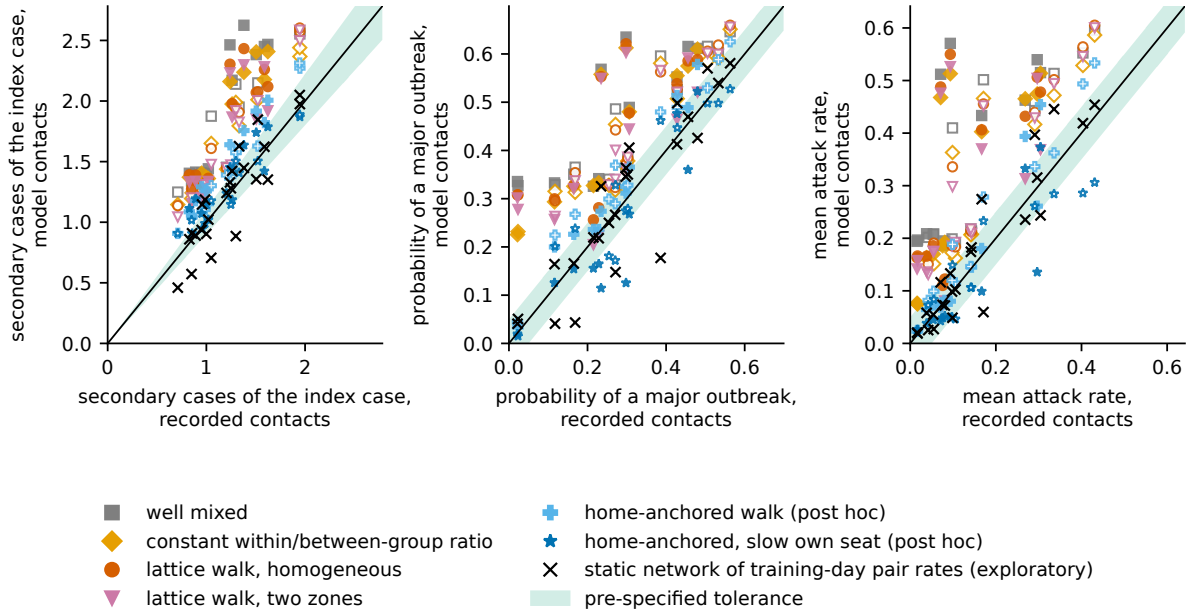


Figure S17: SIR outcomes on model-generated contacts against the same outcomes on the recorded contacts of the test days: six datasets, four scenarios each. Open symbols: the three development datasets; filled: the three confirmatory ones. Shaded: tolerance of the protocol. Points above the diagonal are over-predictions. The lattice walk, in both forms, lies with the two models without space; the walks anchored to home sites were developed after that failure, and the static network was added during the re-check.

Result: the walk fails on all six datasets. The homogeneous walk passes one or two of the seven families on each dataset, namely those it was calibrated on, the total contact time and the mean duration; the two-zone walk passes two on each (Table S31; Fig. S23). What it does not reproduce is who meets whom. People in the walk have too many partners on five of the six datasets: 6.0 per person and day against 4.6 in the office of 2013, 34 against 15 in the high school, 90 against 47 in the primary school. Contact within the recorded groups is 10 to 33 % of contact time in the walk and 57 to 94 % in the data. Long contacts, the burstiness of the gaps between contacts (0.08–0.25 against 0.36–0.61) and the repetition of the same pairs on the next day are missing as well. The real training days, used as a predictor of the test days, reproduce the burstiness on all six datasets, the long contacts on four (not in the office of 2013 and the hospital ward) and the repetition of pairs on three of the four datasets where it is defined (not in the hospital ward; it is undefined at the primary school and the conference).

The consequence for an epidemic is systematic. Because the walk spreads the same contact time over too many, too equal partners, the index case infects more people on the walk’s contacts than on the recorded ones in all 24 scenarios, by 15 to 86 %, and by 25 to 72 % on average per setting; the mean attack rate is too high by 0.10 to 0.29 (Fig. S17). No scenario is passed on any dataset. The total contact time of model and data agrees within 2.5 % in these runs, so the excess is not an artefact of the contact rate. The re-check reproduced the result with its own data loader, walk simulator and epidemic code, on files downloaded again from the source.

The two models without space do neither better nor worse: both pass no scenario, with ratios of 1.25 to 1.79 (well mixed) and 1.20 to 1.63 (constant group ratio). The outcome is A0, with the sentence that the protocol reserved for this case: the lattice walk adds nothing over a non-spatial contact model. The conclusion is, if anything, understated. A static network with no space and no walk, in which each pair keeps the contact rate it had on the training days, meets the epidemic criterion on three of the six datasets and passes 12 of the 24 scenarios, where neither pre-specified walk meets it on any dataset (the anchored walk with a slow own seat, designed

afterwards on the development datasets, meets it on one of them). This comparison was added during the re-check after the results were known, with a crude estimator whose contact time is 2 to 9 % low, so it is exploratory; and the tolerance is tight, since the real training days used as a predictor of the test days meet the criterion on four of the six datasets only. Neither point touches the result for the walk, whose error is several times the tolerance.

One further property is against the mass-action picture of the model. In the walk, as in the well-mixed model, the number of contacts grows as the square of the number of people present (fitted exponents 1.95 to 2.20). In the data the exponent is 0.45, 0.95, 0.76 and 0.81 in the two offices, the high school and the conference, and 2.8 in the hospital ward: in four of five settings a person’s contact rate does not grow with occupancy. This comparison is descriptive: occupancy varies with the time of day, so the exponent mixes density with schedule, and 20-minute intervals without a contact were left out of the fit, which biases the observed exponent on the sparse datasets.

Walkers anchored to home sites: neither supported nor refuted. The protocol provided for a second stage. On the three development datasets the smallest changes that repair the failed families were sought, keeping a Markov walk with same-site contact: a home site for each person with a drift towards it, homes of the same group clustered in space, and either a slow seating zone, which is the heterogeneous mobility D_x of the model, or a walker that is slow on its own seat only, which is not. The three variants, with their automatic fitting procedure, were frozen before the three confirmatory datasets were touched. On those datasets none meets the epidemic criterion (the variant with a slow own seat passes 2, 0 and 0 of 4 scenarios; the variant with a slow seating zone passes none of 12), and the formal outcome is again A0.

This outcome must not be read as a test that anchoring failed. In the re-check synthetic contact data were generated from the anchored models themselves and the frozen fitting procedure was applied to them: it failed the epidemic criterion in six of seven replicates, with degenerate fits in four. A procedure that cannot recover its own truth cannot refute a model, so the confirmatory stage is uninformative. What remains is a description made after the fact. Over all 24 scenarios the mean absolute error of the attack rate falls from 0.17 for the homogeneous walk to 0.04–0.05 for the anchored variants, and the error of the index case’s secondary cases by 47 to 70 %; two of the three variants still over-predict the secondary cases in all 24 scenarios, by 6 to 36 %. These figures use the recorded group labels and are not at matched contact time: the ratio of model to recorded contact time is 0.96 to 1.05 and 0.92 to 1.06 for the two variants with clustered homes, and 0.98 to 1.11 for the variant with a slow own seat except in the high school, where it is 2.34; the reduction of 70 % belongs to that variant and includes the high school. They indicate a direction, home anchoring with clustered homes, and are not a result; no variant reproduces burstiness or the repetition of pairs between days. A random walk in which people slow down near attractive individuals was proposed and compared with SocioPatterns records by Starnini et al. [61]; the anchored walks found here by trial are close to that idea, and what is added is the epidemic consequence and the comparison with models without space.

Scope. The result holds for one identification of the walk: a site is a contact range, steps are synchronous, and the lattice has as many sites as the contact rate requires. The calibrated hop probabilities, 0.20 to 0.42 per 20-second step, correspond to a mobility of about 220 to 450 sites² per day if a site is identified with a cell of Section 2: between $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$, which was not tested against contacts, and walking mobility. Other site sizes and the continuous-time walk were not tried. The size of the over-prediction depends on the design with a nominal reproduction number, which forces a high transmission probability per contact interval on sparse datasets; it is smallest in the primary school (ratios 1.15 to 1.34). The sensors record proximity, not position, so the occupancy of zones could not be tested, and neither could a measured mobility field.

Table S32: The kernel fixed by tracer measurements, with nothing fitted to outbreaks except one ε_k per event, on the stratum events of the first dataset. RR: predicted and observed near/far risk ratio; p : two-sided predictive p of the observed near count under the tracer kernel and under M0; last three columns: log score of the tracer kernel minus that of M0 and of the constant ratios 2 and 3.

Event	Near vs far	RR pred. / obs.	p	p , M0	−M0	−ratio 2	−ratio 3
Hunan coach	3/19 vs 4/26	1.29 / 1.03	1.00	1.00	−0.05	+0.42	+1.15
Zhejiang bus	14/33 vs 9/34	1.03 / 1.60	0.22	0.20	+0.09	−0.63	+0.70
Marin classroom	8/10 vs 4/14	1.77 / 2.80	0.42	0.036	+2.47	−0.34	−0.59
Flight VN54	11/12 vs 1/8	1.66 / 7.33	0.14	7.7×10^{-4}	+4.46	−0.10	−1.37
Restaurant ($K = 2$)	2/16 vs 0/63	1.81 / 19.24 ^a	0.095	0.039	+0.89	−0.15	−0.67
Pooled					+7.87	−0.80	−0.78

^a With a continuity correction (no case at the remote tables). K : number of cases counted as acquired at the two neighbouring tables; with the source’s own count, $K = 5$, the tracer kernel is rejected ($p = 0.002$).

S9.3 Tracer measurements

Measured mixing coefficients. If the non-local term of M2 describes the transport of aerosol, its mixing coefficient can be measured without any outbreak, by releasing a tracer and recording its concentration. Published measurements exist for four of the settings of Section 7: ethane tracer gas at eight seats of the Hunan coach itself, with a tracer-validated flow simulation for all seats [55]; fluorescent 1 μm particles released from a breathing mannequin in the forward cabin of a wide-body aircraft [36]; salt aerosol along the saloon of an inter-city rail carriage [76]; and tracer gas at seven tables of the Guangzhou restaurant, with the source’s flow simulation at the others [42]. For the classroom and the open-plan office no measurement exists, and the coefficient was taken from a regression of the turbulent diffusion coefficient on the air change rate measured in two houses [11], whose measured values are 3.6 to 47 $\text{m}^2 \text{h}^{-1}$ at below 0.2 to above 5 air changes per hour; it is extrapolated to rooms of several hundred cubic metres at 2 to 12 air changes per hour. The estimators were fixed before any coefficient was computed: a least-squares fit of the logarithm of the steady concentration of M2 to the logarithm of the tracer reading, on the lattice of the event.

In the three vehicles the measured fields, summarised by a diffusion coefficient, are of order 100 to 1,000 $\text{m}^2 \text{h}^{-1}$ (Fig. S18d): 1,259 in the coach (5th percentile of a bootstrap 708; the eight points are nearly flat and bound the coefficient from below only), 708 and 112 for the two releases in the cabin (geometric mean 282), and 891 and 355 for a release in the middle and at the end of the carriage. For the classroom and the office no tracer measurement exists, and no coefficient was derived from the restaurant’s readings, which enter as a kernel directly; the regression extrapolated to the volume and ventilation of the classroom and the office gives about 170 to 185 $\text{m}^2 \text{h}^{-1}$ (38 to 753). The coefficients fitted to outbreaks in the first test are 25 $\text{m}^2 \text{h}^{-1}$ for vehicles and 0.5 $\text{m}^2 \text{h}^{-1}$ for rooms, and the flight VN54 alone prefers 0.06 to 0.1. The measured vehicle values are thus 4.5 to about 50 times the vehicle value of the first test, and the extrapolated room values a few hundred times its room value. The second test is different: its aircraft coefficient, 126 $\text{m}^2 \text{h}^{-1}$ (52–438), lies inside the range measured in cabins, and its room coefficient, 794 $\text{m}^2 \text{h}^{-1}$ (302–4,390), lies above the extrapolated room values. Three reservations on the cabin value: the fit uses positive readings only, and estimators that keep the zero readings give 56 to 450 $\text{m}^2 \text{h}^{-1}$; the two fits move from 708 and 112 to 891 and 178 $\text{m}^2 \text{h}^{-1}$ if four additional sensors, one of them with an outlying reading, are left out; and the measured field is not diffusive (it is biased forwards and cut off aft of row 8), so that a single coefficient summarises it loosely. The decay lengths $\sqrt{D_{\text{air}}/\Lambda}$ are about 2 to 5 m in the cabin and 5 to 14 m in the carriage and the coach.

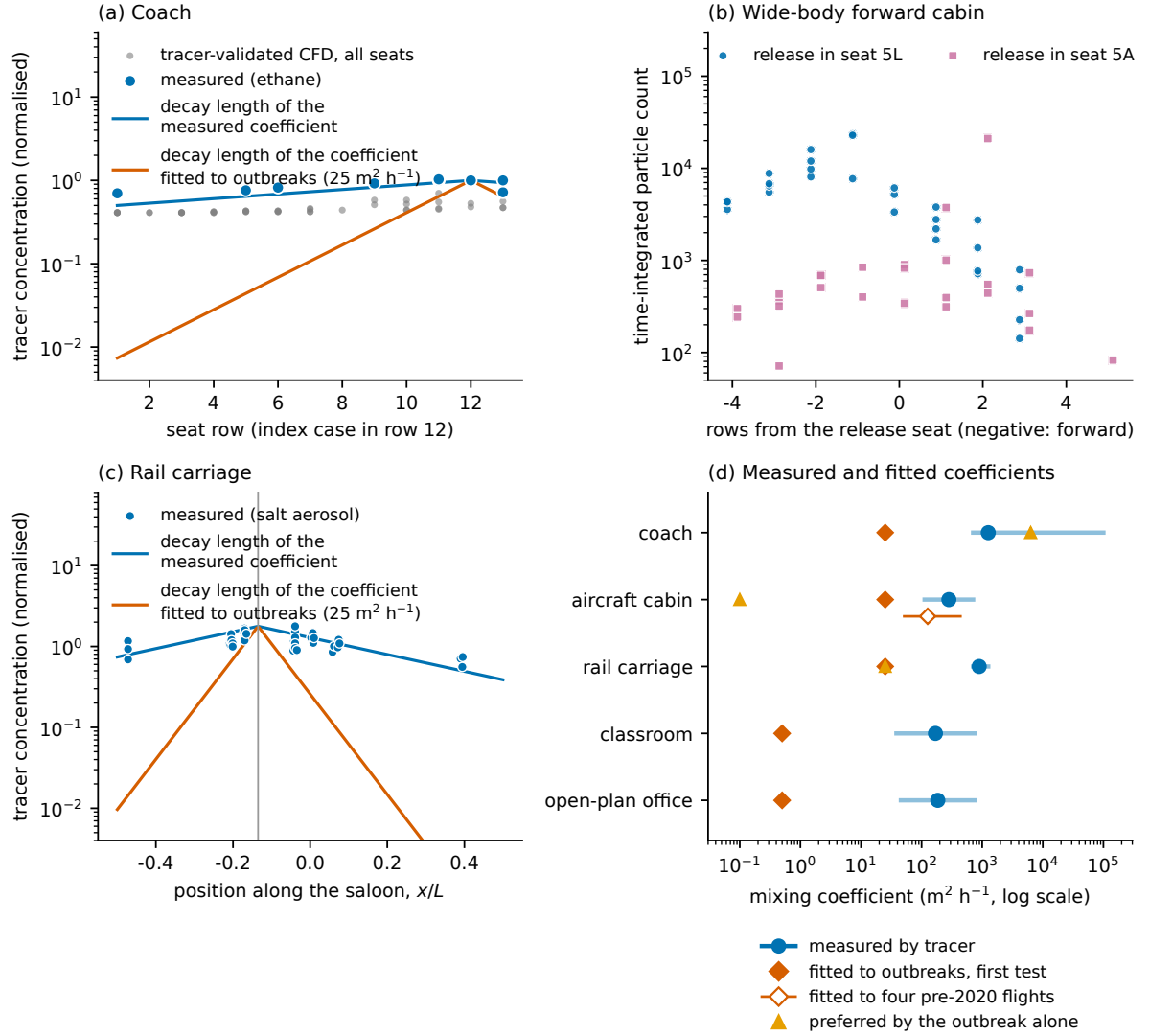


Figure S18: Tracer measurements. (a) Coach of the Hunan outbreak: ethane measured at eight seats and the tracer-validated flow simulation at all seats [55]. (b) Forward cabin of a wide-body aircraft: time-integrated particle counts by row for two releases, zero readings not shown [36]. (c) Rail carriage: salt aerosol along the saloon, release at the vertical line [76]; points digitised from the published figure. The lines in (a) and (c) are exponentials with the decay length $\sqrt{D_{\text{air}}/\Lambda}$ of the measured coefficient and of the $25 \text{ m}^2 \text{ h}^{-1}$ fitted to the train cohort; they illustrate the two length scales and are not the fitted lattice fields. (d) Mixing coefficients: measured (with range), fitted to outbreaks in the first test, fitted to the four pre-2020 flights of the second test (with 90% interval), and preferred by the outbreak of that setting taken alone.

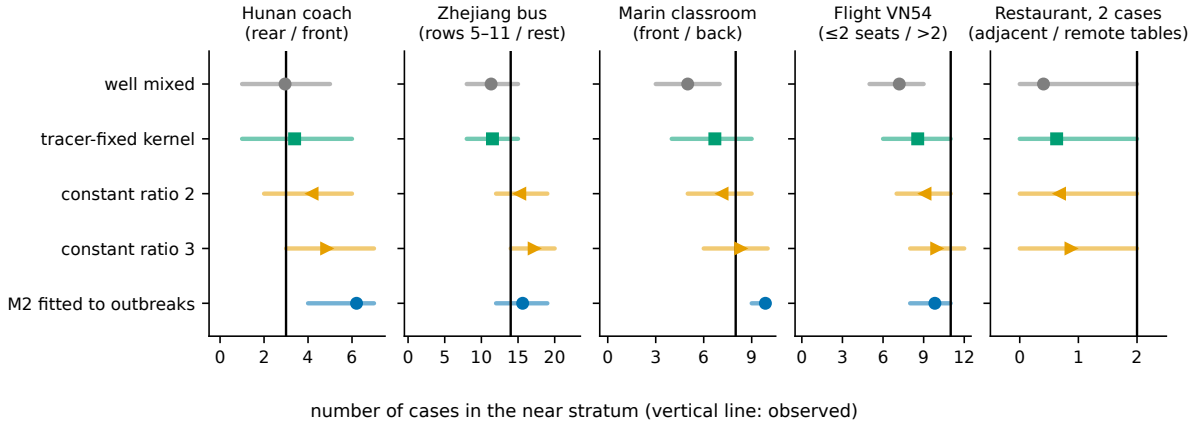


Figure S19: Number of cases in the near stratum, given the event total: mean and 95% predictive interval under the well-mixed model, the kernel fixed by tracer measurements, the constant ratios 2 and 3, and M2 with the coefficient fitted to outbreaks in the first test (at its posterior mode; the restaurant was a calibration event). The tracer kernel removes the over-prediction on the coach by coming close to the well-mixed model. The gradients of flight VN54 and of the restaurant are not reproduced: the formal non-rejection of the flight depends on zero sensor readings (see text), and the restaurant is shown at the lower bound of its case count.

Confronted with the outbreaks. With the kernel fixed in this way, the seven events of the first dataset were predicted with one free parameter each, the infectiousness (Table S32, Fig. S19). The trains and the restaurant, calibration records before, are test events here.

Coach and call centre: no longer rejected, and not informative. For the coach the measured kernel predicts a rear/front ratio of 1.29 where 1.03 was observed ($p = 1.00$); the kernel fitted to outbreaks had predicted 11. For the call centre it predicts little desk-scale clustering, and the probability of a join-count z as small as observed is 0.10 (Fig. S11c). In both cases the measured kernel is close to the well-mixed model, which fits as well or better ($p = 1.00$ and 0.26); for the call centre the kernel still predicts clustering of the wrong sign (median $z = +0.78$ against -0.64 observed). The two failures of Section 7.4 are removed only in the sense that the model has stopped predicting a gradient.

Bus: inconclusive. Predicted ratio 1.03 against 1.60 observed, $p = 0.22$; the well-mixed model has $p = 0.20$.

Classroom: the one gain. The room relation predicts a front/back ratio of 1.77 where 2.8 was observed ($p = 0.42$), and the well-mixed model is rejected ($p = 0.036$). The result holds when the coefficient is doubled or halved and under two other dose-response laws, and the implied emission, 317 quanta per hour (173–642), lies within the range of the single events of Table S22, unlike the median 8×10^7 that the first test needed. Constant ratios of 2 and 3 do as well, and the coefficient is extrapolated from two houses, with the teacher’s position assumed.

Flight VN54: not reproduced. The predicted ratio is 1.66 against 7.3 observed: with 11 of 12 near passengers infected and 1 of 8 far, the exponential dose-response law needs a dose ratio of about 19, and the measured kernels give 1.5 to 3.9. The formal test does not reject ($p = 0.14$), but only because the predictive distributions of two releases that contradict each other are averaged, and because three sensors that map onto far seats read exactly zero in one of them. With the zeros treated as missing, a variant listed in the protocol, $p = 0.003$; the flight is rejected in four of the six listed variants. The tracer tests were made in another airframe.

Restaurant: not reproduced at the reported count. The tracer concentration at the 13 remote tables was on average 0.55 of that at the index table (0.32 to 0.86); four of these values are measured and nine are the source’s flow simulation, the lowest among them. With two cases counted as acquired at the neighbouring tables the kernel is not rejected ($p = 0.095$); with the five that the source reports, against none among 63 remote patrons, it is ($p = 0.002$).

Trains: failure. The kernel measured in the carriage is nearly flat over three rows and expects 14.6 cases in the row of the index case where 50 were observed (deviance 112.8 on 21 degrees of freedom, $p < 0.0005$). Every model fails on this table: the well-mixed model (125.6), constant ratios of 2 and 3 for the index row (81.5 and 64.1) and the kernel that was fitted to the table in the first test (72.3).

Pooled. Over the five stratum events the tracer kernel scores +7.87 against the well-mixed model (exact one-sided $p = 1.4 \times 10^{-6}$; +3.41 without the flight) and -0.80 and -0.78 against the constant ratios 2 and 3, which themselves beat the well-mixed model by more (+8.67 and +8.65). The outcome of the rule is S4, one adequacy failure (the trains) and no gain over the generic gradient; on the defensible reading of the flight it is S5, two failures. The formal outcome S4 does not discriminate between the truths considered in the power analysis (Table S38, Section S10.4). A comparison added during the re-check, after the results were known, gives the probability of the pooled pattern alone (better than well mixed, not better than both constant ratios): 0.23 if the tracer kernel is true, 0.84 under a constant ratio of 2 and about 1.00 under a ratio of 3, so that the pooled evidence favours a generic ratio over the kernel by a factor of about four (3.7 to 4.4).

An exploratory extension to four flights. After these results were known, the same cabin coefficients (112 and $708 \text{ m}^2 \text{ h}^{-1}$, equal weights) were applied to the four held-out flights of Section 7.5. The extension has a protocol of its own, written with the near/far counts and the outcome of the second outbreak test known, and it reverses two exclusions of the main protocol; it is exploratory and takes no part in any decision. The tracer kernel is not rejected on any of the four flights ($p = 0.054, 0.26, 0.25$ and 1.00), scores +10.4 against the well-mixed model and +2.8 against a constant ratio of 2 ($p = 0.005$), and ties with a ratio of 3 (-0.16). The absence of a rejection is at the edge: with the geometric-mean coefficient that the main protocol specifies, the flight to Naha is rejected ($p = 0.044$), and each of the two coefficients taken alone fails one flight ($p = 0.025$ and 0.002). The tracer measurements were made in wide-body cabins, and two of the four flights were on Boeing 737 aircraft (Table S19), so the transfer of the coefficient is itself untested. What survives, as an exploratory result, is small: the two cabin coefficients (112 and $708 \text{ m}^2 \text{ h}^{-1}$), measured without any outbreak, beat the well-mixed model and tie with a constant ratio of 3.

Reading. Measured mixing is fast. A kernel with the measured coefficient is therefore close to well mixed in the coach and the carriage and mild in the rooms, and it cannot produce the steep gradients of the train cohort, of flight VN54 or of the restaurant. Those gradients need something other than diffusion at the scale of the enclosure: exposure within about a metre, companions, directed airflow. The tracer kernel is not validated as a spatial predictor; its one gain over the well-mixed model, the classroom, is matched by a constant ratio. Only the coach and the restaurant have a tracer test in the enclosure of the outbreak; the cabin, the carriage and the rooms rest on transfers that are themselves untested.

S9.4 Zone-level structure

The zone model takes these three shares as given and fits the overall rate β and two parameters of an external hazard from the city, in a daily chain-binomial model whose transmission kernel is the gamma density with mean 1.7 and standard deviation 1.0 days used by Endo et al. [13], chosen by them to give the mean serial interval of 2.2 days of influenza A(H3N2) [71]. Its competitors are fitted in the same way: a well-mixed school, independent classes, and a generic two-level model with one fitted ratio of own-class to rest-of-school risk, the analogue of the constant-ratio model. A model with three free shares serves as a ceiling. The rule has three tiers: (1) under two-fold cross-validation by school, the zone model has a higher held-out log-likelihood than the well-mixed school and than independent classes, with the lower 95 % bootstrap bound above

Table S33: Zone-level test: influenza in 29 primary schools, with mixing shares measured in another school. Differences are of held-out log-likelihood (two-fold cross-validation by school), with 95 % bootstrap intervals over schools. The last two rows are post hoc, added during the re-check.

Criterion	Result	Outcome
Tier 1: zone model minus well-mixed school	+798 (562 to 1,073)	met
Tier 1: zone model minus independent classes	+211 (110 to 340)	met
Shares own class / grade / school: measured; fitted	0.76 / 0.10 / 0.13; 0.66 / 0.10 / 0.24	
Tier 2a: equality of the shares (likelihood ratio, limit 5.99)	42.7	not met
Tier 2b: total-variation distance (limit 0.15)	0.103 (0.047 to 0.168)	met, not securely
Tier 3: between-class heterogeneity, observed and interval of the zone model	3.38; 2.65 to 3.61	met; independent classes also contain it
Zone model minus two-level model with one fitted ratio	+18.8 (−8.8 to +54.0)	not distinguishable
Zone model minus zone model with guessed shares 0.7 / 0.1 / 0.2	−19 (−38 to −3)	the guess is better
Tier 1 without the lower bound on the external hazard	+745 (531 to 999)	unchanged

zero; (2a) the freely fitted shares do not fit better than the measured ones (likelihood ratio at most 5.99), and (2b) they lie within a total-variation distance of 0.15 of them; (3) the observed heterogeneity of attack rates between classes lies inside the 95 % predictive interval of the zone model. The verdict ‘right structure within tolerance’ needs tiers 1, 2b and 3. On synthetic epidemics a true zone model earned it in 80 to 100 % of replicates, and a well-mixed school or independent classes in none. The direction of tier 1 was foreseeable: the source reports that transmission in these schools is dominated by the class.

Result: the structure, not the numbers. (Model definitions and power: Section S10.5.) Table S33 and Fig. S20 give the outcome, which the re-check reproduced from data downloaded again, with full-data likelihoods equal to six digits. The zone model predicts held-out schools far better than a well-mixed school (+798 units of log-likelihood, ahead in 27 of 29 schools) and than independent classes (+211). The share between classes of the same grade agrees to two decimals (0.10). Exact agreement of the three shares is rejected (likelihood ratio 42.7): the measured own-class share is larger than the fitted one, 0.76 against 0.66, and the fitted school-level share is almost twice the measured one, 0.24 against 0.13. This comparison is confounded. The model has no household term, whereas the source attributes 41 % of the pupils’ infections to households (51 % to school, 8 % to the community); siblings attend the same school in different grades, so infections between siblings at home appear as transmission across the school and inflate the fitted school-level share. The shortfall of the measured school share is therefore probably overstated by this confounding; how much cannot be estimated from the released table, which has no household identifier, and other unmodelled sources (the external hazard, contact time as a proxy for transmission) act in directions that are not known. The source itself, with households and the community in its model, estimated from the same data that transmission to classmates makes up about two-thirds of the within-school reproduction number when a grade has a single class [13]; the fitted own-class share here, 0.66, is of that size. The fitted shares lie within the tolerance (0.103), but not securely: the bootstrap interval of the distance reaches 0.168, 11 % of resamples exceed the limit, and so does one of the sensitivity analyses listed in the protocol (no winter break: 0.162). Among the models of the protocol the third tier excludes only the well-mixed school; the interval of the generic two-level model, simulated afterwards by the re-check (2.32 to

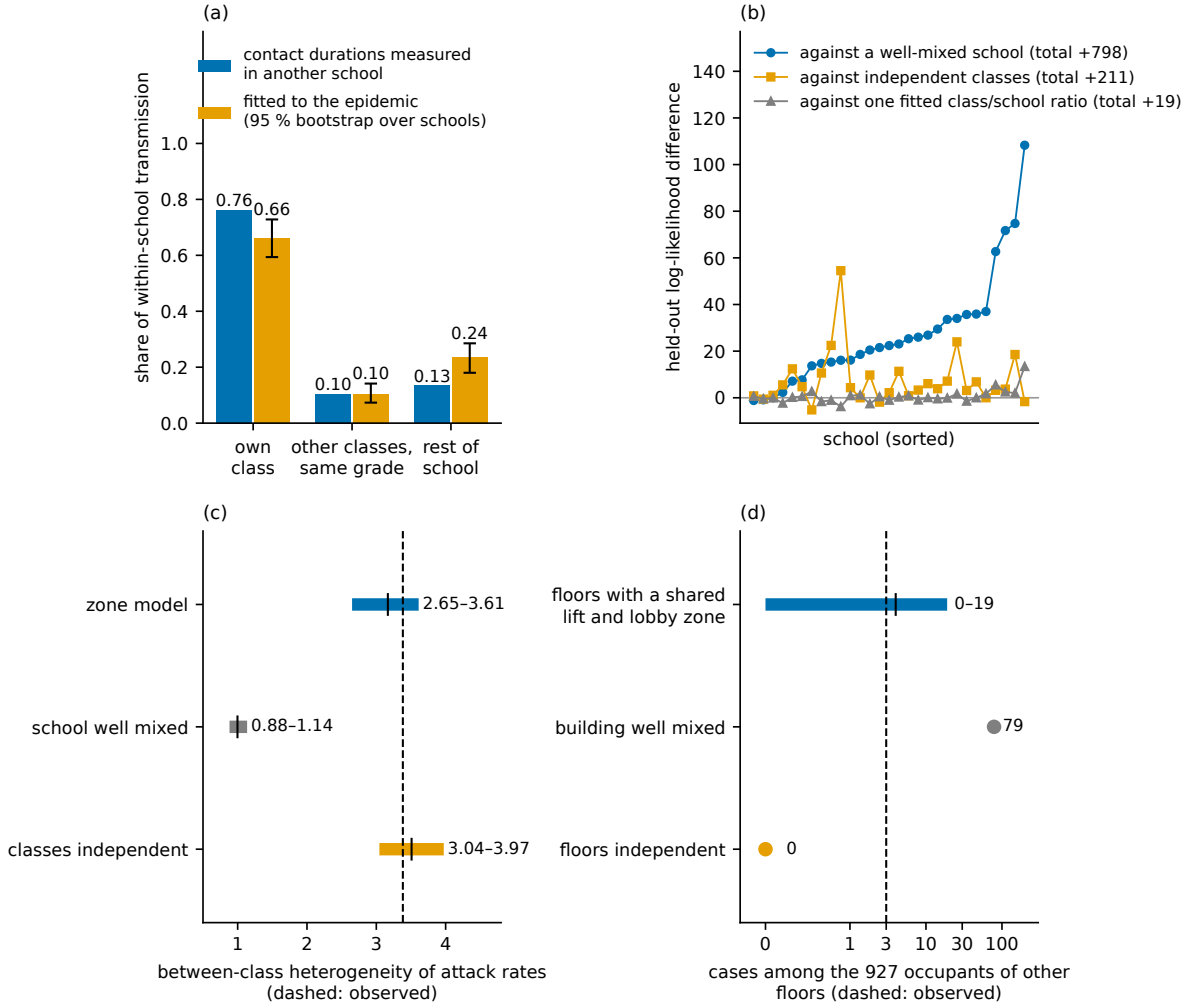


Figure S20: Zone level. (a) Shares of within-school transmission by level: measured contact durations in a primary school in Lyon and shares fitted to the influenza epidemic in Matsumoto. (b) Held-out log-likelihood of the zone model minus that of three competitors, by school. (c) Heterogeneity of attack rates between classes: 95 % predictive interval of three models. (d) Seoul call-centre building: cases among the occupants of the other floors predicted from the 11th floor; the bar is a prior-predictive interval over an assumed time in the shared lifts and lobby, not a measurement.

3.11), does not contain the observed 3.38, a small point in favour of the zone model. The verdict of the rule, ‘right structure within tolerance, not exact’, is therefore reported as a weak pass.

Two comparisons bound what the pass means. First, the generic two-level model with one fitted ratio is not beaten: the zone model is ahead by 18.8 out of sample, with an interval that includes zero, and the generic model fits the full data better by 13.6. What the measurement supplies is a ratio that the generic model has to fit: about 62 implied by the Lyon shares in the Matsumoto class structure, against 37 fitted. Second, a comparison added during the re-check asks whether measured mixing does better than a guess, in two runs that differ only in the lower bound of 10^{-9} on the external hazard of the protocol. With that bound, a zone model with the round shares 0.7, 0.1 and 0.2 predicts held-out schools better than the measured shares by 19 units (3 to 38), and the guess two-thirds, one-sixth, one-sixth by 8.9 (–12.1 to 34.5, not significant); without it, the guess 0.7, 0.1, 0.2 is better by 34 (11 to 66) and 0.6, 0.2, 0.2 by 25 (–17 to 76). Guesses with 80 % or more in the own class and equal thirds do worse (all three significantly), and so, in point estimate only, does 0.5, 0.25, 0.25 (by 30, –30 to 83; run without the bound). These guesses were chosen after the fitted shares were known. With that reservation,

the measurement is not shown to matter beyond the statement that most transmission is within the class. Had contact been counted as distinct pairs per day instead of duration, a choice the protocol excluded in advance, the shares would have been 0.43, 0.14 and 0.43 and the tolerance would have been missed (0.23). And at the ratio that the data favour, a world in which the generic model is true earns the full verdict in about 38 % of synthetic replicates (6 of 16), far above the 0 to 7.5 % obtained when the ratio is drawn at random.

The frequency-dependent normalisation of eq. (22) fits better than a density-dependent transfer of the measured contact times, by 37 units of log-likelihood on the full data, in a sensitivity analysis that the re-check did not re-run. The source had already reported, from the same data, that the within-school reproduction number hardly depends on class size or the number of classes, that is, frequency-dependent mixing [13]; the comparison here agrees with it and adds nothing independent of it.

A building by floor: inconclusive. The Seoul call-centre building had 94 cases among the 216 people on the 11th floor and 3 among the 927 occupants of the other floors [56]. A zone model with one zone per floor and a shared zone of lifts and lobby, its rate calibrated on the 11th floor alone, predicts a median of 4 cases elsewhere (interval 0–19); a well-mixed building would have had 79. The time spent in the shared zone is not measured, however: it is a prior of 3 to 30 minutes a day, placed after a rough calculation on the observed counts, and with 30 to 120 minutes the same model would be inconsistent with the data. The check shows that a well-mixed building is excluded, which needs no model, and nothing more. For the two wings of the 11th floor, and for the statements of Section 4.3 on early growth and on the concentration of risk in high-efficiency zones, no outbreak with an independent measurement of mixing was found (a cruise ship, a warship, the meat-processing plant, prisons and nursing homes were considered), and no test was run.

S9.5 All events with a stratum test

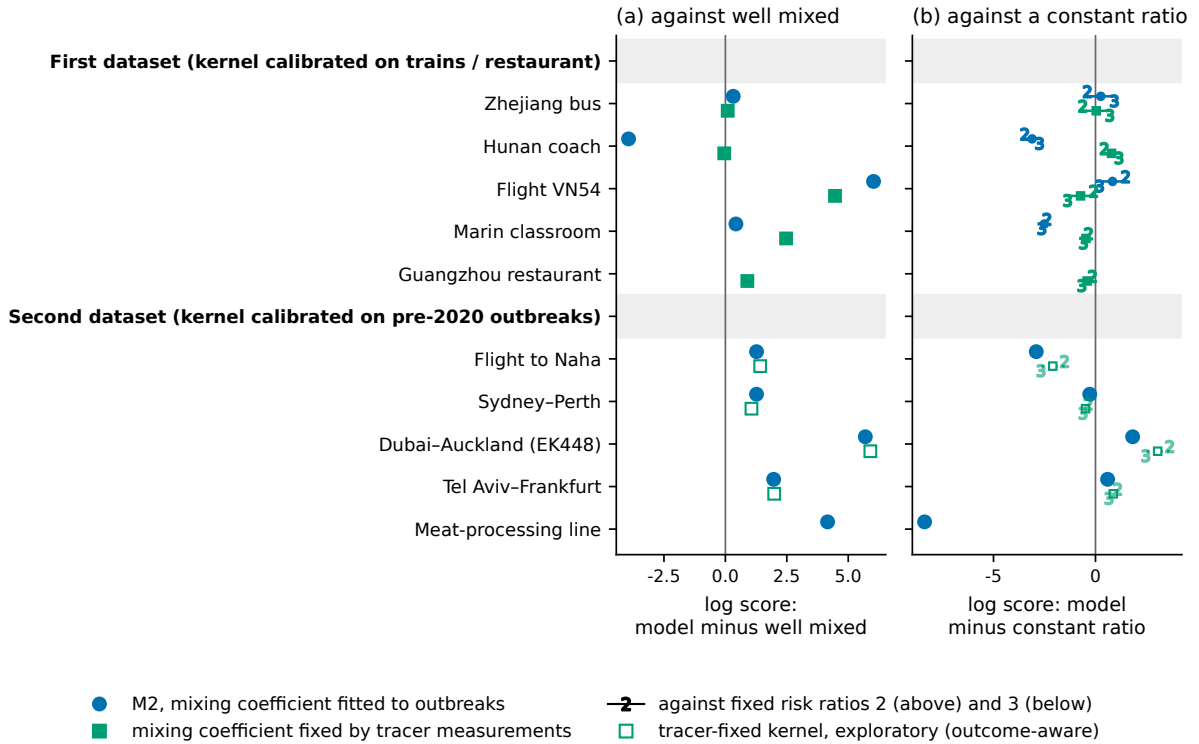


Figure S21: Every event with a stratum test, in both datasets: difference of log scores (a) between the spatial kernel and the well-mixed model and (b) between the spatial kernel and a constant near/far risk ratio. Positive values favour the kernel. Circles: M2 with the mixing coefficient fitted to outbreaks (first set: train cohort for vehicles, restaurant for the classroom; second dataset: the six pre-2020 events). In (b) the first dataset is compared with the fixed ratios 2 and 3 (digits; introduced after unblinding) and the second with the ratio calibrated on the same events as the kernel. Squares: the kernel with its coefficient taken from tracer measurements, against the ratios 2 and 3; open squares: its exploratory, outcome-aware application to four flights of the second dataset. The restaurant is a calibration event of the first test and has no circle; the processing line has no tracer measurement. The kernel is to the right of zero in (a) for all events but the coach, and scattered about zero in (b).

S10 Protocols of the second outbreak test and of the further analyses; register of all tests

This section documents the four analyses of Section 7 of the main text other than the first outbreak test: what was fixed in advance and the evidence for when; the decision rules and their power; the data and what may be redistributed; and a register of every test of Section 7, which separates what was pre-specified from what was added afterwards, by the analysis itself or by its re-check. The four protocols, their power addenda, freeze records and post-hoc logs are released with the code, under the file names they were given (PREREGISTRATION.md and so on), as redacted copies (Section S8); as in Section S8, they are called protocols here and their contents pre-specified. ‘The re-check’ of an analysis is its re-implementation with separately written code, within the same project and on the same machine, as described in Section 1; where it corrected a result, the corrected result is the one reported. Where the frozen protocol texts and logs speak of a ‘verifier’, they mean this same-project re-check, not an independent party, and by ‘the analyst’ they mean the first analysis (both passes were made by the author). The frozen protocol texts call the four analyses avenues 1 to 4, and the first outbreak test with its dataset and the later analyses rounds 1 and 2 (two successive sets of analyses); the article does not use these labels.

Table S34: The four protocols of the second outbreak test and of the further analyses. Times are British Summer Time on 1 October 2026, from the freeze records (self-reported) and from the modification times of the files. Hashes: number of files whose SHA-256 is stored in the freeze records; all were recomputed for this article on the unredacted originals and match.

Analysis	Frozen	Hashes	First result that uses outcomes	Known beforehand
Further outbreaks (Section 7.5)	protocol 09:42:01; calibration and power 09:52:41	4 + 6	held-out strata scored 09:53:08	every count, including the held-out ones; the outcome of the first test
Contact records (Section 7.7)	protocol 09:23:27; modified walks 10:34:24, before the confirmatory datasets	8 + 26	first dataset 09:24:00 (the run takes 19 s); first confirmatory dataset 10:35:34	raw counts per day and group labels; the published properties of such data; expectation of failure written down
Tracer measurements (Section 7.8)	protocol 09:18:16; kernels, predictions and power 09:44:43	23 + 10	evaluation 09:49:08	the outbreak counts and the failures of the first test; the tracer sources had been opened
extension to four flights	protocol 10:24:56; predictions 10:26:01	7 + 4	evaluation 10:26:09	the near/far counts and the result of the second outbreak test: outcome-aware
Zone level (Section 7.9)	protocol 09:48:58	14	schools 09:56:44 (run 09:49–09:56); floors 10:21:08	total of cases; qualitative conclusion of the source; for the floors, the counts

S10.1 What was frozen, and the evidence for when

Table S34 gives the times. The limits are those of the first test (Section S8.1) and apply to all four analyses. (i) Every protocol is *data-aware* and *model-blind*. What was fixed in advance is the list of events or datasets, the models, the split into calibration and hold-out, the metrics, the decision rule and the list of sensitivity analyses, each with a power analysis; the data themselves had been read. (ii) The freezes are self-attested. The records are files on the author’s machine, with matching hashes and no external time stamp or version-control history; the hashes show that the files have not changed since, not when they were written. In the outbreak analysis the record was itself rewritten when the power addendum was added, and the clock times written into its post-hoc log are one to two hours ahead of the file times. (iii) Parts of the code were written after the freeze: in the outbreak analysis the modules that build the events and run the comparison (the last one four seconds before the hold-out output), in the tracer analysis the evaluation script. The intervals are short because each analysis was prepared and run in one or two continuous stretches of work. (iv) In the outbreak analysis the constant-ratio model was coded as $p_j = 1 - \exp(-\varepsilon \varrho)$ in the near zone, not as the $p_j = \varrho p_0$ of the protocol. With the form of the protocol the pooled Δ_C is -7.19 in the re-check’s re-implementation, against -9.20 ; the outcome is the same. (v) The extension of the tracer analysis to four flights was written, run and evaluated within 73 seconds, with the outcomes known; its freeze is nominal, and it is reported as exploratory throughout.

S10.2 Further outbreaks: rule, power, numerical correction

The rule of Table S35 was written to remove the weakness found in the first test: a constant-ratio world passes B1 with probability 0.65 to 0.73, so that B1 alone is no evidence for the kernel, and passes B1 and B2 together with probability 0.03 to 0.05. The re-check recomputed the power from 400,000 draws of its own (0.927, 0.048 and 0.010 for B1 and B2 together under the three truths; 0.822 for the four flights).

Table S35: (a) Decision rule of the second outbreak test, applied to M2 (and, in separate columns of the analysis, to M1 in closed form and to an exploratory variant with a well-mixed fraction). (b) Probability of each criterion and of the full claim V1 when the data are generated by the model in the first column, at its calibrated posterior (200,000 replicates), as computed before any held-out stratum count was scored and, in brackets, after the numerical correction of the room kernel.

(a) <i>Criteria and outcomes</i>				
B1	better than well mixed	$\Delta_0 > 0$ with one-sided $p < 0.05$ under M0		
B2	better than the constant ratio	$\Delta_C > 0$ with one-sided $p < 0.05$ under the constant-ratio model; mirror: $\Delta_C < 0$ with $p < 0.05$ under M2		
B3	adequacy	two-sided predictive $p \geq 0.05$ for every held-out event		
V1	B1, B2, B3, and power of (B1 and B2) at least 0.8	‘predicts held-out distance gradients better than both a well-mixed model and a constant near/far risk ratio’		
V2	B1 and B3 with at most one failure; neither B2 nor its mirror	‘better than well mixed; not distinguishable from a constant ratio’		
V3	B1; mirror of B2	‘better than well mixed, but a constant ratio predicts better’		
V4	B1 not met	‘no out-of-sample support for the kernel’		
V5	two or more failures of B3	a list of the events reproduced and of those not		
<i>Not covered:</i> B1 and B2 met with exactly one B3 failure was given no outcome by the protocol (probability 0.16 if M2 is true); the code classes it with V2. The observed outcome, V5, does not depend on this.				
(b) Truth	B1	B2	B1 and B2	V1
well mixed	0.007 (0.015)	0.096 (0.169)	0.005 (0.010)	0.003 (0.008)
constant ratio	0.652 (0.732)	0.029 (0.052)	0.028 (0.049)	0.020 (0.039)
M2	0.989 (0.982)	0.961 (0.935)	0.956 (0.926)	0.784 (0.762)
M2, four flights only	0.980	0.822	0.821	–

The numerical correction. The propagator of the first test writes $P_t(y|x)$ on a rectangle with reflecting walls as a sum over cosine modes. Its absolute error reaches about 10^{-10} of the largest exposure in an event, so that receivers whose true exposure is smaller than that carry round-off noise of either sign. Replacements by the corrected propagator occur for mixing coefficients up to about 2.5 to $16 \text{ m}^2 \text{ h}^{-1}$, in the cabins and in the rooms of the second test alike. The posterior of the room coefficient first computed (‘mode $0.02 \text{ m}^2 \text{ h}^{-1}$, bimodal’) came from two such noise values in the likelihood of the ward inpatients; it was found by inspection of a diagnostic plot after the held-out events had been scored. In the corrected code each one-dimensional propagator is a positive sum of image terms, and the image value replaces the mode sum wherever the latter is below 10^{-8} of the event maximum; where both are resolved, at coefficients at which a replacement occurs, they agree to about 10^{-4} or better (largest relative difference 8.4×10^{-5} , in the ward inpatients). The primary analysis, the power and the secondary analyses were then re-run unchanged (the analyses stored in `10_corrected.json`); the 15 sensitivity variants were not, and are reported with the kernel as first frozen. For the flights no reported digit changes (their posterior lies above the affected range). For the rooms the posterior becomes $794 \text{ m}^2 \text{ h}^{-1}$ (302–4,390), the score of the processing line -15.24 instead of -16.22 and its predictive p 1.4×10^{-5} instead of 5×10^{-6} ; the pooled scores are $+14.35$ and -9.15 instead of $+13.37$ and -10.13 ; and the outcome is V5 in both cases. Both versions are stored. The re-check confirmed the defect and the correction with a third propagator, built by uniformisation and sharing no code with either: it agrees with the image sum to 10^{-14} at small coefficients, while the mode sum returns non-positive exposures at up to 32 of the 74 ward receivers. On the restaurant calibration of the first test the same defect changes the log-likelihood by about 10^{-7} at most at every coefficient in the posterior range, so that calibration is not affected. The hold-out predictions of

Table S36: Mixing coefficient preferred by each event of the second dataset taken alone (profile likelihood of the stratum counts given the total; interval where the log-likelihood is within 1.92 of its maximum; grid from 10^{-2} to $10^4 \text{ m}^2 \text{ h}^{-1}$), and gain in log-likelihood over the largest coefficient of the grid ($10^4 \text{ m}^2 \text{ h}^{-1}$), which in a cabin is close to, but not yet, the well-mixed model.

Event	Pathogen	D_{air} ($\text{m}^2 \text{ h}^{-1}$)	Interval	Gain
Hong Kong–Beijing, CA112	SARS-CoV-1	126	40–794	3.87
Chicago–Honolulu	<i>M. tuberculosis</i>	16	≤ 316	3.49
Los Angeles–Auckland	influenza A(H1N1)pdm09	lower edge	≤ 794	2.33
Cancun–Birmingham	influenza A(H1N1)pdm09	upper edge	≥ 100	0.00
Flight to Naha	SARS-CoV-2	501	≥ 100	1.58
Sydney–Perth	SARS-CoV-2	200	≥ 40	1.33
Dubai–Auckland, EK448	SARS-CoV-2	13	1–126	6.75
Tel Aviv–Frankfurt	SARS-CoV-2	lower edge	≤ 631	2.90
Ward 8A, students	SARS-CoV-1	lower edge	≤ 398	5.55
Ward 8A, inpatients	SARS-CoV-1	1,995	≥ 501	0.31
Meat-processing line	SARS-CoV-2	40	16–126	9.56

Table S37: The six contact datasets (SocioPatterns, RFID proximity at 20 s resolution) and four statistics of the test days, observed and under the homogeneous lattice walk calibrated on the training days. Role: development (dev.) or confirmatory (conf.) dataset for the modified walks. Days: training + test. Sites: size of the lattice fixed by the calibration. Exponent: of the number of contacts per 20 minutes against the number of people present (not comparable for the primary school, where occupancy hardly varies; the conference has no groups).

Setting	Role	People	Days	Sites	observed / walk			
					Partners per day	Within-group share	Burstiness	Exponent
Office, 2013 [17]	dev.	92	5+5	1,273	4.6 / 6.0	0.82 / 0.24	0.36 / 0.23	0.45 / 1.97
Primary school [63]	dev.	242	1+1	644	46.6 / 90.4	0.73 / 0.10	0.58 / 0.08	n.a. / 2.12
Hospital ward [70]	dev.	75	3+2	120	16.1 / 27.6	0.57 / 0.33	0.61 / 0.25	2.83 / 2.20
Office, 2015 [16]	conf.	217	5+5	1,388	8.6 / 15.5	0.80 / 0.14	0.45 / 0.21	0.95 / 2.06
High school [46]	conf.	327	3+2	1,089	14.6 / 33.8	0.94 / 0.12	0.51 / 0.15	0.76 / 2.12
Conference [16]	conf.	403	1+1	1,219	26.5 / 23.4	– / –	0.50 / 0.17	0.81 / 1.95

the first test were affected in the classroom, where the mode sum returns non-positive exposures at up to 11 of the 24 desks for coefficients of $0.1 \text{ m}^2 \text{ h}^{-1}$ and below; they were re-run with the corrected propagator for this article (Section 7.4).

S10.3 Contact records: data and statistics

Table S37 lists the datasets. The first three were used to develop the modified walks and the last three were not touched by any model or statistic until the modifications and their fitting code had been frozen. Clock offsets, the removal of days with fewer than 1 % of the records and the definition of a person’s presence (from the first to the last own contact of the day) are fixed in the frozen loader. The seven families of statistics, with their tolerances, are: total contact time (ratio in 0.8–1.25); mean contact duration and share of contact time in contacts of at least five minutes; burstiness of the gaps between a person’s contacts (absolute difference at most 0.10) and recurrence of pairs; mean and coefficient of variation of the number of partners per person and day; coefficient of variation of individual contact time; share of a day’s pairs already in contact on the previous day; share of contact time within the recorded groups. A family is passed if all its statistics are. The power analysis was made on synthetic data before the freeze, with four replicates per combination of generating model and candidate: a true or nesting model passed every family and the epidemic criterion in 20 of 20 replicates, and a wrong non-nested model

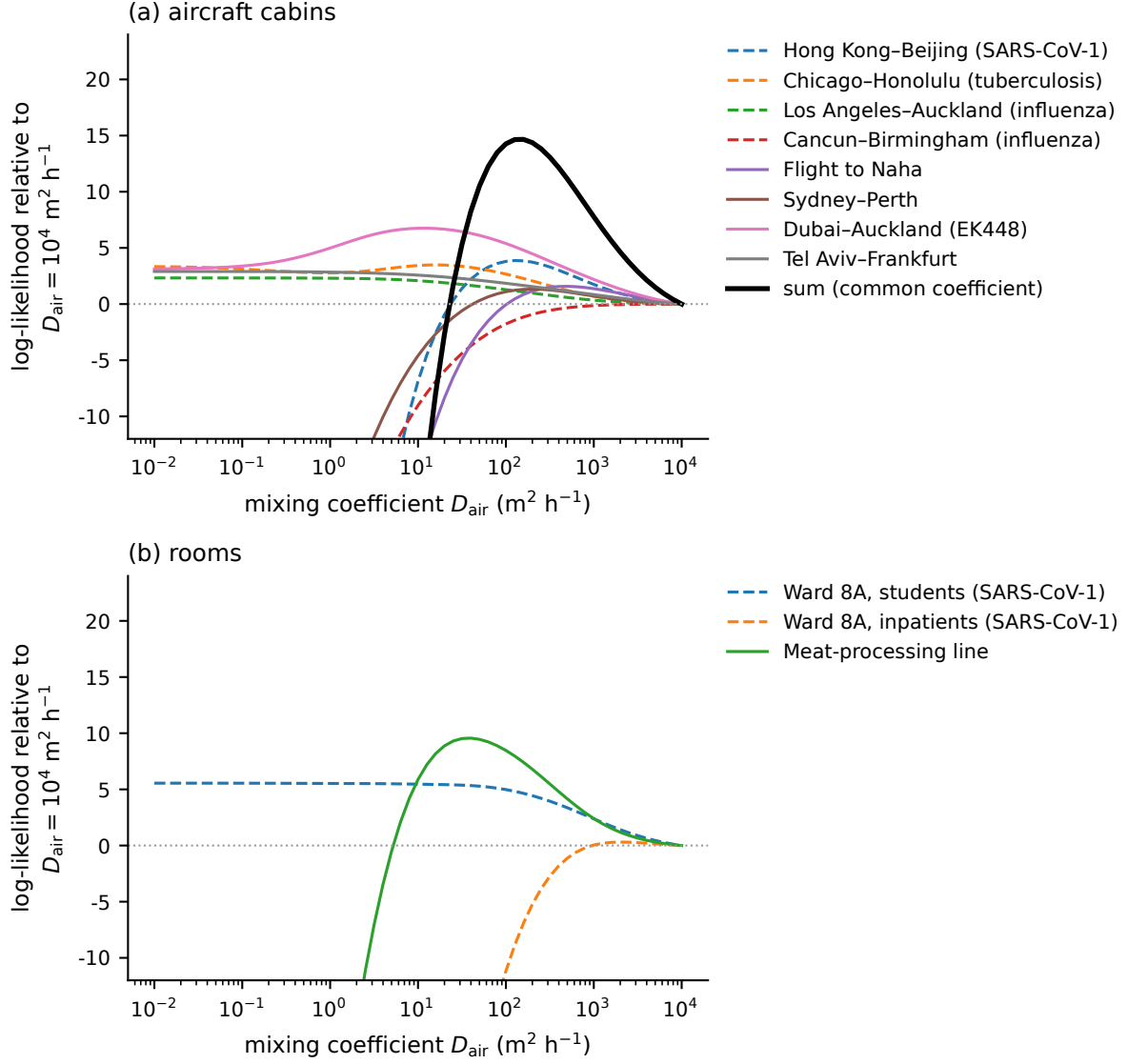


Figure S22: Log-likelihood of the mixing coefficient of M2 for each event of the second dataset taken alone, relative to the largest coefficient of the grid ($10^4 \text{ m}^2 \text{ h}^{-1}$). Solid: held-out events; dashed: calibration events. The thick line in (a) is the sum over the eight flights (common coefficient).

reached the per-dataset criterion in none of 44. The re-check qualified this in three ways: the 44 cases are not independent; the analysis covers the base models only, and the modified walks fail on their own synthetic truth (Section 7.7); and on real data the tolerance of the epidemic criterion is at the level of the noise, since two runs of 4,000 epidemics on the same recorded network differ by up to 7% in the secondary cases of the index case, against a tolerance of 10%. The rule is therefore adequate for the failure of the lattice walk, whose errors are 25 to 72%, and weak for anything close to the tolerance.

S10.4 Tracer measurements: rule and power

The rule (Table S38) is safe against a false claim and weak at confirming a true one: if the tracer kernel were true it would earn the full claim S1 with probability 0.21 only, because on the coach and the bus it predicts almost what the well-mixed model predicts. The outcome obtained, S4, has probability 0.20 under the kernel, 0.40 if outbreaks are well mixed and 0.46 or 0.20 under a constant ratio of 2 or 3, so that it does not discriminate. Every model fails on the train table,

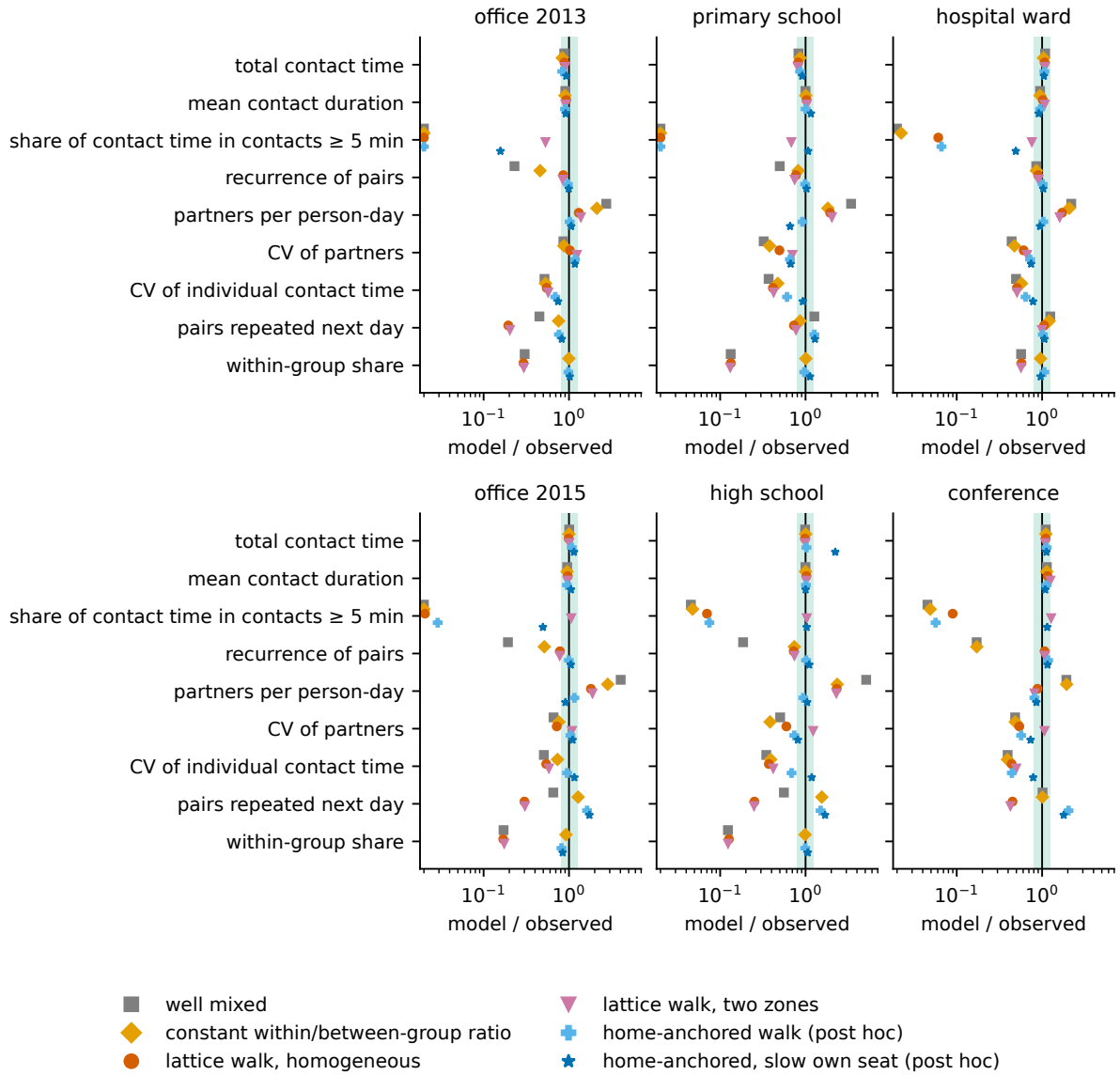


Figure S23: Contact statistics of the test days: value under each model divided by the observed value (shaded: tolerance 0.8 to 1.25; values below 0.02 are drawn at the left edge). The lattice walk matches what it was calibrated on, the total contact time and the mean duration, and misses long contacts, the number of partners, the differences between people, the repetition of pairs and the group structure. The anchored variants were developed on the first three datasets (top row) after that failure.

and the rule counts this as a failure of the kernel without asking whether a competitor passes. For the restaurant the rule fails the kernel only if it is rejected at every count from two to five cases, which is lenient; the main text reports the result at each count. Two departures from the protocol were forced by the data and are logged: the forward cabin has unmasked breathing releases at three seats, not four, so that the flight uses two measured kernels in place of three; and the minibus of the Hunan outbreak has 15 passengers with a simulated tracer value, not the 17 of the first test. A full re-run of the stored analysis into a separate directory reproduced every result file without a difference; the re-check then re-derived the coefficients with its own solver (identical grid optima) and re-typed the tracer values from the sources.

Table S38: Tracer analysis. (a) Decision rule over seven events (coach, bus, flight, trains, restaurant, classroom, call centre); D1: number of events rejected at the 5 % level; D2: pooled log score above that of M0 with $p < 0.05$; D3: above that of the constant ratios 2 and 3, each with $p < 0.05$. (b) Probability of each outcome when the data are generated by the model in the first column, computed after the kernels had been estimated and before any comparison with outbreak counts.

S1	no D1 failure, D2 and D3: ‘with the kernel fixed by physical measurement the framework predicts the spatial patterns, and better than a generic gradient’
S2	no D1 failure, D2, not D3: ‘compatible with all events and better than well mixed; not shown to be better than a generic gradient’
S3	no D1 failure, not D2: ‘compatible, not discriminating’
S4	exactly one D1 failure: a sentence that names it
S5	two or more D1 failures: the hypothesis fails

(b) Truth	S1	S2	S3	S4	S5
tracer kernel	0.213	0.117	0.453	0.196	0.021
well mixed	0.009	0.004	0.488	0.401	0.099
constant ratio 2	0.017	0.204	0.112	0.455	0.211
constant ratio 3	0.000	0.022	0.002	0.204	0.773

S10.5 Zone level: models and power

A susceptible pupil of class c has onset on day t with probability $1 - \exp(-\lambda_c(t))$, $\lambda_c(t) = \beta x_c(t) + \alpha_0 + \alpha_1 P_{\text{city}}(t)$, where $P_A(t)$ is the kernel-weighted number of earlier onsets in the set A per pupil of A and the last term is the same quantity for the other schools of the city. The kernel is a gamma density with mean 1.7 and standard deviation 1.0 days on daily bins, as in Endo et al. [13], who chose it to give the mean serial interval of 2.2 days of influenza A(H3N2) [71]. The three sets are exclusive: for a pupil of class c , P_{class} counts the other pupils of the class (denominator $n_c - 1$), P_{grade} the pupils of the other classes of the grade and P_{school} the pupils of the other grades. The zone model has $x_c = 0.762 P_{\text{class}} + 0.104 P_{\text{grade}} + 0.134 P_{\text{school}}$, with the shares measured in Lyon; the well-mixed school has $x_c = P_{\text{whole school}}$; independent classes have $x_c = P_{\text{class}}$; the generic model has one fitted ratio ϱ of own-class to rest-of-school risk; and the ceiling has three free shares. Within-school terms are switched off during the winter break. In the 36 grades that have a single class (453 pupils) the grade term is set to zero and not redistributed, so that the shares of the zone model do not sum to one there; this was fixed in the protocol and its effect was not assessed. The table holds the 2,548 pupils with an episode reported in the prospective survey of the source (96 % of the cases known to the schools) and the 8,375 who reported none in a later survey, 10,923 or 83 % of the 13,217 enrolled; the missing pupils are mostly non-cases, so the denominators of the classes are incomplete. The source excluded three small schools (71 pupils) from its own analysis; they are kept here. The model has no household term, although the source attributes 41 % of the pupils’ infections to households (Section 7.9). Power was computed before the freeze on synthetic epidemics on the real class structure (40 replicates per truth, two strengths of the school signal). The verdict of the rule was earned in 100 % and 80 % of replicates when the zone model is true, in none when the school is well mixed or the classes independent, in 7.5 % and 0 % under the generic model with a ratio drawn log-uniformly between 1 and 1,000, and in 2.5 and 5 % under random shares. In the re-check the generic model turned out to be a more dangerous competitor than these averages suggest: with its ratio set to the fitted value, about 37, it earns the verdict in 6 of 16 replicates. The tolerance of 0.15 is a judgement, two to three times the sampling noise under a true zone model; held-out differences depend slightly on a lower bound of 10^{-9} on the external hazard, which is active in one fold (Table S33).

S10.6 Register of tests

Table S39 lists every test of Section 7 of the main text, and the secondary and sensitivity analyses of their protocols that are reported there and in Sections S7–S10, with their status and their outcome after the re-checks. ‘Protocol’ means that the test and its criterion were fixed before any model was run on the data concerned; ‘listed’ that the analysis is on the list of secondary or sensitivity analyses of a protocol, which may never upgrade a primary outcome; ‘after’ that it was added after results were known, by the analysis or, where marked, by its re-check.

Table S39: Register of the tests of Section 7 of the main text and Sections S7–S10.

Test	Status	Outcome after the re-check
<i>First outbreak test (Section 7.4 of the main text and Sections S7.2–S7.6)</i>		
Flight VN54, near/far split	protocol	pass (M2 $p = 0.61$, M0 $p = 0.0008$); 20 passengers; a constant ratio of 2 to 3 also passes
Classroom, front/back split	protocol	not a real prediction: rejected at the calibrated value ($p = 0.0009$); $p = 0.068$ over the posterior
Bus, zone split	protocol	compatible ($p = 0.38$), non-discriminating
Coach, rear/front split	protocol	fail ($p = 0.007$)
Call centre, desk-scale clustering	protocol (quadrature after, check)	fail (0.007 to 0.013 by quadrature; 0.013 to 0.024 with the prior extended to $10^6 \text{ m}^2 \text{ h}^{-1}$; with the re-check’s mode-sum propagator; its round-off acts only through the outbreak-size weights, and with zero weight at or below $0.1 \text{ m}^2 \text{ h}^{-1}$ the second value would be 0.015; the 40-draw estimate of the protocol had Monte Carlo error large enough to read as borderline)
Pooled test against M0	protocol	weak pass ($+2.83$, $p = 2.5 \times 10^{-4}$, after the numerical correction of the classroom; $+3.87$ before), resting on the flight; constant ratios of 2 to 3 score higher (after, check)
Classroom with the round-off-safe propagator	after	$p = 0.068$ instead of 0.11; outcome W4 unchanged
Totals attainable by M1	protocol (rule P4 after)	mean number of cases below the observed one on bus, flight and classroom; observed total reached with probability 0.0002, 0.025 and 0.081, so unattainable under rule P4 on the bus only
Level checks and negative controls	protocol	met, with almost no power
Common coefficient, four vehicle events	after	rejected (approximate $p = 0.023$, LR 9.52); for the two rooms not rejected (approximate $p = 0.080$)
Default parameters of Sections 2–6 against the bus and the choir	after	expected infections at most 0.10 and 0.16 against 23 and 52 (32 confirmed) cases; reached only for β of about 16 and 14 (12) day^{-1}
<i>Second outbreak test (Section 7.5)</i>		
B1: M2 against M0, five held-out events	protocol	pass ($+14.35$)
B2: M2 against the constant ratio, pooled	protocol	fail (-9.15), nine-tenths from the processing line, whose near zone was chosen by the source on the same data
B2 on the four flights	protocol	inconclusive (-0.77 ; power 0.82)
B3: adequacy of M2	protocol	fail on 2 of 5 events; outcome V5
M1 in closed form	protocol	V5, as M2; by the protocol a pass of the closed form, which overstates the reach of M1, does not bear on M1: against M0 pass ($+15.45$); against the constant ratio pooled -8.06 (mirror $p = 1.5 \times 10^{-5}$), on the flights -4.63 ; fails on 3 of 5 events
Train kernel applied to eight flights	listed	fail: four flights rejected
Exchange of calibration and hold-out	listed	weak pass on flights ($+3.58$, $p = 0.004$); fail in the ward
Train kernel applied to the rooms	listed	rejected on the ward inpatients ($p = 2 \times 10^{-5}$) and the processing line ($p = 4 \times 10^{-4}$)
Leave one event out within each class	listed	flights not separated ($\Delta_C = -0.06$; $p = 0.046$ under the constant ratio, mirror $p = 0.044$ under M2: improbable under either); rooms no better than well mixed, constant ratio better (-14.0)

Test	Status	Outcome after the re-check
Seat-level log score, three held-out flights	listed	constant ratio -76.4 , M2 -78.6 , well mixed -83.5
Published ratio 2.4 in place of the calibrated one	listed	flights $+1.18$ ($p = 0.028$); pooled -4.44 (mirror $p = 0.002$; room kernel as first frozen)
Two-scale variant (M2 plus a well-mixed fraction)	protocol, exploratory	V3: $+17.5$ against well mixed, -5.95 against the constant ratio
Common coefficient within a class	listed	rejected (approximate p : rooms 1.6×10^{-4} , flights 0.034)
Fifteen sensitivity variants	listed	V5 in 12, V3 in 3 (kernel as first frozen); on flights Δ_C from -3.42 (constant ratio significantly better in three) to $+1.21$ (M2 significantly better in one)
Near/far risk ratios	listed	description: 3.98 ($2.32-6.82$; random effects); no gradient for influenza
Near/far ratio without the processing line	after	3.30 ($1.99-5.45$)
Numerical correction of the room kernel	after	outcome unchanged
Other near zones and strata; ratio in the form of the protocol	after, check	Δ_C on flights from -0.78 to $+7.8$ in these specifications; with the listed variants the sign is not determined
Exponential kernel	after, check	as good as M2 on flights, better on the processing line
<i>Contact records (Section 7.7)</i>		
Lattice walk, structure and epidemic outcomes, six datasets	protocol	fail (A0) on all six
Walk against models without space	protocol	adds nothing; a static network of pair rates does better (after, check)
Modified walks on three confirmatory datasets	protocol (second freeze)	formally A0; uninformative, since the fit does not recover its own truth (after, check)
Error sizes of the modified walks	after	description only
Density scaling of contacts	listed	walk: exponent about 2; data below 1 in four of five settings
Power of the rule	protocol	adequate for the walk, weak near the tolerance
<i>Tracer measurements (Section 7.8)</i>		
Coach	protocol	not rejected; non-discriminating
Call centre	protocol	not rejected; non-discriminating; wrong sign of clustering
Bus	protocol	inconclusive
Classroom	protocol	weak pass; constant ratios do as well
Flight VN54	protocol	not reproduced (formal $p = 0.14$; rejected in four of six listed variants)
Restaurant	protocol	not reproduced at the reported count
Trains	protocol	fail; every model fails
Pooled rule	protocol	S4 as computed, S5 on the defensible reading of the flight; not better than constant ratios
Probability of the pooled pattern under each truth	after, check	0.23 under the tracer kernel, 0.84 and 1.00 under ratios 2 and 3
Measured against fitted coefficients	listed	vehicle values 4.5 to about 50 times the first test's vehicle value; second-test aircraft value inside the cabin range, which does not make it a measurement of air mixing
Four further flights	after	exploratory, outcome-aware: weak
<i>Zone level (Section 7.9)</i>		
Tier 1: against well-mixed school and independent classes	protocol	pass
Tier 2a: equality of shares	protocol	fail
Tier 2b: tolerance; tier 3: dispersion	protocol	weak pass
Against a generic model with one fitted ratio	protocol (not decisional)	not beaten: zone model ahead by 18.8 (-8.8 to 54.0) out of sample, generic model better by 13.6 on the full data

Test	Status	Outcome after the re-check
Against guessed shares	after, check	guess 0.7/0.1/0.2 better by 19 (3 to 38); guesses chosen after the fitted shares were known; school-level share confounded by households
Frequency- against density-dependent transfer	listed	frequency-dependent fits better; not re-run by the check
Building by floor	protocol (low power, not blind)	inconclusive
Wings of one floor; early growth; hotspot maps	–	not run: no independent measurement of mixing found

S10.7 Data, licences and what is redistributed

Outbreaks. The seat maps and stratum counts of the eleven events were transcribed by eye from the published figures and tables into the code and are released as tables, with the reference and the table or figure of origin of every number (`data/outbreaks2/`). Copies of the source texts and figures are not redistributed. *Contact records.* The SocioPatterns files are distributed by their authors under Creative Commons licences, some of which exclude commercial use; they are not redistributed here. A script downloads them from the source and checks their SHA-256 against the values recorded when the protocol was frozen. *Tracer measurements.* The cabin measurements are published on figshare under CC BY 4.0 [36]; the released tables are derived from those workbooks. The values for the coach and the carriage were read or digitised from the supplementary figures of the sources and are released as tables of numbers. *Schools.* The anonymised pupil-level data of Endo et al. [13] are published by their authors under the MIT licence; the release contains a tabular copy, with that licence, and the script that produced it from the original file.

S11 Verification of the theorems, the simulator and the analyses

The statements of Sections 3 and 4 of the main text are proved in Sections S2 and S3. This section records the numerical tests that were run in addition, as a guard against errors in the proofs and in the code, and the tests of the stochastic simulator used in Sections 5 and 6. Most tests were carried out twice, by two sets of programs: the programs that produced the results of the article, and a second set written separately for checking (the directories named `verify` in the repository), which was also used to search for counter-examples. The second set does not call the first, except that the schedule test of Section S5.10 and a timing benchmark run the production simulator; other exceptions are stated. Monte Carlo uncertainties are standard errors unless a 95 % interval is given in brackets.

S11.1 Scope of the re-checks

Each strand of this work was re-checked with separately written code, within the same project and on the same machine (Section 1 of the main text). The re-check re-derived the proofs, re-implemented the computations, including a second exact simulator, retyped the outbreak counts from the sources and searched for counter-examples. Its core re-implementations were written separately; where it re-ran the analyses, compared kernels or added comparisons, it reused modules of the analyses (data loaders of the contact and tracer analyses, event geometries, the transcribed seat maps, the simulator for timing benchmarks and schedules). It was not carried out by anyone independent of the project. The re-checks of the outbreak dataset and of the five tests are described where their results are reported (Sections S6–S10); this section describes those of the theorems and of the simulator.

Table S40: Random-instance tests of the statements of Sections 3 and 4. Deviations are relative unless stated; a ‘violation’ is the amount by which an inequality fails (positive values of the order of 10^{-14} are rounding errors of double precision).

Statement	Test	Worst case over all instances
Theorem 3.1(b)	$ s(\mathbf{M} + \mathbf{F}/R_0 - \gamma\mathbf{1}) $; 400 instances (103 symmetric, 119 departure-site, 96 non-reversible, 82 dense)	1.3×10^{-14}
Theorem 3.1(c)	sign of $R_0 - 1$ against sign of λ_1 ; λ_1 after replacing β by β/R_0 ; same 400 instances	0 sign failures; $ \lambda_1 \leq 1.3 \times 10^{-14}$
Theorem 3.1(a)	$\mathbf{V}^{-1}$ against Gauss–Laguerre quadrature of $\int_0^\infty e^{-\gamma t} e^{\mathbf{M}t} dt$ (one instance, 18 cells); Monte Carlo sojourn times of one infective, 4×10^6 infectious periods on a 3×4 lattice	3.1×10^{-6} ; largest $ z = 1.57$, root-mean-square $z = 1.05$ over the 12 cells
Theorem S2.4, Corollary S2.5	Rayleigh value against $\rho(\mathbf{K})$; largest diagonal modal term against R_0 ; monotonicity of the truncated modal formula; 150 symmetric and 100 departure-site instances	8.0×10^{-15} and 1.3×10^{-14} ; diagonal term never above R_0 (closest: 0.04% below); 0 monotonicity failures
Theorem 3.2	both bounds on R_0 and on λ_1 ; 2,000 instances	violation $\leq 4.0 \times 10^{-14}$; for non-constant q the gaps to the lower and upper bound are at least 1.1×10^{-6} and 1.2×10^{-3} ; constant q : 9.5×10^{-15}
Theorem 3.3	$R_0(D_0)$ at 41 values of D_0 from 10^{-4} to 10^4 , limits at $D_0 = 10^{\pm 9}$, convexity of $\lambda_1(D_0)$; 300 instances	strictly decreasing in every instance with non-constant q ; largest increase in the other instances 2.7×10^{-12} ; limits to 3.8×10^{-6} and 2.6×10^{-7} ; convexity violation 1.3×10^{-15}
Proposition 3.4	ratio of the errors of each expansion at two values of D_0 a factor 10 apart (100 for a second-order remainder)	median 99.9 (fast mixing, 227 instances) and 99.5 (slow mixing, 299 instances)
Proposition 3.5	878 random zones in 300 instances of all four families	no violation
Theorem 3.6	900 increases of one edge rate by a factor 1–20, symmetric rates	largest increase of R_0 : 7.3×10^{-16}
Theorem 4.1, Proposition 3.7(ii)	uniform q : 500 instances, square rooms of side 3 to 40, and 300 instances with random contact kernels	8.3×10^{-15} ; $R_0 = 3.571429$ for every side; kernels: 5.8×10^{-15} , no violation of $\beta q_{\min}/\gamma \leq R_0 \leq \beta q_{\max}/\gamma$
Theorem 4.2	400 instances with random door cells and exit rates; corridors of 3 to 100 cells with absorbing ends	part (3): 2.1×10^{-14} ; parts (2) and (4): violation $\leq 1.9 \times 10^{-15}$; absorbing limit (exit rate 10^9): 1.3×10^{-6} ; corridor formula: absolute error below 10^{-15}
Theorem 4.4	200 instances: long-time profile against $\mathbf{u}$; iterated generations against $\mathbf{w}$; $\sum_x e_x$; elasticity against central finite differences; reversible formula	5.9×10^{-14} ; 4.5×10^{-14} ; $ \sum_x e_x - 1 \leq 4.4 \times 10^{-16}$; 3.9×10^{-5} ; 2.5×10^{-14}

S11.2 Random-instance tests of the theorems

The tests of Table S40 use random instances drawn with a fixed seed and without any tuning: connected lattices of 3–8 by 3–8 cells with a random fraction of the cells (up to about 30%) removed, and four families of movement generators, namely symmetric rates built from log-normal cell mobilities, the departure-site rule (reversible with a non-uniform stationary distribution), non-reversible rates (independent log-normal rates in the two directions of every edge), and dense irreducible generators of order 3–11 that are not lattices. The efficiency q is log-normal, zoned with two to four levels, uniform with about 30% of zeros, or uniform on $[0.5, 2]$; $\beta \in [0.02, 3] \text{ day}^{-1}$ and $\gamma \in [0.05, 2] \text{ day}^{-1}$. All computations are in double precision with dense eigen-solvers; the whole table takes 38 s. Figure S24 shows the instances of the threshold and of the bounds tests.

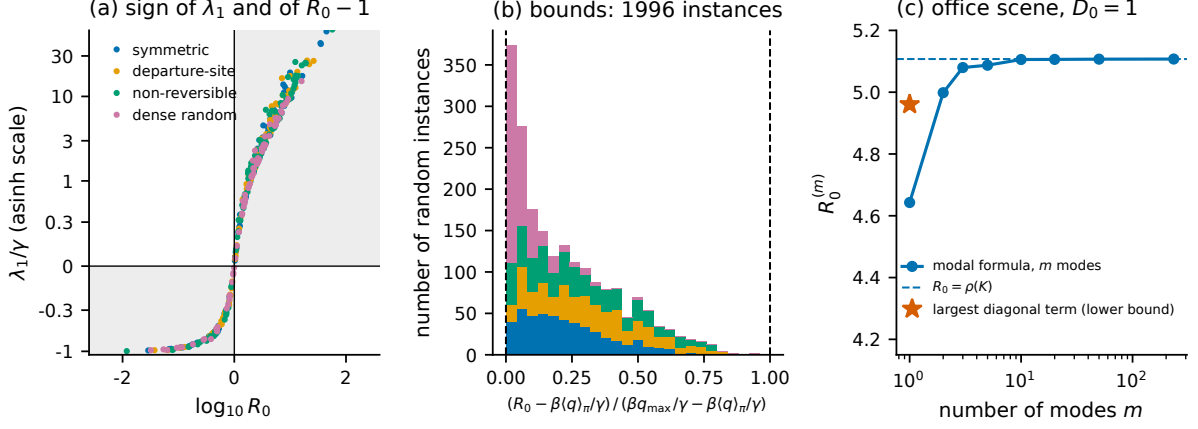


Figure S24: Numerical checks of Section 3. (a) Growth rate λ_1/γ against R_0 for the 400 random instances of the threshold test, by generator family; all points lie in the two quadrants allowed by Theorem 3.1(c). (b) Position of R_0 between the lower bound $\beta\langle q\rangle_\pi/\gamma$ (0) and the upper bound $\beta q_{\max}/\gamma$ (1) of Theorem 3.2 for the 1,996 of the 2,000 instances in which q is not constant (stacked by family, colours as in (a)). (c) Office scene, $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$, $\beta = 0.5 \text{ day}^{-1}$, $\gamma = 0.14 \text{ day}^{-1}$: largest eigenvalue of the leading block of the modal matrix of Corollary S2.5 against the number m of modes retained, which rises from $\beta\langle q\rangle/\gamma = 4.643$ to $R_0 = 5.107$ (dashed); the star is the largest diagonal term, 4.961, which is a lower bound (Corollary S2.5).

The table also shows why symmetric rates are assumed in Theorem 3.6: under the departure-site rule, tripling the mobility of one cell raised R_0 in 201, lowered it in 98 and left it unchanged (to 10^{-10}) in one of 300 instances.

S11.3 Re-implementation and searches for counter-examples

The second set of programs has its own generators and eigen-solvers and other seeds. It repeated the tests on 600 instances for the threshold (residual 2.1×10^{-14} , 0 sign failures), 3,000 for the bounds (violation 9×10^{-16}), 400 instances and 49 mobility values for the monotone decrease (no increase), 1,800 zones (no violation) and 1,200 edge changes (1.4×10^{-14}), with the same conclusions for the leaky room and the hotspot maps. (One check of the absorbing limit first stopped on a floating-point tolerance at exit rate 10^9 and passed when repeated with 10^6 .)

Random instances rarely come near equality, so an optimiser (Nelder–Mead, random restarts) was used to *minimise* $R_0/(\beta\langle q\rangle_\pi/\gamma) - 1$ and to *maximise* $R_0(D_2) - R_0(D_1)$, $D_2 > D_1$, over non-reversible dense generators of order 3–5 and over q . The first searches reported apparent violations of -6×10^{-2} for the lower bound and $+5 \times 10^{-4}$ for the monotonicity. Both were numerical artefacts. In the first, the stationary vector had been taken from an ill-conditioned eigen-solve and the optimiser had pushed the rates over 30 orders of magnitude; with the stationary vector from a linear solve and rates between e^{-12} and e^{12} , the minimum over 120 optimised instances is -7×10^{-13} . In the second, the increases occurred at condition numbers of $\mathbf{V}$ near 10^{12} , where a generator assembled in double precision is no longer exactly conservative. In a new search in which every optimised instance was evaluated in 60-digit arithmetic [66] with a generator whose column sums vanish exactly, the largest value of $R_0(D_2) - R_0(D_1)$ over 80 instances (rates between e^{-8} and e^8) was 8×10^{-59} . We repeated both searches with other seeds and evaluated every optimised instance in 60-digit arithmetic. For the lower bound, the double-precision minimum over 120 instances was -5×10^{-14} , and the exact values were all positive, the smallest being 2×10^{-18} . For the monotonicity, with rates confined to $[e^{-c}, e^c]$ for $c = 4, 8$ and 15 (40 instances each), double precision reported increases up to 1.2×10^{-6} , whereas the exact difference was positive in a single instance, by 1.5×10^{-62} , which is zero at the working precision. The practical lesson is one of numerical hygiene: a generator assembled in double

precision loses exact conservation, $\mathbf{1}^\top \mathbf{M} = 0$, once the products of mobility scale and hop rates span more than about ten orders of magnitude, and its stationary vector should be obtained from a linear solve; otherwise the theorems appear to fail, in the searches above by amounts between 10^{-8} and 10^{-2} . The tests of Table S40 use log-normal rates of unit order and mobility scales between 10^{-4} and 10^4 ($10^{\pm 9}$ only for the two limits, whose deviations are correspondingly larger).

S11.4 Nonlinear mean-field model

The threshold statement concerns the linearisation, so the full system (3) was integrated as well (LSODA, relative tolerance 10^{-9}), from a spatially uniform seed of 10^{-6} of the population. These runs used one of the two alternative office layouts of Section 3 (same zone fractions as the office scene; $R_0 = 5.109$ against 5.107 at the default parameters) and a 20×13 two-zone room, not the canonical scenes, and were not repeated on them. With β rescaled so that $R_0 = 0.80$ and 0.95 , the prevalence never exceeds the seed and the final attack fraction is 3.9×10^{-6} and 9.9×10^{-6} ; for $R_0 = 1.05, 1.20$ and 2.00 it is 5.1 %, 19.8 % and 72.6 %. In the two-zone room (workspace $q = 1.5$ on 69.2 % of the floor at mobility $0.3 D_0$, corridor band $q = 0.5$ at $2 D_0$, $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$) with β chosen so that the well-mixed estimate $\beta \langle q \rangle / \gamma$ equals 0.900, the reproduction number is $R_0 = 1.076$ and an outbreak occurs: the attack fraction is 12.4 %, with 97.2 % of the infections generated in the workspace, whereas the same room with q replaced by its mean has no outbreak (attack fraction 1.0×10^{-5}). The second implementation returned the same values to the digits quoted.

S11.5 The simulator

The individual-based model of Section 2.3 is a continuous-time Markov chain and is simulated without time discretisation by the direct method of Gillespie [18, 19] (Fig. S25). The production code is a C routine called from Python. It keeps, for every cell y , the exposure field $E_y = \sum_{i \text{ infective}} h(x_i, y)$ and the propensity $n_S(y) E_y$, updates them incrementally at each hop, and recomputes them from scratch after every infection and recovery and after every 20,000 hops, which bounds the accumulation of rounding errors. Every replicate has its own xoshiro256** stream [8] derived from the seed of the batch and the replicate index.

Because the infection tree is recorded, secondary cases are counted and not fitted; with $g_{\max} = 0$ only the index case transmits, which measures R_1 without interference from later cases. The *exposure* of the index case is the time integral of the rate at which it would infect if everybody within its contact neighbourhood were susceptible; its expectation for an index case born in cell x is the offspring number $n_{\text{off}}(x)$ of (7), so that it equals R_0 when x is drawn from the incidence map $\mathbf{w}$. This identity ties the simulator to the next-generation matrix without any approximation.

S11.6 Validation of the simulator

Table S41 lists seven tests; Fig. S26 shows three of them.

Tests A to D and G establish that the code simulates the stated model; test C in addition shows that the simulator and $\mathbf{K}$ describe the same model, so that the gap between $\bar{R}_1$ and the mean-field value in test D (ratios 0.17 to 0.93) is a property of discrete individuals and not of the code. Test F shows the convergence to the mean-field model, and how far the default parameters are from it.

Second simulator. The checking programs contain a separate exact simulator: it recomputes all rates from scratch at every event, selects an infection as an (infective, kernel cell) pair, uses another random-number generator, and has its own scene parser, next-generation matrix and pair

State. Cells $x_i \in \Omega$ and states $\sigma_i \in \{S, I, R\}$ of the N people; time t .

Initialisation. Positions independent and uniform on Ω (the stationary law); person 1 is the index case, infective at $t = 0$, at its own position or in a prescribed or randomly drawn cell (several initial infectives in test F of Table S41).

Loop, while at least one person is infective and $t < t_{\max}$:

1. Compute the total rates of hops, $H = \sum_i \sum_y \omega_{x_i \rightarrow y}$, of recoveries, γn_I , and of infections, $A = \chi(t) \sum_{i \in I} \sum_{j \in S} h(x_i, x_j)$, where n_I is the number of infectives, the sums in A run over infective people i and susceptible people j , h is the pair hazard (5) and $\chi(t)$ the schedule multiplier ($\chi \equiv 1$ without schedule).
2. Draw a waiting time Δt from the exponential distribution with rate $H + \gamma n_I + A$. If $t + \Delta t$ lies beyond the end of the current schedule segment, move t to that boundary, switch to the rates of the next segment and return to step 1 (exact, because the rates are constant within a segment).
3. Set $t \leftarrow t + \Delta t$ and choose the type of event with probabilities proportional to H , γn_I , A :
hop: a person i chosen with probability proportional to its exit rate moves to a neighbour y chosen with probability proportional to $\omega_{x_i \rightarrow y}$;
recovery: a uniformly chosen infective becomes removed;
infection: a pair (i, j) is chosen with probability proportional to $h(x_i, x_j)$; j becomes infective, and its infector i , generation $g_j = g_i + 1$, time and cell are recorded. With the option $g_{\max}$, a person of generation above $g_{\max}$ is counted as infected but does not transmit.

Output. Final size, infection tree, numbers of susceptible and infective people on a time grid, and the exposure $\int_0^\infty \chi(t) \sum_{j \neq 1} h(x_1, x_j) \mathbf{1}[\sigma_1 = I] dt$ of the index case.

Figure S25: Exact event-driven simulation of one epidemic. The per-cell rule of Remark S4.1 is obtained by replacing the infection rate A by $\sum_y \beta q_y n_S(y) n_I(y) / n_{\text{tot}}(y)$, with $n_S(y)$, $n_I(y)$ and $n_{\text{tot}}(y)$ the numbers of susceptible people, of infectives and of all people in cell y .

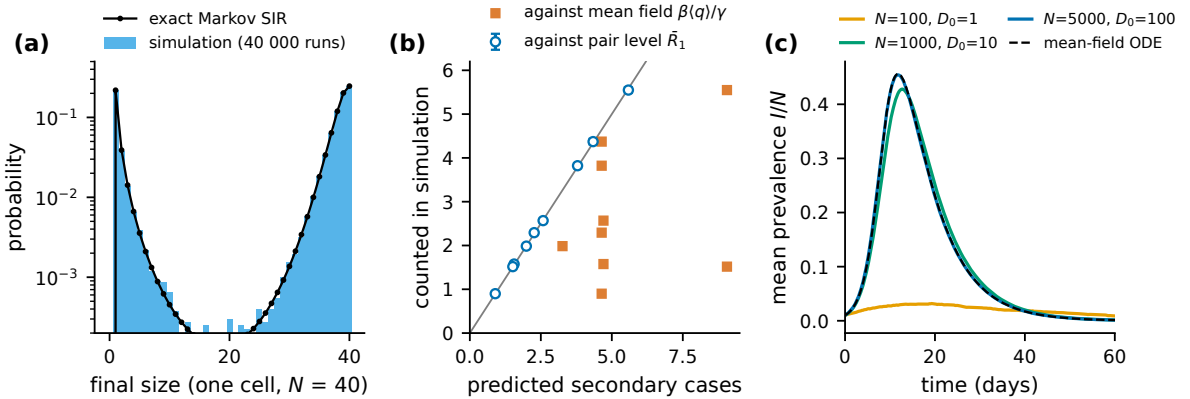


Figure S26: Validation of the simulator. (a) Test B: final-size distribution of the one-cell model with 40 people, simulation (40,000 runs) and exact law. (b) Test D: counted secondary cases of a uniformly placed index case ($g_{\max} = 0$, 20,000 runs per point; the 95 % intervals are about the size of the symbols) against $\bar{R}_1$ from the pair solve (circles) and against the mean-field value $\beta \langle q \rangle / \gamma$ (squares), for the nine configurations of Table S41. (c) Test F: mean prevalence in the office scene with 1 % initially infective for three combinations of the number of people N and the mobility scale D_0 ($\text{cell}^2 \text{ day}^{-1}$), and the mean-field solution for $D_0 = 100$ (dashed).

Table S41: Tests of the simulator. $\beta = 0.5 \text{ day}^{-1}$, $\gamma = 0.14 \text{ day}^{-1}$; scenes with their default occupancy and $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$ unless stated; z is a difference in units of its standard error.

Test	Result
A C routine against a separately written pure-Python implementation of Fig. S25 (own code and generator); 4×3 lattice with one barrier, two q zones, two mobility zones, 10 people; 20,000 against 1,500 runs; same-cell kernel, kernel of radius one cell, the per-cell rule of Remark S4.1, and $g_{\max} = 0$	mean final size 6.101 against 6.065, 6.776 against 6.755, 3.157 against 3.106, 3.190 against 3.240; $ z \leq 1.31$ for the eight means compared; χ^2 of the final-size histograms 3.6 to 10.4 on 9 degrees of freedom ($p \geq 0.32$)
B one cell, 40 people: Markovian SIR epidemic with pair hazard $\beta/39$, whose final-size law is computed exactly by recursion on the embedded jump chain [cf. 3]; 40,000 runs	mean final size 27.769 ± 0.084 against 27.808; probability of at most 3 cases 0.2725 against 0.2719; $\chi^2 = 39.8$ on 39 degrees of freedom ($p = 0.44$); larger samples in the text
C exposure identity; office scene (same-cell kernel) and classroom (1 m kernel); 40,000 runs for random index cells, 10,000 for fixed cells	office: index from $\mathbf{w}$, 5.055 (4.996–5.115) against $R_0 = 5.107$; uniform index, 4.681 (4.625–4.737) against $\beta\langle q\rangle/\gamma = 4.643$; cells with largest and smallest $n_{\text{off}}(x)$, 5.449 (5.318–5.581) against 5.335 and 3.962 (3.864–4.060) against 3.995. Classroom: 5.855 (5.787–5.922) against 5.866; 4.697 (4.640–4.753) against 4.706; 6.939 (6.787–7.090) against 6.892; 3.010 (2.936–3.084) against 3.029. Largest $ z = 1.72$
D counted secondary cases with $g_{\max} = 0$ and a uniformly placed index case against $\bar{R}_1$ from the exact pair solve (12); nine configurations, 20,000 runs each	office, $D_0 = 0.1, 1, 10, 100$: 0.899, 2.293, 3.821, 4.372 against 0.895, 2.266, 3.793, 4.341; supermarket 1.985 against 1.990; classroom 1.576 against 1.559 (same cell) and 2.568 against 2.581 (1 m); metro 1.517 against 1.514 and 5.549 against 5.584. All nine values of $\bar{R}_1$ lie inside the 95 % intervals; $ z \leq 1.52$; differences at most 1.2 %. The mean-field values $\beta\langle q\rangle/\gamma$ are 4.643, 3.263, 4.706 and 9.056
E per-cell rule of Remark S4.1, 100 people, 20,000 runs	secondary cases of the index case 0.690 (0.676–0.704) in the office scene and 0.382 (0.372–0.392) in a uniform 20×20 room; mean final size 2.6 and 1.6 people (Remark S4.1)
F mean-field limit: office scene, 1 % initially infective; mean prevalence curve against the lattice system (3)	largest absolute difference of the curves 0.423 ($N = 100$, $D_0 = 1$; 200 runs), 0.058 ($N = 1000$, $D_0 = 10$; 200 runs), 0.004 ($N = 5000$, $D_0 = 100$; 40 runs); final size 0.9898 against 0.9900 in the last case
G schedules, office scene: (i) three identical segments per day against no schedule, 20,000 runs each; (ii) room occupied during the first third of each day, 40,000 runs	(i) mean final size 23.61 against 23.69 people ($ z = 0.31$), secondary cases of the index 1.872 against 1.861 ($ z = 0.57$), fraction of major outbreaks 0.5001 against 0.4996; (ii) exposure 1.631 (1.609–1.653) against $\beta\langle q\rangle \times 2.493 \text{ days} = 1.620$

solver. On the lattice of test A it gives 2.1818 ± 0.0045 counted secondary cases against 2.1782 from its pair solve (same-cell kernel; 200,000 runs), 2.5138 ± 0.0050 against 2.5163 (radius-one kernel), and exposures 4.419 ± 0.012 and 4.425 ± 0.011 against 4.416. In square rooms of side 4 to 20 with $D = 0.51, 1$ and $10 \text{ cell}^2 \text{ day}^{-1}$ its counts agree with its pair solve within 1.65 standard errors in seven configurations (60,000 to 200,000 runs each). Re-run for this section on the office scene, it gives an exposure of 4.627 ± 0.009 for a uniform index case against 4.643 (400,000 runs; $z = -1.7$) and 5.105 ± 0.014 for an index case drawn from $\mathbf{w}$ against $R_0 = 5.107$ (200,000 runs). In about 25 configurations in all the two simulators agree within sampling error, the largest differences being 1.6 to 2.3 standard errors (outbreak probability and peak day in the metro scene, outbreak probability in the office at $D_0 = 10$), and the stored results that were re-run from their seeds were reproduced bit for bit. Two items were not re-simulated with the second simulator: the scene results at $D_0 = 100$ and the run with 5,000 people of test F, which was only repeated with the same code and another seed (largest difference 0.007).

The exact final-size law with larger samples. On the one-cell model the second simulator’s own batch of 40,000 runs gave a mean of 27.872 ± 0.084 and $\chi^2 = 55.7$ on 39 degrees of freedom ($p = 0.040$). To find out whether this signals a defect, both simulators were run 4×10^6 times. The second simulator then agreed with the exact law (mean 27.812 ± 0.008 , $\chi^2 = 45.4$, $p = 0.22$), but the mean of the production code lay 2.6 standard errors above the exact value (27.830 ± 0.008 ; $\chi^2 = 30.2$, $p = 0.84$). A further 2×10^7 runs of each with other seeds gave 27.805 ± 0.004 (0.7 standard errors below; $\chi^2 = 44.2$, $p = 0.26$) for the production code and 27.811 ± 0.004 ($\chi^2 = 32.2$, $p = 0.77$) for the second simulator. Over all 2.4×10^7 runs the means are 27.8096 ± 0.0034 and 27.8110 ± 0.0034 against the exact 27.8082, with $\chi^2 = 34.1$ ($p = 0.69$) and 37.4 ($p = 0.54$). We conclude that both deviations were sampling fluctuations; they are reported because they were observed.

S11.7 Reproducibility

All random numbers come from generators with fixed seeds that are set in the source of each script (for example 20261001 for the theorem tests and one seed per validation test), so that re-running a script reproduces its stored output. The computations used Python 3.14.6 with NumPy 2.5.3 [22], SciPy 1.18.1 [72], Matplotlib 3.11.2 [28] and mpmath 1.3.0 [66]; the C routines were compiled with Apple clang 21 at optimisation level 2. On one core of a laptop (Apple M4 Pro) the production simulator needs 0.5 ms for one epidemic in the office scene (100 people, $D_0 = 1 \text{ cell}^2 \text{ day}^{-1}$, about 5,900 events), against 1.8 s for the pure-Python reference implementation; the validation tests take about 8 minutes, the theorem tests 38 seconds, the script of this section about 9 minutes, and all simulation experiments of the article 3 to 4 hours, with a peak memory use below 1.5 GB. In the repository [79], `code` holds the simulator (the C routine, its Python front end, the reference implementation and the deterministic solvers), one script per experiment, the theorem tests, the checking programs and one script per section of this article; `data` holds the output of every script, mostly as JSON files, which are the primary record from which all tables and figures are generated; and `reproduce.md` gives the command for each of them.

References

- [1] Linda J. S. Allen, Benjamin M. Bolker, Yuan Lou, and Andrew L. Nevai. Asymptotic profiles of the steady states for an SIS epidemic patch model. *SIAM Journal on Applied Mathematics*, 67(5): 1283–1309, 2007. doi: 10.1137/060672522.
- [2] Lee Altenberg. Resolvent positive linear operators exhibit the reduction phenomenon. *Proceedings of the National Academy of Sciences*, 109(10):3705–3710, 2012. doi: 10.1073/pnas.1113833109.
- [3] Håkan Andersson and Tom Britton. *Stochastic Epidemic Models and Their Statistical Analysis*. Springer, New York, 2000. doi: 10.1007/978-1-4612-1158-7.
- [4] Nicolas Bacaër and Souad Guernaoui. The epidemic threshold of vector-borne diseases with seasonality. *Journal of Mathematical Biology*, 53(3):421–436, 2006. doi: 10.1007/s00285-006-0015-0.
- [5] M. G. Baker, C. N. Thornley, C. Mills, S. Roberts, S. Perera, J. Peters, A. Kelso, I. Barr, and N. Wilson. Transmission of pandemic A/H1N1 2009 influenza on passenger aircraft: retrospective cohort study. *BMJ*, 340:c2424, 2010. doi: 10.1136/bmj.c2424.
- [6] Abraham Berman and Robert J. Plemmons. *Nonnegative Matrices in the Mathematical Sciences*. Society for Industrial and Applied Mathematics, Philadelphia, PA, 1994. doi: 10.1137/1.9781611971262.
- [7] Derdei Bichara, Yun Kang, Carlos Castillo-Chavez, Richard Horan, and Charles Perrings. SIS and SIR epidemic models under virtual dispersal. *Bulletin of Mathematical Biology*, 77(11):2004–2034, 2015. doi: 10.1007/s11538-015-0113-5.
- [8] David Blackman and Sebastiano Vigna. Scrambled linear pseudorandom number generators. *ACM Transactions on Mathematical Software*, 47(4):1–32, 2021. doi: 10.1145/3460772.
- [9] Jing Cai, Wenjie Sun, Jianping Huang, Michelle Gamber, Jing Wu, and Guiqing He. Indirect virus transmission in cluster of COVID-19 cases, Wenzhou, China, 2020. *Emerging Infectious Diseases*, 26(6):1343–1345, 2020. doi: 10.3201/eid2606.200412.

- [10] Ting Chen, Songxue Guo, and Ping Zhong. Epidemic characteristics of the COVID-19 outbreak in Tianjin, a well-developed city in China. *American Journal of Infection Control*, 48(9):1068–1073, 2020. doi: 10.1016/j.ajic.2020.06.006.
- [11] Kai-Chung Cheng, Viviana Acevedo-Bolton, Ruo-Ting Jiang, Neil E. Klepeis, Wayne R. Ott, Oliver B. Fringer, and Lynn M. Hildemann. Modeling exposure close to air pollution sources in naturally ventilated residences: association of turbulent diffusion coefficient with air change rate. *Environmental Science & Technology*, 45(9):4016–4022, 2011. doi: 10.1021/es103080p.
- [12] Joel E. Cohen. Convexity of the dominant eigenvalue of an essentially nonnegative matrix. *Proceedings of the American Mathematical Society*, 81(4):657–658, 1981. doi: 10.1090/s0002-9939-1981-0601750-2.
- [13] Akira Endo, Mitsuo Uchida, Naoki Hayashi, Yang Liu, Katherine E. Atkins, Adam J. Kucharski, and Sebastian Funk. Within and between classroom transmission patterns of seasonal influenza among primary school students in Matsumoto city, Japan. *Proceedings of the National Academy of Sciences*, 118(46):e2112605118, 2021. doi: 10.1073/pnas.2112605118.
- [14] Yuki Furuse, Eiichiro Sando, Naho Tsuchiya, Reiko Miyahara, Ikkoh Yasuda, Yura K. Ko, Mayuko Saito, Konosuke Morimoto, Takeaki Imamura, Yugo Shobugawa, et al. Clusters of coronavirus disease in communities, Japan, January–April 2020. *Emerging Infectious Diseases*, 26(9):2176–2179, 2020. doi: 10.3201/eid2609.202272.
- [15] Daozhou Gao and Chao-Ping Dong. Fast diffusion inhibits disease outbreaks. *Proceedings of the American Mathematical Society*, 148(4):1709–1722, 2020. doi: 10.1090/proc/14868.
- [16] Mathieu Génois and Alain Barrat. Can co-location be used as a proxy for face-to-face contacts? *EPJ Data Science*, 7(1):11, 2018. doi: 10.1140/epjds/s13688-018-0140-1.
- [17] Mathieu Génois, Christian L. Vestergaard, Julie Fournet, André Panisson, Isabelle Bonmarin, and Alain Barrat. Data on face-to-face contacts in an office building suggest a low-cost vaccination strategy based on community linkers. *Network Science*, 3(3):326–347, 2015. doi: 10.1017/nws.2015.10.
- [18] Daniel T. Gillespie. A general method for numerically simulating the stochastic time evolution of coupled chemical reactions. *Journal of Computational Physics*, 22(4):403–434, 1976. doi: 10.1016/0021-9991(76)90041-3.
- [19] Daniel T. Gillespie. Exact stochastic simulation of coupled chemical reactions. *The Journal of Physical Chemistry*, 81(25):2340–2361, 1977. doi: 10.1021/j100540a008.
- [20] Thomas Günther, Manja Czech-Sioli, Daniela Indenbirken, Alexis Robitaille, Peter Tenhaken, Martin Exner, Matthias Ottinger, Nicole Fischer, Adam Grundhoff, and Melanie M. Brinkmann. SARS-CoV-2 outbreak investigation in a German meat processing plant. *EMBO Molecular Medicine*, 12(12):EMMM202013296, 2020. doi: 10.15252/emmm.202013296.
- [21] Lea Hamner, Polly Dubbel, Ian Capron, Andy Ross, Amber Jordan, Jaxon Lee, Joanne Lynn, Amelia Ball, Simranjit Narwal, Sam Russell, Dale Patrick, and Howard Leibrand. High SARS-CoV-2 attack rate following exposure at a choir practice — Skagit County, Washington, March 2020. *MMWR. Morbidity and Mortality Weekly Report*, 69(19):606–610, 2020. doi: 10.15585/mmwr.mm6919e6.
- [22] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, et al. Array programming with NumPy. *Nature*, 585(7825):357–362, 2020. doi: 10.1038/s41586-020-2649-2.
- [23] Rebecca B. Hershow, Karen Wu, Nathaniel M. Lewis, Alison T. Milne, Dustin Currie, Amanda R. Smith, Spencer Lloyd, Brian Orleans, Erin L. Young, Brandi Freeman, et al. Low SARS-CoV-2 transmission in elementary schools — Salt Lake County, Utah, December 3, 2020–January 31, 2021. *MMWR. Morbidity and Mortality Weekly Report*, 70(12):442–448, 2021. doi: 10.15585/mmwr.mm7012e3.
- [24] Vicki Stover Hertzberg and Howard Weiss. On the 2-row rule for infectious disease transmission on aircraft. *Annals of Global Health*, 82(5):819–823, 2016. doi: 10.1016/j.aogh.2016.06.003.
- [25] Sebastian Hoehl, Onur Karaca, Niko Kohmer, Sandra Westhaus, Jürgen Graf, Udo Goetsch, and Sandra Ciesek. Assessment of SARS-CoV-2 transmission on an international flight and among a tourist group. *JAMA Network Open*, 3(8):e2018044, 2020. doi: 10.1001/jamanetworkopen.2020.18044.
- [26] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, Cambridge, 2nd edition, 2013. doi: 10.1017/CBO9781139020411.
- [27] Maogui Hu, Hui Lin, Jinfeng Wang, Chengdong Xu, Andrew J. Tatem, Bin Meng, Xin Zhang, Yifeng Liu, Pengda Wang, Guizhen Wu, Haiyong Xie, and Shengjie Lai. Risk of coronavirus disease 2019 transmission in train passengers: an epidemiological and modeling study. *Clinical Infectious Diseases*, 72(4):604–610, 2021. doi: 10.1093/cid/ciaa1057.

- [28] John D. Hunter. Matplotlib: a 2D graphics environment. *Computing in Science & Engineering*, 9(3): 90–95, 2007. doi: 10.1109/MCSE.2007.55.
- [29] Sukbin Jang, Si Hyun Han, and Ji-Young Rhee. Cluster of coronavirus disease associated with fitness dance classes, South Korea. *Emerging Infectious Diseases*, 26(8):1917–1920, 2020. doi: 10.3201/eid2608.200633.
- [30] Guanyu Jiang, Can Wang, Lu Song, Xing Wang, Yangyang Zhou, Chunnan Fei, and He Liu. Aerosol transmission, an indispensable route of COVID-19 spread: case study of a department-store cluster. *Frontiers of Environmental Science & Engineering*, 15(3):46, 2021. doi: 10.1007/s11783-021-1386-6.
- [31] Anthea L. Katelaris, Jessica Wells, Penelope Clark, Sophie Norton, Rebecca Rockett, Alicia Arnott, Vitali Sintchenko, Stephen Corbett, and Shopna K. Bag. Epidemiologic evidence for airborne transmission of SARS-CoV-2 during church singing, Australia, 2020. *Emerging Infectious Diseases*, 27(6):1677–1680, 2021. doi: 10.3201/eid2706.210465.
- [32] Tosio Kato. Superconvexity of the spectral radius, and convexity of the spectral bound and the type. *Mathematische Zeitschrift*, 180(3):265–273, 1982. doi: 10.1007/bf01318910.
- [33] Tosio Kato. *Perturbation Theory for Linear Operators*. Classics in Mathematics. Springer, Berlin, Heidelberg, reprint of the 2nd (1980) edition, 1995. ISBN 978-3-540-58661-6. doi: 10.1007/978-3-642-66282-9.
- [34] Thomas A. Kenyon, Sarah E. Valway, Walter W. Ihle, Ida M. Onorato, and Kenneth G. Castro. Transmission of multidrug-resistant *Mycobacterium tuberculosis* during a long airplane flight. *New England Journal of Medicine*, 334(15):933–938, 1996. doi: 10.1056/NEJM199604113341501.
- [35] Nguyen Cong Khanh, Pham Quang Thai, Ha-Linh Quach, Ngoc-Anh Hoang Thi, Phung Cong Dinh, Tran Nhu Duong, Le Thi Quynh Mai, Ngu Duy Nghia, Tran Anh Tu, La Ngoc Quang, et al. Transmission of SARS-CoV 2 during long-haul flight. *Emerging Infectious Diseases*, 26(11):2617–2624, 2020. doi: 10.3201/eid2611.203299.
- [36] Sean M. Kinahan, David B. Silcott, Blake E. Silcott, Ryan M. Silcott, Peter J. Silcott, Braden J. Silcott, Steven L. Distelhorst, Vicki L. Herrera, Danielle N. Rivera, Kevin K. Crown, et al. Aerosol tracer testing in Boeing 767 and 777 aircraft to simulate exposure potential of infectious aerosol such as SARS-CoV-2. *PLoS ONE*, 16(12):e0246916, 2021. doi: 10.1371/journal.pone.0246916.
- [37] David Koh. Migrant workers and COVID-19. *Occupational and Environmental Medicine*, 77(9): 634–636, 2020. doi: 10.1136/oemed-2020-106626.
- [38] Keun-Sang Kwon, Jung-Im Park, Young Joon Park, Don-Myung Jung, Ki-Wahn Ryu, and Ju-Hyung Lee. Evidence of long-distance droplet transmission of SARS-CoV-2 by direct air flow in a restaurant in Korea. *Journal of Korean Medical Science*, 35(46):e415, 2020. doi: 10.3346/jkms.2020.35.e415.
- [39] Tracy Lam-Hine, Stephen A. McCurdy, Lisa Santora, Lael Duncan, Russell Corbett-Detig, Beatrix Kapusinszky, and Matthew Willis. Outbreak associated with SARS-CoV-2 B.1.617.2 (Delta) variant in an elementary school — Marin County, California, May–June 2021. *MMWR. Morbidity and Mortality Weekly Report*, 70(35):1214–1219, 2021. doi: 10.15585/mmwr.mm7035e2.
- [40] Fan-Yun Lan, Christian Suharlim, Stefanos N. Kales, and Justin Yang. Association between SARS-CoV-2 infection, exposure risk and mental health among a cohort of essential retail workers in the USA. *Occupational and Environmental Medicine*, 78(4):237–243, 2021. doi: 10.1136/oemed-2020-106774.
- [41] Wenning Li, Jianhua Gong, Jieping Zhou, Hongkui Fan, Cheng Qin, Yujiang Gong, and Weidong Hu. The analysis of patterns of two COVID-19 outbreak clusters in China. *International Journal of Environmental Research and Public Health*, 19(8):4876, 2022. doi: 10.3390/ijerph19084876.
- [42] Yuguo Li, Hua Qian, Jian Hang, Xuguang Chen, Pan Cheng, Hong Ling, Shengqi Wang, Peng Liang, Jiansen Li, Shenglan Xiao, et al. Probable airborne transmission of SARS-CoV-2 in a poorly ventilated restaurant. *Building and Environment*, 196:107788, 2021. doi: 10.1016/j.buildenv.2021.107788.
- [43] James O. Lloyd-Smith, Sebastian J. Schreiber, P. Ekkehard Kopp, and Wayne M. Getz. Superspreading and the effect of individual variation on disease emergence. *Nature*, 438(7066):355–359, 2005. doi: 10.1038/nature04153.
- [44] Jianyun Lu, Jieni Gu, Kuibiao Li, Conghui Xu, Wenzhe Su, Zhisheng Lai, Deqian Zhou, Chao Yu, Bin Xu, and Zhicong Yang. COVID-19 outbreak associated with air conditioning in restaurant, Guangzhou, China, 2020. *Emerging Infectious Diseases*, 26(7):1628–1631, 2020. doi: 10.3201/eid2607.200764.
- [45] Kaiwei Luo, Zhao Lei, Zheng Hai, Shanliang Xiao, Jia Rui, Hao Yang, Xinping Jing, Hui Wang, Zhengshen Xie, Ping Luo, et al. Transmission of SARS-CoV-2 in public transportation vehicles: a case study in Hunan Province, China. *Open Forum Infectious Diseases*, 7(10):ofaa430, 2020. doi: 10.1093/ofid/ofaa430.
- [46] Rossana Mastrandrea, Julie Fournet, and Alain Barrat. Contact patterns in a high school: a

- comparison between data collected using wearable sensors, contact diaries and friendship surveys. *PLoS ONE*, 10(9):e0136497, 2015. doi: 10.1371/journal.pone.0136497.
- [47] Shelly L. Miller, William W. Nazaroff, Jose L. Jimenez, Atze Boerstra, Giorgio Buonanno, Stephanie J. Dancer, Jarek Kurnitski, Linsey C. Marr, Lidia Morawska, and Catherine Noakes. Transmission of SARS-CoV-2 by inhalation of respiratory aerosol in the Skagit Valley Chorale superspreading event. *Indoor Air*, 31(2):314–323, 2021. doi: 10.1111/ina.12751.
 - [48] Kenji Mizumoto, Katsushi Kagaya, Alexander Zarebski, and Gerardo Chowell. Estimating the asymptomatic proportion of coronavirus disease 2019 (COVID-19) cases on board the Diamond Princess cruise ship, Yokohama, Japan, 2020. *Eurosurveillance*, 25(10):2000180, 2020. doi: 10.2807/1560-7917.es.2020.25.10.2000180.
 - [49] Elliott W. Montroll and George H. Weiss. Random walks on lattices. II. *Journal of Mathematical Physics*, 6(2):167–181, 1965. doi: 10.1063/1.1704269.
 - [50] P. A. P. Moran. The interpretation of statistical maps. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 10(2):243–251, 1948. doi: 10.1111/j.2517-6161.1948.tb00012.x.
 - [51] Michael R. Moser, Thomas R. Bender, Harold S. Margolis, Gary R. Noble, Alan P. Kendal, and Donald G. Ritter. An outbreak of influenza aboard a commercial airliner. *American Journal of Epidemiology*, 110(1):1–6, 1979. doi: 10.1093/oxfordjournals.aje.a112781.
 - [52] Edward A. Nardell, Joann Keegan, Sally A. Cheney, and Sue C. Etkind. Airborne infection: theoretical limits of protection achievable by building ventilation. *American Review of Respiratory Disease*, 144(2):302–306, 1991. doi: 10.1164/ajrccm/144.2.302.
 - [53] G. Nicholls, B. Atkinson, K. van Veldhoven, I. Nicholls, M. Coldwell, A. Clarke, C. J. Atchison, A. I. Raja, A. M. Bennett, D. Morgan, et al. An outbreak of SARS-CoV-2 in a public-facing office in England. *Occupational Medicine*, 74(7):475–485, 2024. doi: 10.1093/occmed/kqad100.
 - [54] Sonja J. Olsen, Hsiao-Ling Chang, Terence Yung-Yan Cheung, Antony Fai-Yu Tang, Tamara L. Fisk, Steven Peng-Lim Ooi, Hung-Wei Kuo, Donald Dah-Shyong Jiang, Kow-Tong Chen, Jim Lando, et al. Transmission of the severe acute respiratory syndrome on aircraft. *New England Journal of Medicine*, 349(25):2416–2422, 2003. doi: 10.1056/NEJMoa031349.
 - [55] Cuiyun Ou, Shixiong Hu, Kaiwei Luo, Hongyu Yang, Jian Hang, Pan Cheng, Zheng Hai, Shanliang Xiao, Hua Qian, Shenglan Xiao, et al. Insufficient ventilation led to a probable long-range airborne transmission of SARS-CoV-2 on two buses. *Building and Environment*, 207:108414, 2022. doi: 10.1016/j.buildenv.2021.108414.
 - [56] Shin Young Park, Young-Man Kim, Seonju Yi, Sangeun Lee, Baeg-Ju Na, Chang Bo Kim, Jung-il Kim, Hea Sook Kim, Young Bok Kim, Yoojin Park, et al. Coronavirus disease outbreak in call center, South Korea. *Emerging Infectious Diseases*, 26(8):1666–1670, 2020. doi: 10.3201/eid2608.201274.
 - [57] E. C. Riley, G. Murphy, and R. L. Riley. Airborne spread of measles in a suburban elementary school. *American Journal of Epidemiology*, 107(5):421–432, 1978. doi: 10.1093/oxfordjournals.aje.a112560.
 - [58] Z. Schuss, A. Singer, and D. Holcman. The narrow escape problem for diffusion in cellular microdomains. *Proceedings of the National Academy of Sciences*, 104(41):16098–16103, 2007. doi: 10.1073/pnas.0706599104.
 - [59] Ye Shen, Changwei Li, Hongjun Dong, Zhen Wang, Leonardo Martinez, Zhou Sun, Andreas Handel, Zhiping Chen, Enfu Chen, Mark H. Ebell, et al. Community outbreak investigation of SARS-CoV-2 transmission among bus riders in eastern China. *JAMA Internal Medicine*, 180(12):1665–1671, 2020. doi: 10.1001/jamainternmed.2020.5225.
 - [60] Hollie Speake, Anastasia Phillips, Tracie Chong, Chisha Sikazwe, Avram Levy, Jurissa Lang, Benjamin Scalley, David J. Speers, David W. Smith, Paul Effler, et al. Flight-associated transmission of severe acute respiratory syndrome coronavirus 2 corroborated by whole-genome sequencing. *Emerging Infectious Diseases*, 26(12):2872–2880, 2020. doi: 10.3201/eid2612.203910.
 - [61] Michele Starnini, Andrea Baronchelli, and Romualdo Pastor-Satorras. Modeling human dynamics of face-to-face interaction networks. *Physical Review Letters*, 110(16):168701, 2013. doi: 10.1103/PhysRevLett.110.168701.
 - [62] Dietrich Stauffer and Amnon Aharony. *Introduction to Percolation Theory*. Taylor & Francis, London, 2nd edition, 1994. doi: 10.1201/9781315274386.
 - [63] Juliette Stehlé, Nicolas Voirin, Alain Barrat, Ciro Cattuto, Lorenzo Isella, Jean-François Pinton, Marco Quaggiotto, Wouter Van den Broeck, Corinne Régis, Bruno Lina, et al. High-resolution measurements of face-to-face contact patterns in a primary school. *PLoS ONE*, 6(8):e23176, 2011. doi: 10.1371/journal.pone.0023176.
 - [64] Chen Stein-Zamir, Nitza Abramson, Hanna Shoob, Erez Libal, Menachem Bitan, Tanya Cardash,

- Rafael Cayam, and Ian Miskin. A large COVID-19 outbreak in a high school 10 days after schools' reopening, Israel, May 2020. *Eurosurveillance*, 25(29):2001352, 2020. doi: 10.2807/1560-7917.es.2020.25.29.2001352.
- [65] Tara Swadi, Jemma L. Geoghegan, Tom Devine, Caroline McElnay, Jillian Sherwood, Phil Shoemack, Xiaoyun Ren, Matt Storey, Sarah Jefferies, Erasmus Smit, et al. Genomic evidence of in-flight transmission of SARS-CoV-2 despite predeparture testing. *Emerging Infectious Diseases*, 27(3): 687–693, 2021. doi: 10.3201/eid2703.204714.
 - [66] The mpmath development team. mpmath: a Python library for arbitrary-precision floating-point arithmetic (version 1.3.0), 2023. URL <https://mpmath.org/>.
 - [67] Suochen Tian, Min Wu, Zhenqin Chang, Yunxia Wang, Guijie Zhou, Wenming Zhang, Junmin Xing, Hui Tian, Xihong Zhang, Xiuli Zou, et al. Epidemiological investigation and intergenerational clinical characteristics of 24 coronavirus disease patients associated with a supermarket cluster: a retrospective study. *BMC Public Health*, 21(1):647, 2021. doi: 10.1186/s12889-021-10713-z.
 - [68] Joseph H. Tien, Zhisheng Shuai, Marisa C. Eisenberg, and P. van den Driessche. Disease invasion on community networks with environmental pathogen movement. *Journal of Mathematical Biology*, 70(5):1065–1092, 2015. doi: 10.1007/s00285-014-0791-x.
 - [69] Takao Toyokawa, Tomoe Shimada, Takahiro Hayamizu, Tsuyoshi Sekizuka, Yuji Zukeyama, Miyako Yasuda, Yuko Nakamura, Sho Okano, Jun Kudaka, Tetsuya Kakita, et al. Transmission of SARS-CoV-2 during a 2-h domestic flight to Okinawa, Japan, March 2020. *Influenza and Other Respiratory Viruses*, 16(1):63–71, 2022. doi: 10.1111/irv.12913.
 - [70] Philippe Vanhems, Alain Barrat, Ciro Cattuto, Jean-François Pinton, Nagham Khanafer, Corinne Régis, Byeul-a Kim, Brigitte Comte, and Nicolas Voirin. Estimating potential infection transmission routes in hospital wards using wearable proximity sensors. *PLoS ONE*, 8(9):e73970, 2013. doi: 10.1371/journal.pone.0073970.
 - [71] Margaretha Annelie Vink, Martinus Christoffel Jozef Bootsma, and Jacco Wallinga. Serial intervals of respiratory infectious diseases: a systematic review and analysis. *American Journal of Epidemiology*, 180(9):865–875, 2014. doi: 10.1093/aje/kwu209.
 - [72] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17(3):261–272, 2020. doi: 10.1038/s41592-019-0686-2.
 - [73] Wendi Wang and Xiao-Qiang Zhao. Threshold dynamics for compartmental epidemic models in periodic environments. *Journal of Dynamics and Differential Equations*, 20(3):699–717, 2008. doi: 10.1007/s10884-008-9111-8.
 - [74] Dana Weissberg, Jürg Böni, Silvana K. Rampini, Verena Kufner, Maryam Zaheri, Peter W. Schreiber, Irene A. Abela, Michael Huber, Hugo Sax, and Aline Wolfensberger. Does respiratory co-infection facilitate dispersal of SARS-CoV-2? investigation of a super-spreading event in an open-space office. *Antimicrobial Resistance & Infection Control*, 9(1):191, 2020. doi: 10.1186/s13756-020-00861-z.
 - [75] Tze-wai Wong, Chin-kei Lee, Wilson Tam, Joseph Tak-fai Lau, Tak-sun Yu, Siu-fai Lui, Paul K. S. Chan, Yuguo Li, Joseph S. Bresee, Joseph J. Y. Sung, et al. Cluster of SARS among medical students exposed to single patient, Hong Kong. *Emerging Infectious Diseases*, 10(2):269–276, 2004. doi: 10.3201/eid1002.030452.
 - [76] Huw Woodward, Rick J. B. de Kreijl, Emily S. Kruger, Shiwei Fan, Arvind Tiwari, Sarkawt Hama, Simon Noel, Megan S. Davies Wykes, Prashant Kumar, and Paul F. Linden. An evaluation of the risk of airborne transmission of COVID-19 on an inter-city train carriage. *Indoor Air*, 32(10):e13121, 2022. doi: 10.1111/ina.13121.
 - [77] Nicholas Young, Richard Pebody, Gillian Smith, Babatunde Olowokure, Giri Shankar, Katja Hoschler, Monica Galiano, Helen Green, Anders Wallensten, Angela Hogan, et al. International flight-related transmission of pandemic influenza A(H1N1)pdm09: an historical cohort study of the first identified cases in the United Kingdom. *Influenza and Other Respiratory Viruses*, 8(1):66–73, 2014. doi: 10.1111/irv.12181.
 - [78] I. T. S. Yu, T. W. Wong, Y. L. Chiu, N. Lee, and Y. Li. Temporal-spatial analysis of severe acute respiratory syndrome among hospital inpatients. *Clinical Infectious Diseases*, 40(9):1237–1243, 2005. doi: 10.1086/428735.
 - [79] Xiaoxiao Zhouyi. indoor-epidemic-random-walk: code and data, version 1.2.0. <https://github.com/zhouyi-xiaoxiao/indoor-epidemic-random-walk>, 2026.